CS7015 (Deep Learning) : Lecture 12

Visualizing Convolutional Neural Networks, Guided Backpropagation, Deep Dream, Deep Art, Fooling Convolutional Neural Networks

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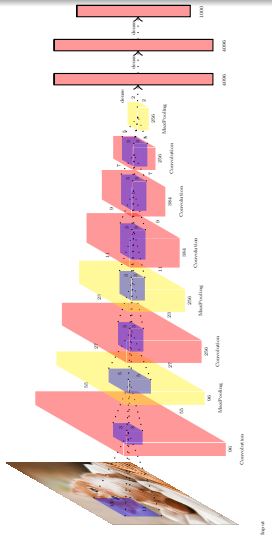
Acknowledgements

- Andrej Karpathy Video Lecture on Visualization and Deep Dream*

*Visualization, Deep Dream, Neural Style, Adversarial Examples

Module : Visualizing patches which maximally activate a neuron



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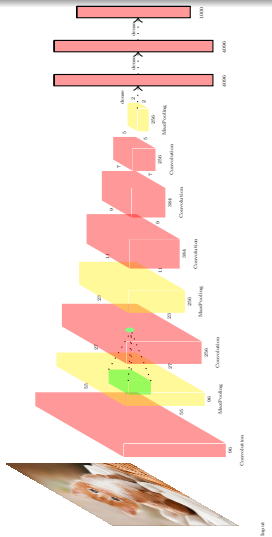

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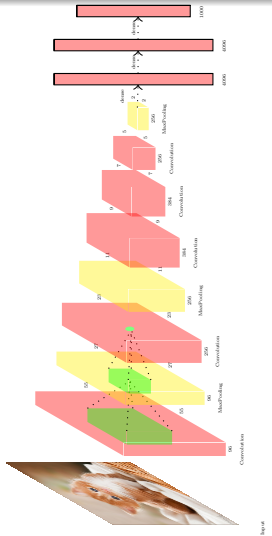

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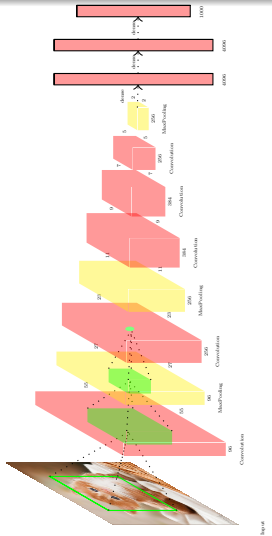

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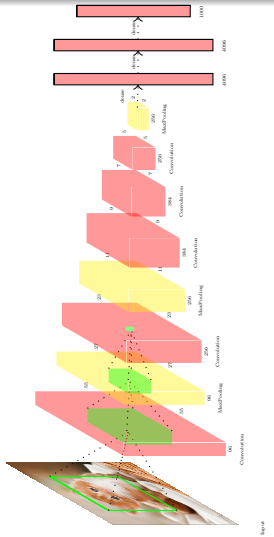


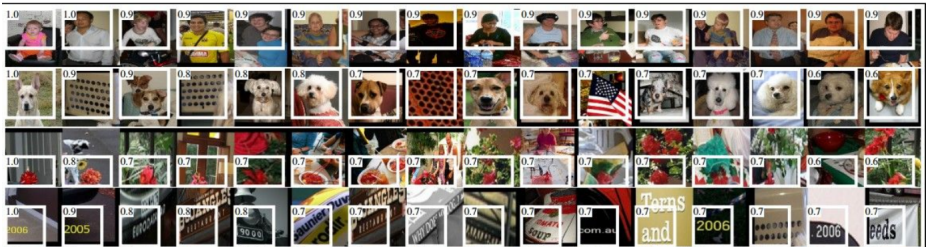
- Consider some neurons in a given layer of a CNN
- We can feed in images to this CNN and identify the images which cause these neurons to fire
- We can then trace back to the patch in the image which causes these neurons to fire
- Let us look at the result of one of such experiment conducted by Grishick et al., 2014

- They consider 6 neurons in the pool5 layer and find the image patches which cause these neurons to fire
- One neuron fires for people faces

- They consider 6 neurons in the pool5 layer and find the image patches which cause these neurons to fire
- Another neuron fires for dog faces

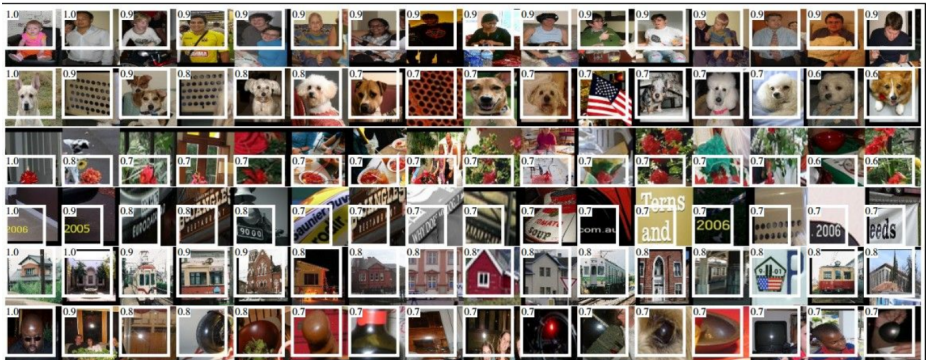
- They consider 6 neurons in the pool5 layer and find the image patches which cause these neurons to fire
- Another neuron fires for flowers

- They consider 6 neurons in the pool5 layer and find the image patches which cause these neurons to fire
- Another neuron fires for numbers

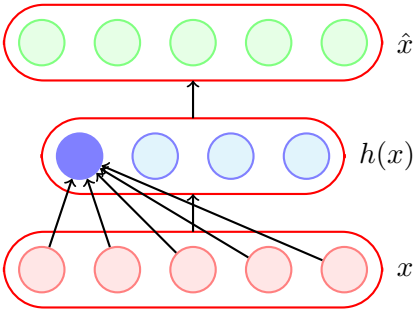


- They consider 6 neurons in the pool5 layer and find the image patches which cause these neurons to fire
- Another neuron fires for houses

- They consider 6 neurons in the pool5 layer and find the image patches which cause these neurons to fire
- Another neuron fires for shiny surfaces

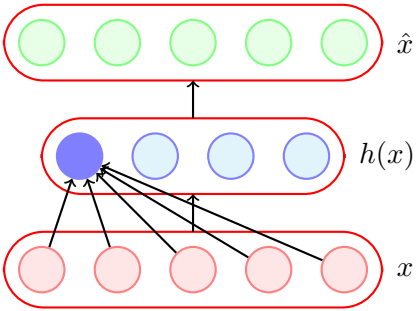


Module : Visualizing filters of a CNN



- Recall that we had done something similar while discussing autoencoders

$$\max_{x} \quad \{w^T x\}$$
$$s.t. \quad ||x||^2 = x^T x = 1$$
$$\text{Solution:} \quad x = \frac{w_1}{\sqrt{w_1^T w_1}}$$



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- We are interested in finding an input which maximally excites a neuron

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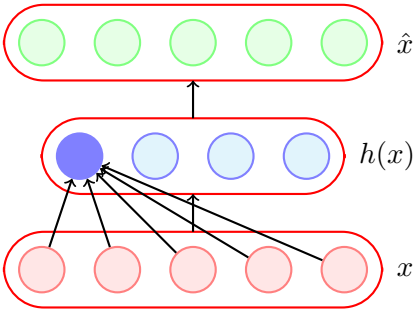

- Recall that we had done something similar while discussing autoencoders
- We are interested in finding an input which maximally excites a neuron
- Turns out that the input which will maximally activate a neuron is $\frac{W}{\|W\|}$

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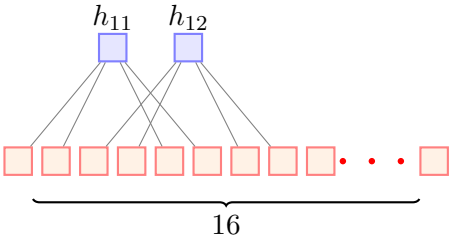
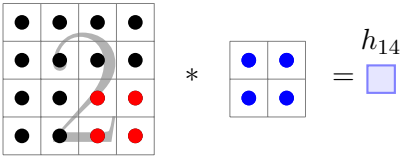
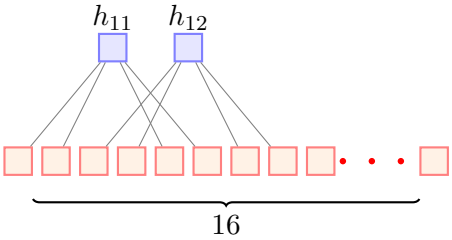
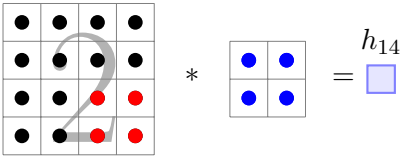




- Now recall that we can think of a CNN also as a feed-forward network with sparse connections and weight sharing





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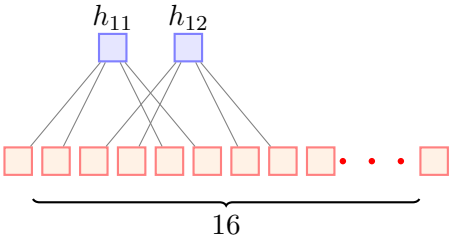

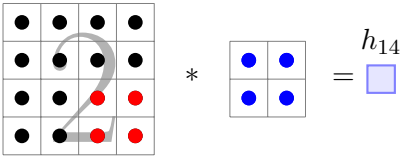

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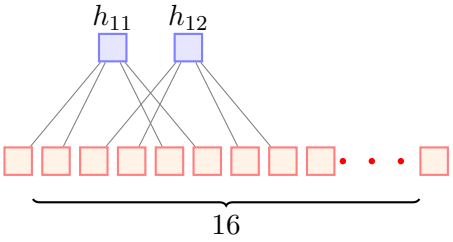
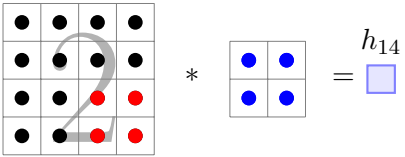
- The solution would be the same ($\frac{W}{\|W\|}$) where W is the filter(2×2, in this case)





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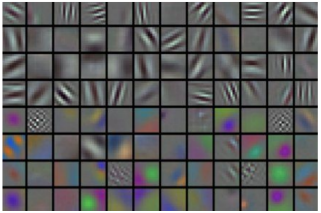
- We can thus think of these filters as pattern detectors



- We can simply plot the $K \times K$ weights (filters) as images & visualize them as patterns

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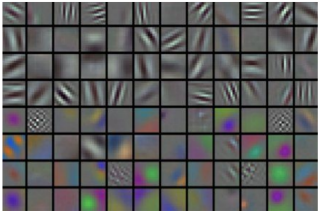

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- The filters essentially detect these patterns (by causing the neurons to maximally fire)

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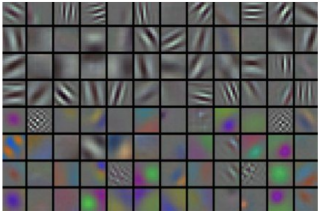

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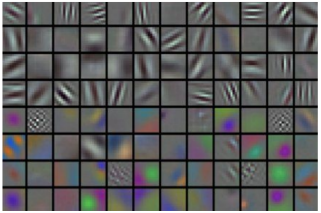

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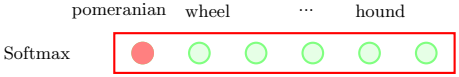
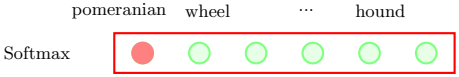
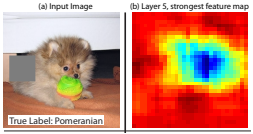
Module : Occlusion experiments



- Typically we are interested in understanding which portions of the image are responsible for maximizing the probability of a certain class





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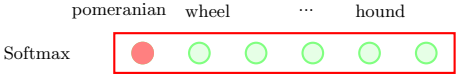

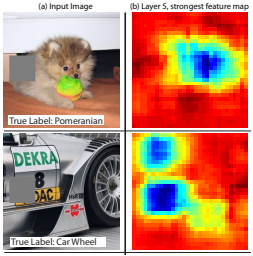

- Typically we are interested in understanding which portions of the image are responsible for maximizing the probability of a certain class
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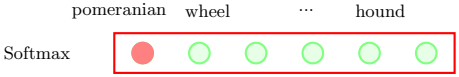
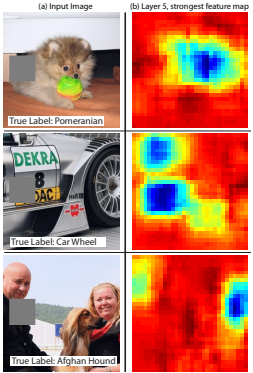
- For example this heat map shows that occluding the face of the dog causes a maximum drop in the prediction probability





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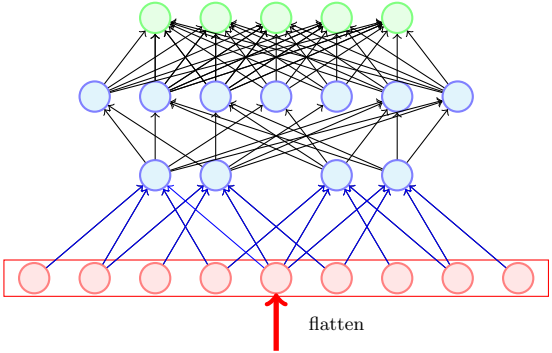
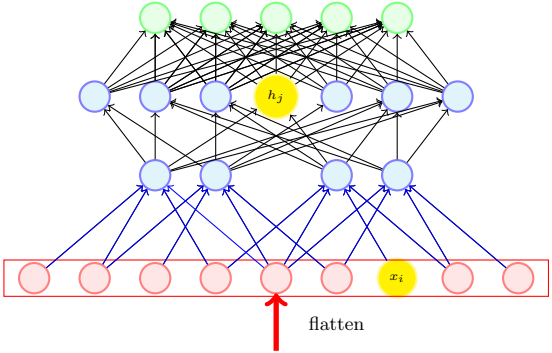
- Similar observations are made for other images

Module : Finding influence of input pixels using backpropagation



- We can think of an image as a $m \times n$ inputs $x_0, x_1, \ldots, x_{m \times n}$



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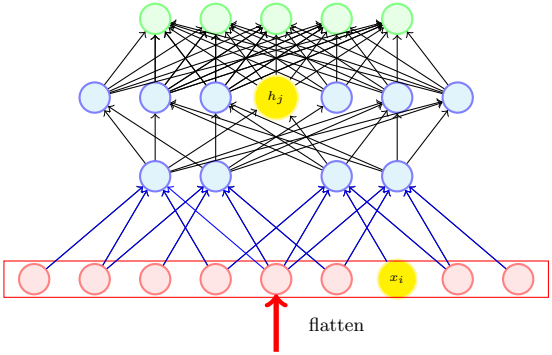
- We are interested in finding the influence of each of these inputs(x_i) on a given neuron(h_j)



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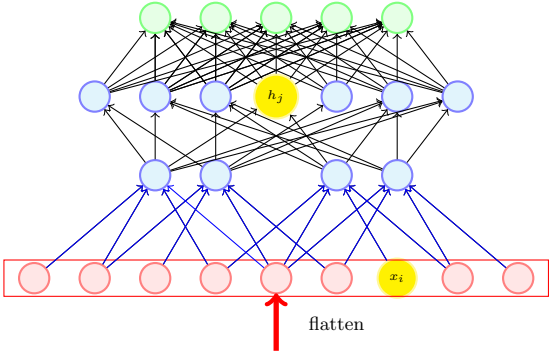

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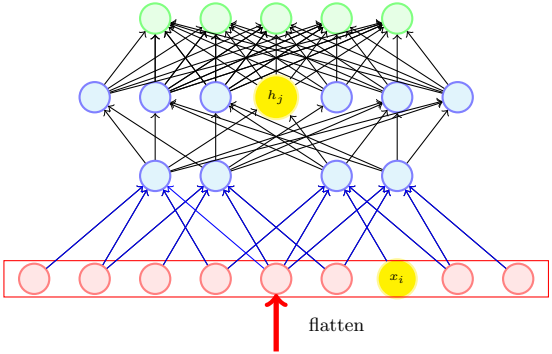
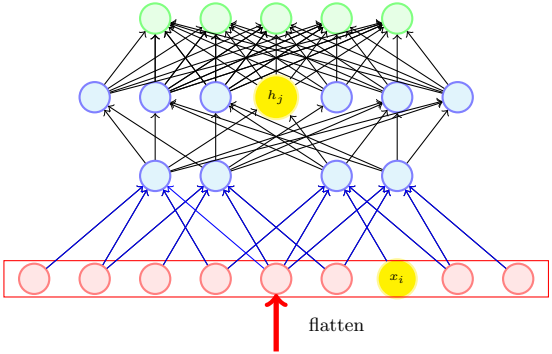
- In other words the gradient $\frac{\partial h_j}{\partial x_i}$ could tell us about the influence



$$\frac{\partial h_j}{\partial x_i} = 0 \qquad \longrightarrow \text{ no influence}$$



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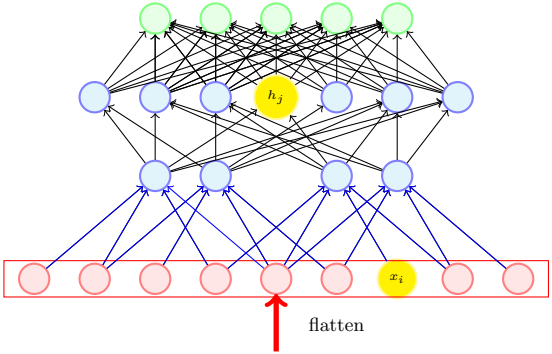

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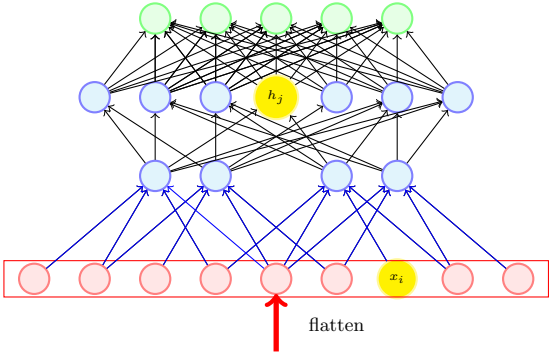

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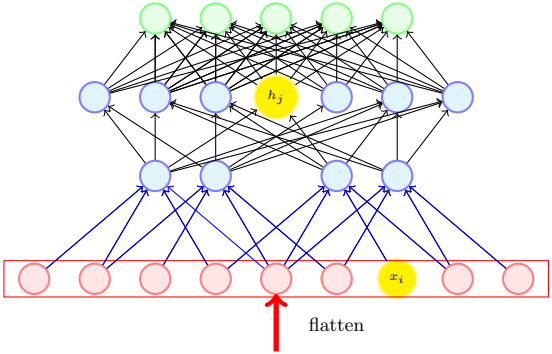

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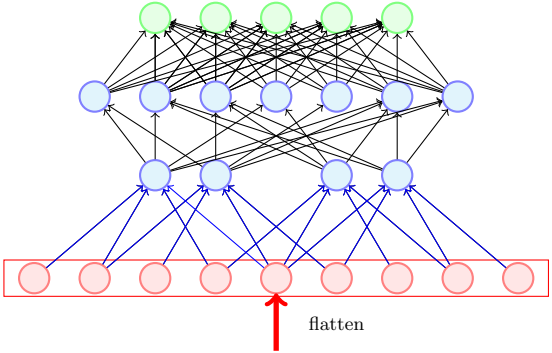
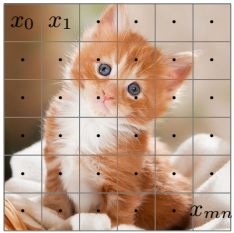
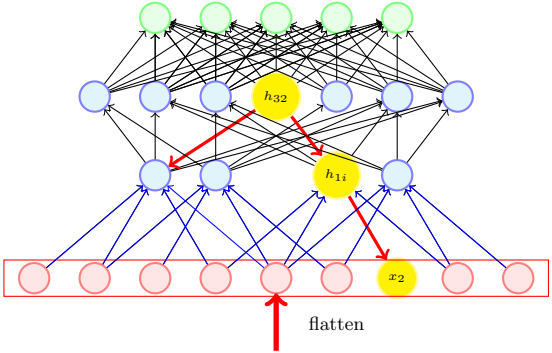
- And then visualize this gradient matrix as an image itself

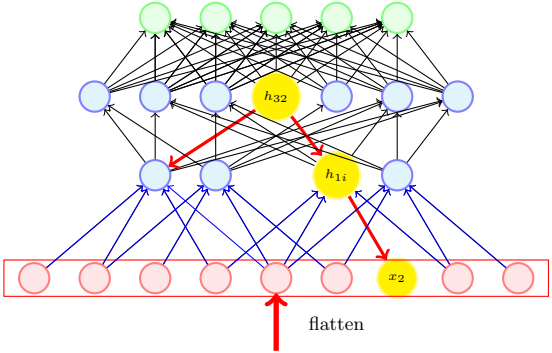


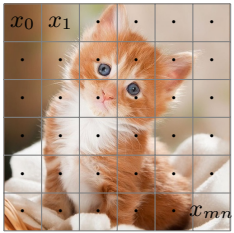


- But how do we compute these gradients?



- But how do we compute these gradients?
- Recall that we can represent CNNs by feedforward neural network





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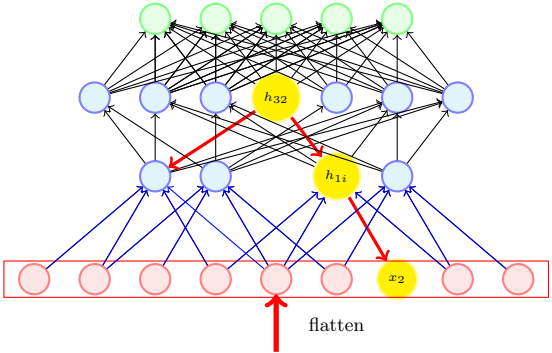

- But how do we compute these gradients?
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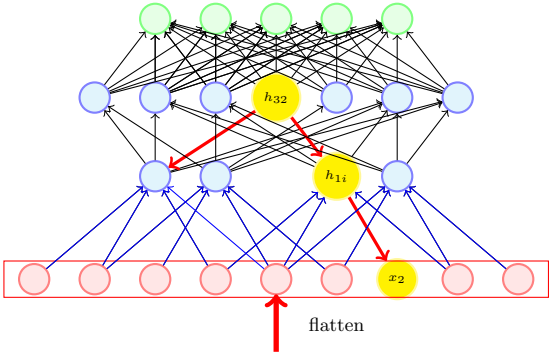
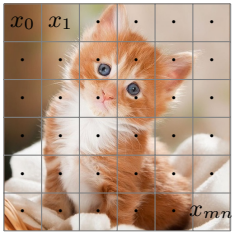
- For example, we know how to back-prop the gradients till the first hidden layer





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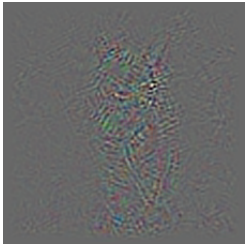
$$\frac{\partial h_{32}}{\partial x_2} = \sum_{i=1}^{3} \frac{\partial h_{32}}{\partial h_{1i}} \frac{\partial h_{1i}}{\partial x_2}$$

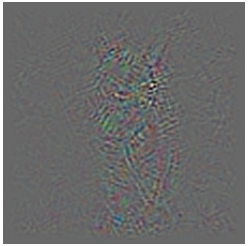
$$h_{1i} = \sum_{j=1}^{4} w_{ji} x_j$$

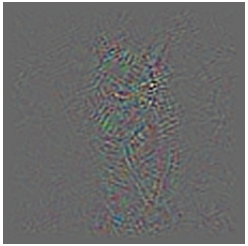
$$\frac{\partial h_{1i}}{\partial x_2} = w_{12}$$

- This is what we get if we compute the gradients and plot it as an image



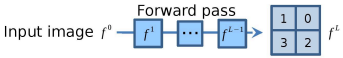
- This is what we get if we compute the gradients and plot it as an image
- The above procedure does not show very sharp influences



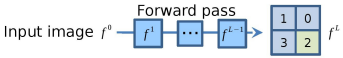


- This is what we get if we compute the gradients and plot it as an image
- The above procedure does not show very sharp influences
- Springenberg et al. proposed “guided back propagation” which gives a better idea about the influences

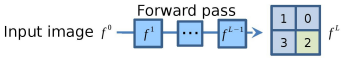
Module : Guided Backpropagation



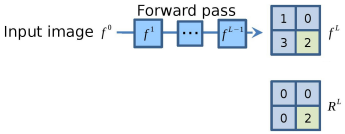
- We feed an input to the CNN and do a forward pass



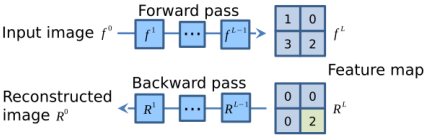
- We feed an input to the CNN and do a forward pass
- We consider one neuron in some feature map at some layer



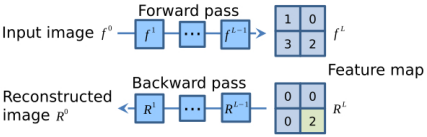
- We feed an input to the CNN and do a forward pass
- We consider one neuron in some feature map at some layer
- We are interested in finding the influence of the input on this neuron



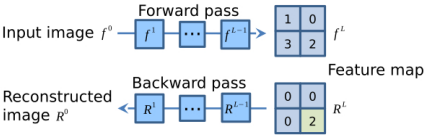
- We feed an input to the CNN and do a forward pass
- We consider one neuron in some feature map at some layer
- We are interested in finding the influence of the input on this neuron
- We retain this neuron and set all other neurons in the layer to zero



- We now backpropogate all the way to the inputs



- We now backpropogate all the way to the inputs
- Recall that during forward pass relu activation allows only positive values to pass & clamps $-ve$ values to zero



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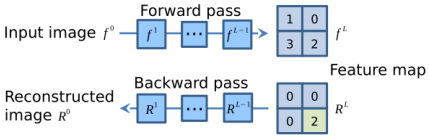
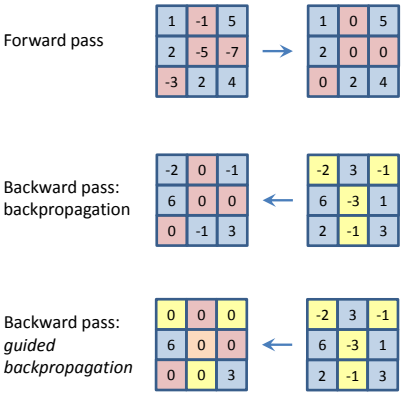
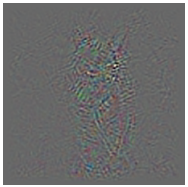


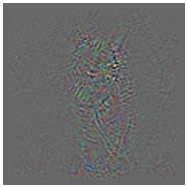


- We now backpropogate all the way to the inputs
- Recall that during forward pass relu activation allows only positive values to pass & clamps $-ve$ values to zero
- Similarly during backward pass no gradient passes through the dead relu neurons
- In guided back propagation any -ve gradients flowing from the upper layer are also set to 0



Backpropagation

- **Intuition:** Neglect all the negative influences (gradients) and focus only on the positive influences (gradients)

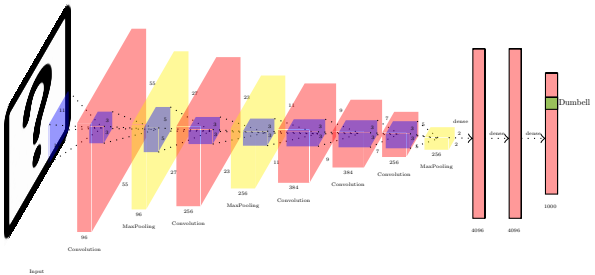


Backpropagation

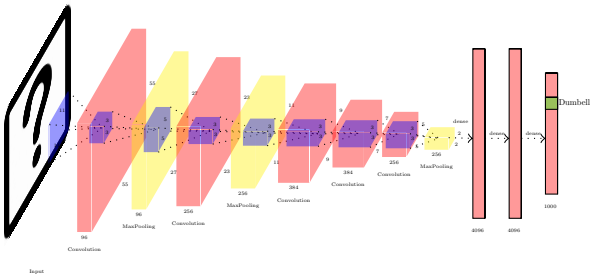
Guided Backpropagation

- **Intuition:** Neglect all the negative influences (gradients) and focus only on the positive influences (gradients)
- This gives a better picture of the true influence of the input

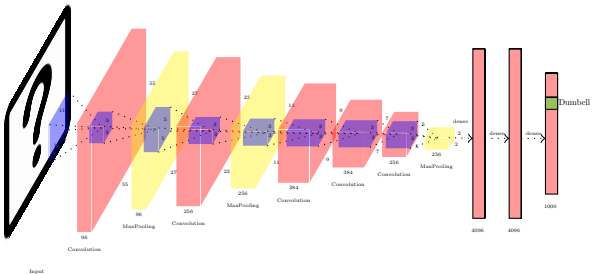
Module : Optimization over images



- Suppose we want to create an image which looks like a dumbell (or an ostrich, or a car, or just anything)



- Suppose we want to create an image which looks like a dumbell (or an ostrich, or a car, or just anything)
- In other words we want to create an image such that if we pass it through a trained ConvNet it should maximize the probability of the class dumbell



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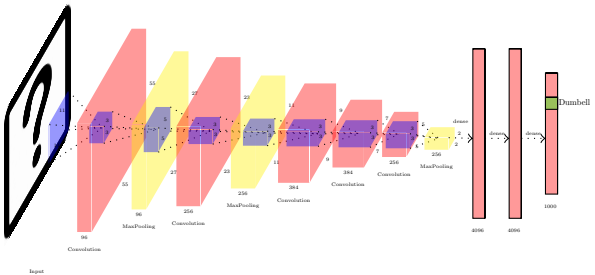
- We could pose this as an optimization problem w.r.t I $(i_0,\ i_1,\ \ldots,\ i_{mn})$



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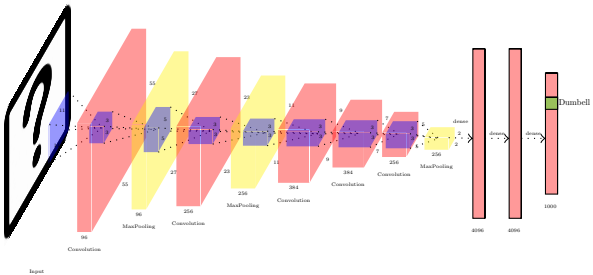
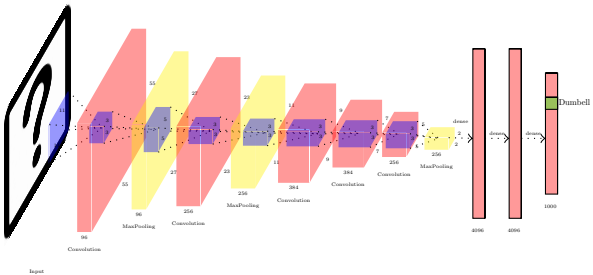
$$\arg\max_I (S_c(I) - \lambda\Omega(I))$$

$S_c(I)$ = Score for class C before softmax

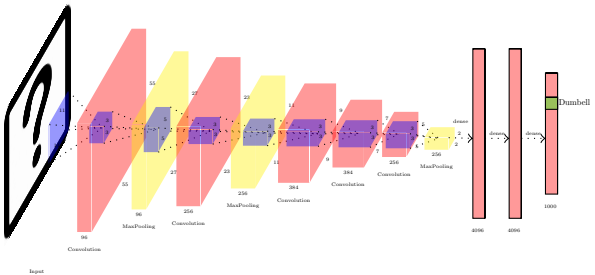
$\Omega(I)$ = Some regularizer to ensure that I looks like an image



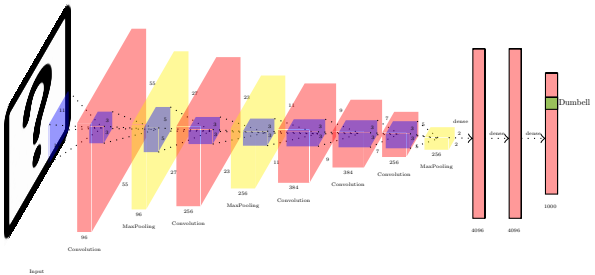
- We can essentially think of the image as a collection of parameters



- We can essentially think of the image as a collection of parameters
- Keep the weights of trained convolutional neural network fixed



- We can essentially think of the image as a collection of parameters
- Keep the weights of trained convolutional neural network fixed
- Now adjust these parameters(image pixels) so that the score of a class is maximized



- We can essentially think of the image as a collection of parameters
- Keep the weights of trained convolutional neural network fixed
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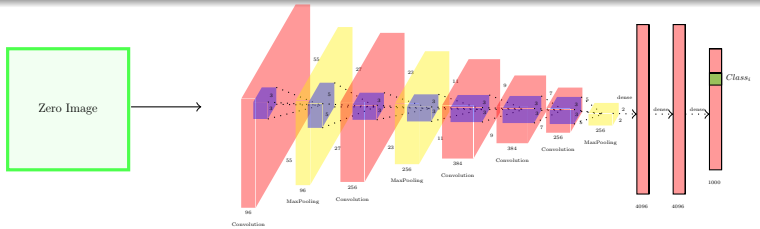
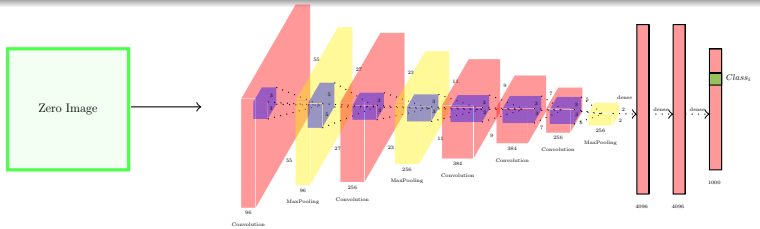
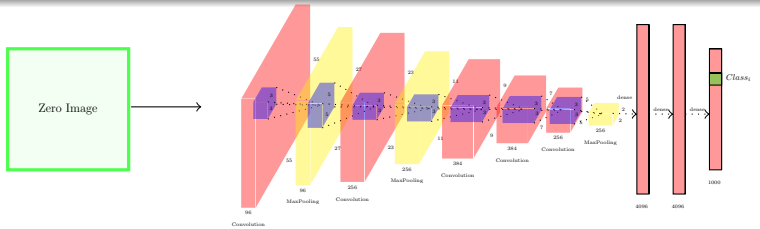
- Let us see how



1. Start with a zero image



1. Start with a zero image
2. Set the score vector to be $[0, 0, \ldots 1, 0, 0]$



1. Start with a zero image
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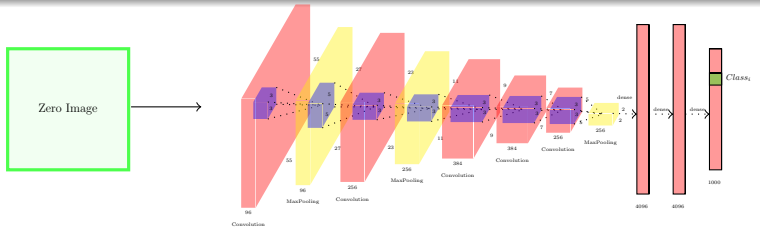
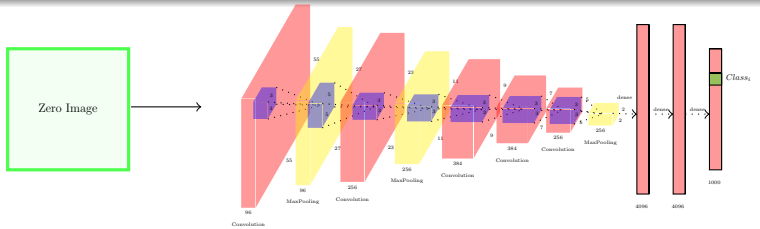
3. Compute the gradient $\frac{\partial S_c(I)}{\partial i_k}$



1. Start with a zero image
2. Set the score vector to be $[0, 0, \ldots 1, 0, 0]$
3. Compute the gradient $\frac{\partial S_c(I)}{\partial i_k}$
4. Now update the pixel $i_k = i_k - \eta \frac{\partial S_c(I)}{\partial i_k}$



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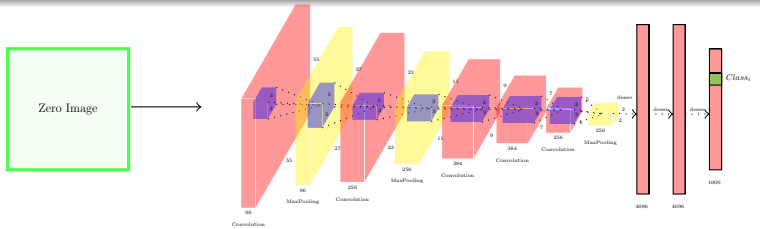
3. Compute the gradient $\frac{\partial S_c(I)}{\partial i_k}$
4. Now update the pixel $i_k = i_k - \eta \frac{\partial S_c(I)}{\partial i_k}$
5. Now again do a forward pass through the network



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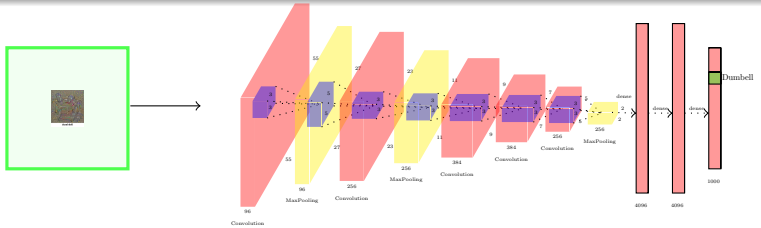
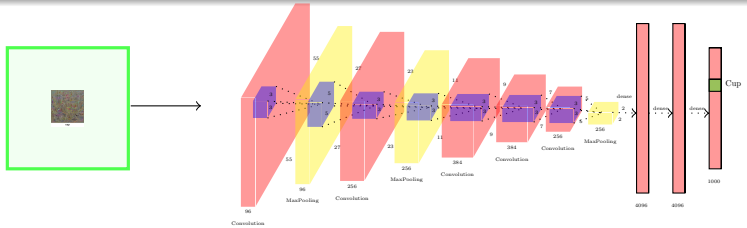
4. Now update the pixel $i_k = i_k - \eta \frac{\partial S_c(I)}{\partial i_k}$
5. Now again do a forward pass through the network
6. Go to step 2



- Lets look at the images obtained for maximizing some class scores



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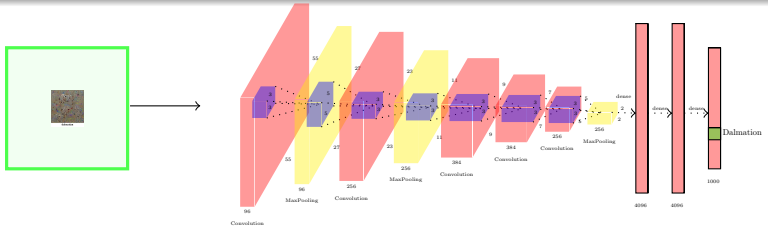

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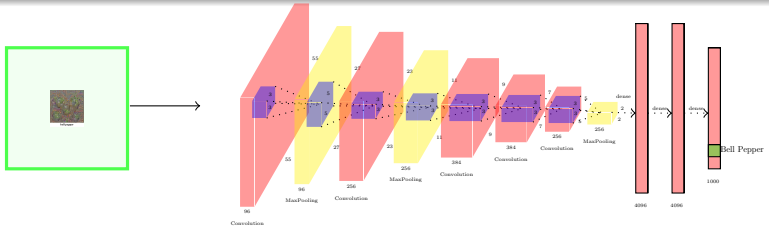

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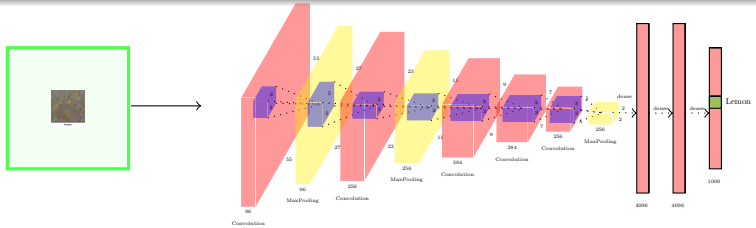

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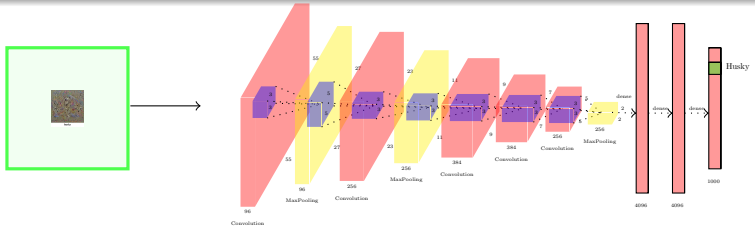

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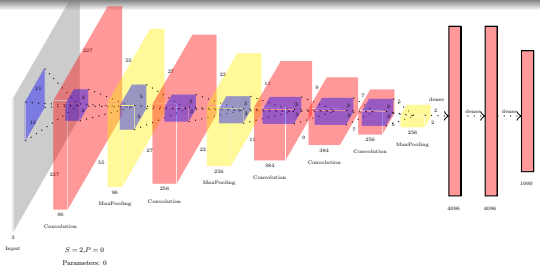
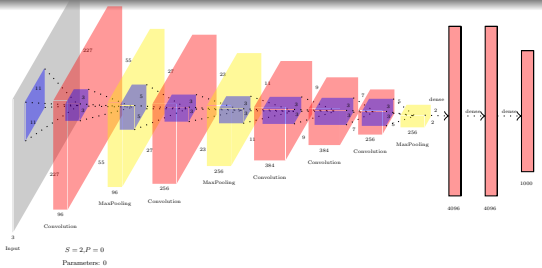


Repeat:

- Feed an image through the network

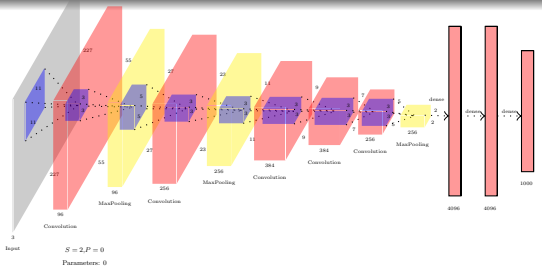
- We can actually do this for any arbitrary neuron in the convnet



- We can actually do this for any arbitrary neuron in the convnet

Repeat:

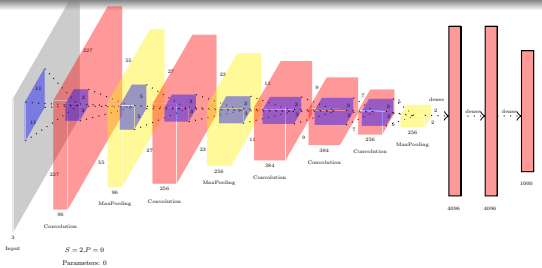
- Feed an image through the network
- Set activation in layer of interest to all zero, except for a neuron of interest



- We can actually do this for any arbitrary neuron in the convnet

Repeat:

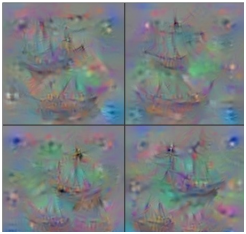
- Feed an image through the network
- Set activation in layer of interest to all zero, except for a neuron of interest
- Backprop to image



- We can actually do this for any arbitrary neuron in the convnet

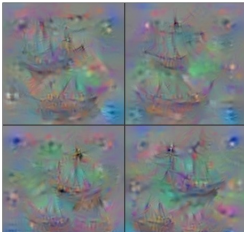
Repeat:

- Feed an image through the network
- Set activation in layer of interest to all zero, except for a neuron of interest
- Backprop to image
- $i_k = i_k - \eta \frac{\partial A(I)}{\partial i_k}$, A(I) is the activation of the i^{th} neuron in some layer



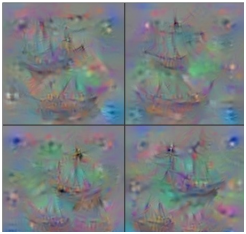
Layer-8

- Let us look at some “updated” images which excite certain neurons in some layer



Layer-8

- Let us look at some “updated” images which excite certain neurons in some layer
- Starting with different initializations instead of using a zero image we can get different insights



Layer-8

- Let us look at some “updated” images which excite certain neurons in some layer
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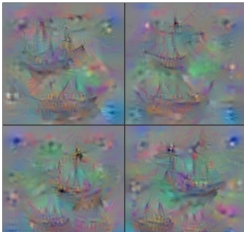
- Each of these 4 images are obtained by focusing on one neuron in layer 8 and starting with different initializations



Layer-8

- Let us look at some “updated” images which excite certain neurons in some layer
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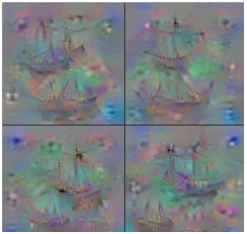
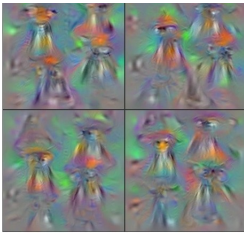
- We can do a similar analysis with other layers



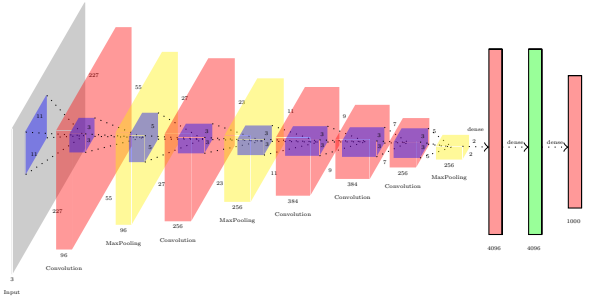
Layer-8



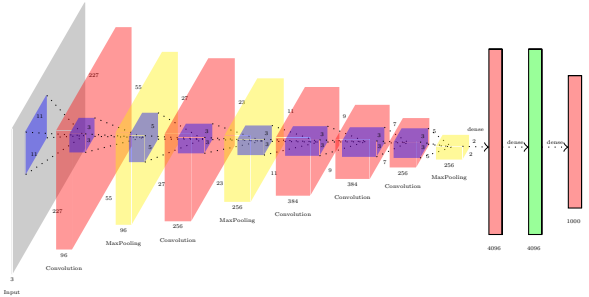
Layer-7

- Let us look at some “updated” images which excite certain neurons in some layer
- Starting with different initializations instead of using a zero image we can get different insights
- Each of these 4 images are obtained by focusing on one neuron in layer 8 and starting with different initializations
- We can do a similar analysis with other layers

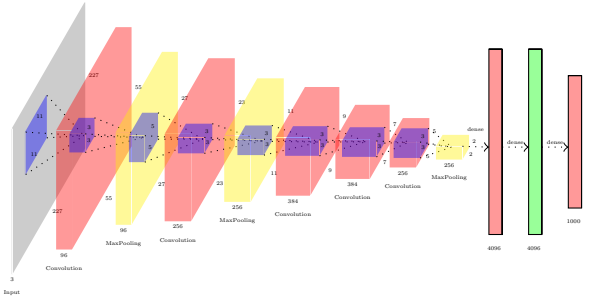
Module : Creating images from embeddings



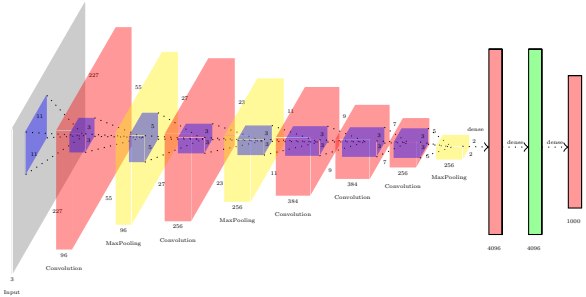
- We could think of the fc7 layer as some kind of an embedding for the image



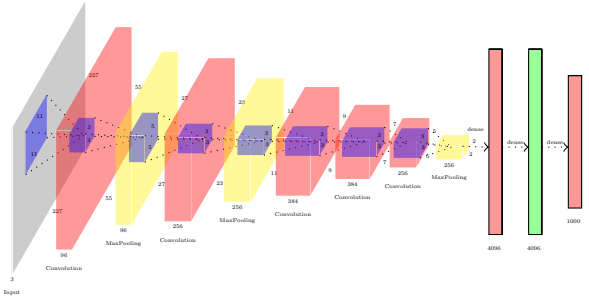
- We could think of the fc7 layer as some kind of an embedding for the image
- **Question:** Given this embedding can we reconstruct the image?



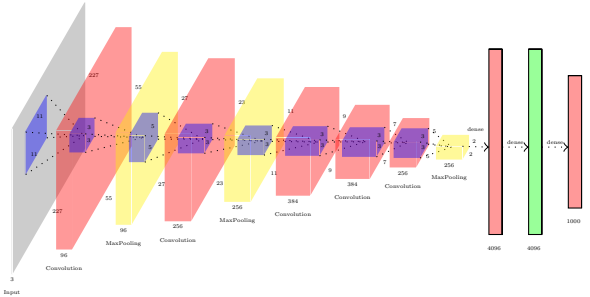
- We could think of the fc7 layer as some kind of an embedding for the image
- **Question:** Given this embedding can we reconstruct the image?
- We can pose this as an optimization problem



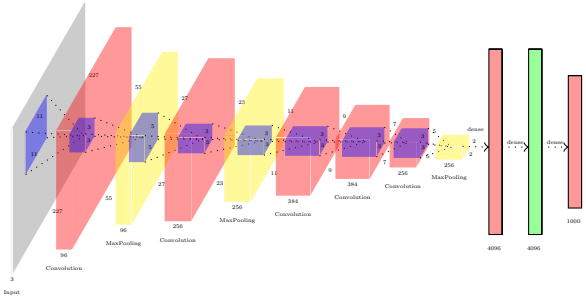
- Find an image such that



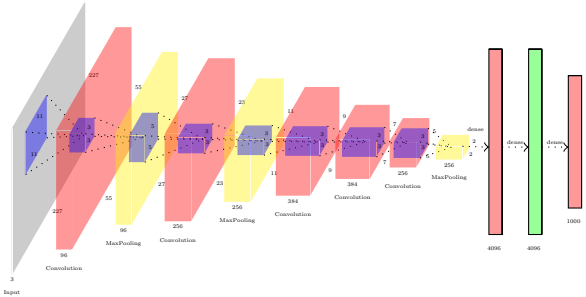
- Find an image such that
- Its embedding is similar to a given embedding



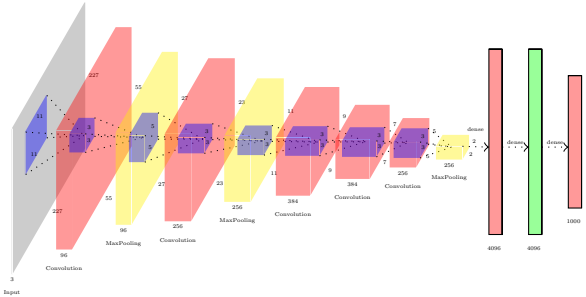
- Find an image such that
- Its embedding is similar to a given embedding
- It looks natural (some prior regularization)



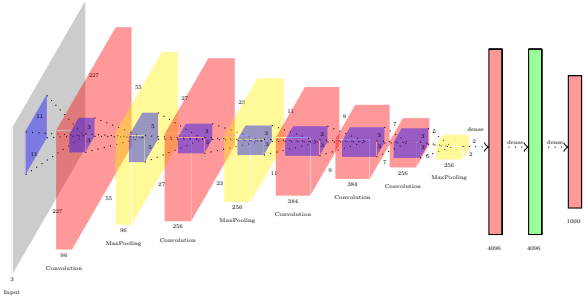
- ϕ_0 :Embedding of an image of interest



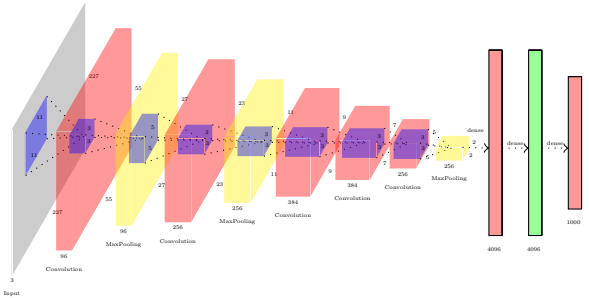
- ϕ_0 :Embedding of an image of interest
- X :Random image (say zero image)



- ϕ_0 :Embedding of an image of interest
- X :Random image (say zero image)
- Repeat

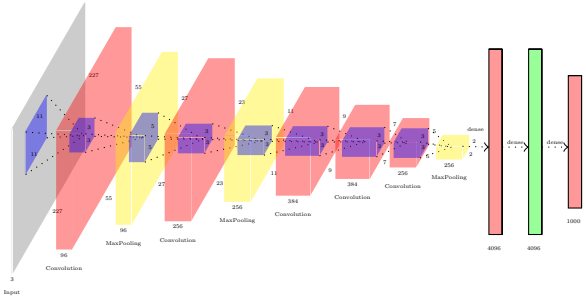


- ϕ_0 :Embedding of an image of interest
- X :Random image (say zero image)
- Repeat
  - Forward pass using X and compute $\phi(x)$.



- ϕ_0 :Embedding of an image of interest
- X :Random image (say zero image)
- Repeat
  - Forward pass using X and compute $\phi(x)$.
  - Compute

$$\mathscr{L}(i) = ||\phi(x) - \phi_0||^2 + \lambda||\phi(x)||_6^6$$



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- Repeat
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  - Compute

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  - $i_k = i_k - \eta\frac{\mathscr{L}(i)}{\partial i_k}$

Original Image

Conv-1

Original Image

Relu-1

Original Image

Mpool-1

Original Image

Norm-1

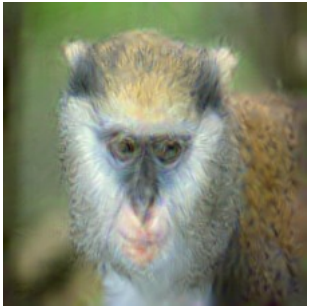
Original Image

Conv-2

Original Image

Relu-2

Original Image



Mpool-2

Original Image

Norm-2

Original Image

Conv-3

Original Image

Relu-3

Original Image

Conv-4

Original Image

Relu-4

Original Image

Conv-5

Original Image

Relu-5

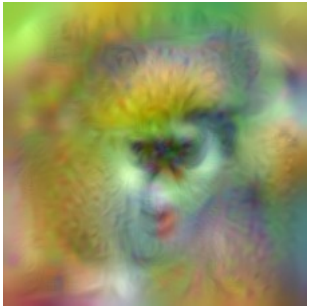
Original Image

Mpool-5

Original Image

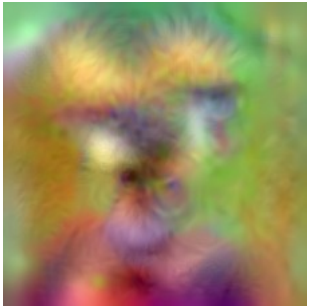
FC-6

Original Image



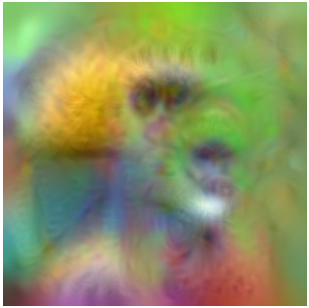
Relu-6

Original Image



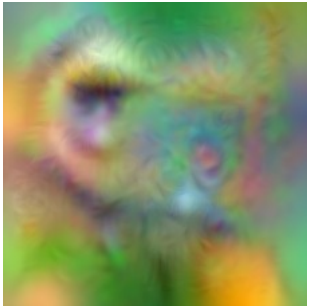
FC-7

Original Image



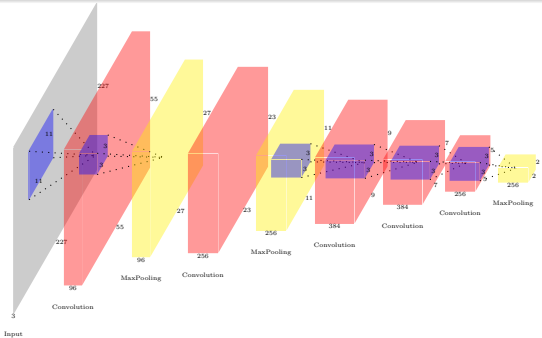
Relu-7

Original Image

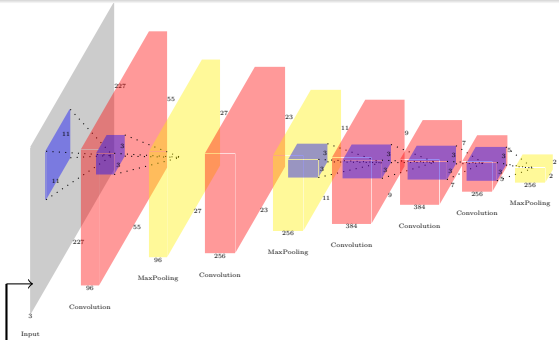


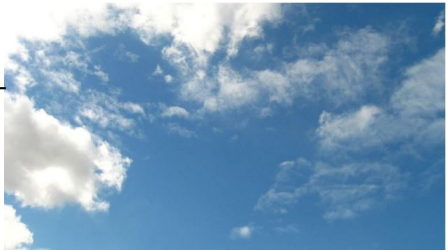
FC-8

Module : Deep Dream

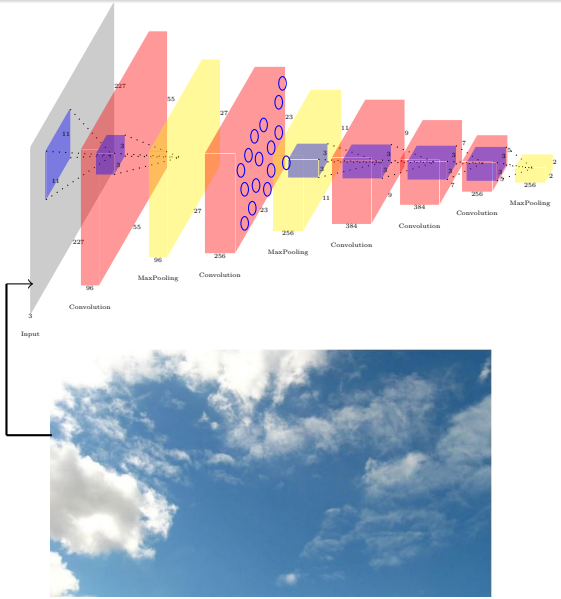


- Suppose instead of starting with a blank (zero) image we start with an actual image.

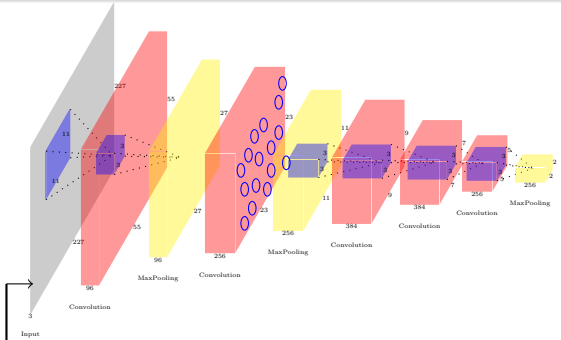


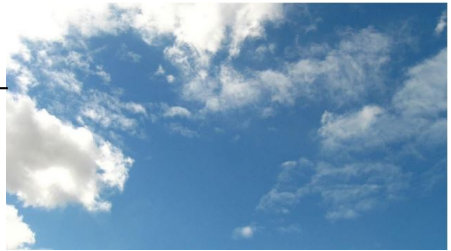


- Suppose instead of starting with a blank (zero) image we start with an actual image.

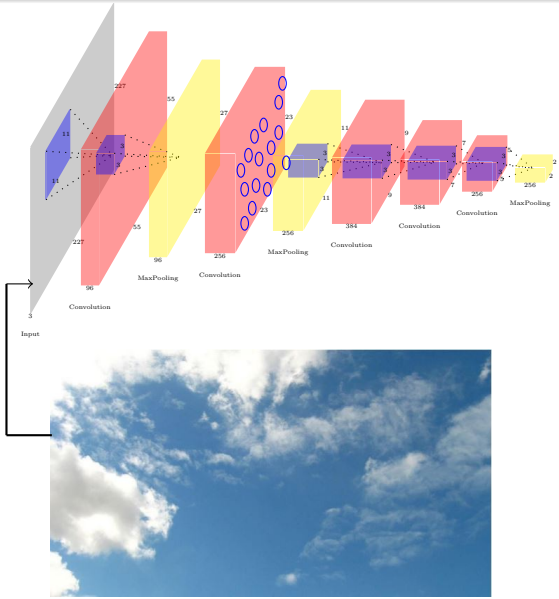


- Suppose instead of starting with a blank (zero) image we start with an actual image.
- We focus on some layer and check the activations of the neurons

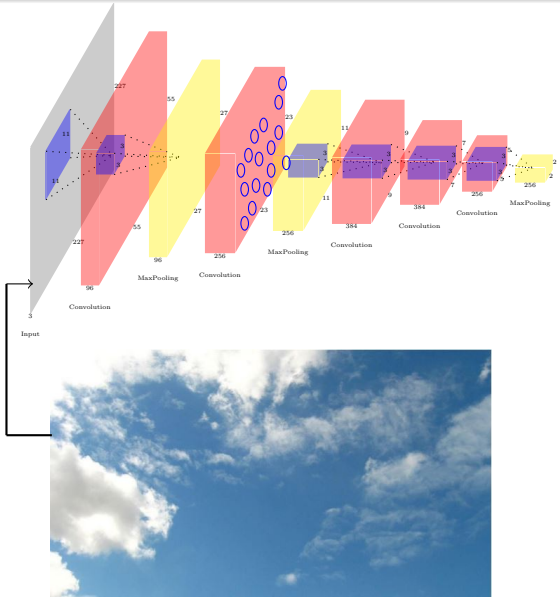




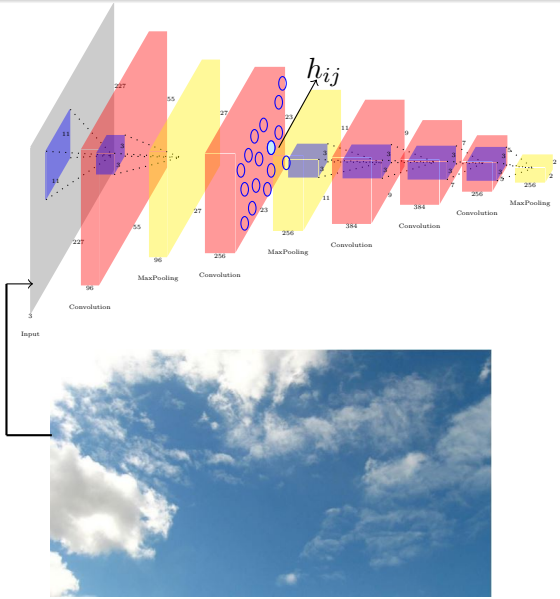
- Suppose instead of starting with a blank (zero) image we start with an actual image.
- We focus on some layer and check the activations of the neurons
- We want to change the image so that these neurons fire even more



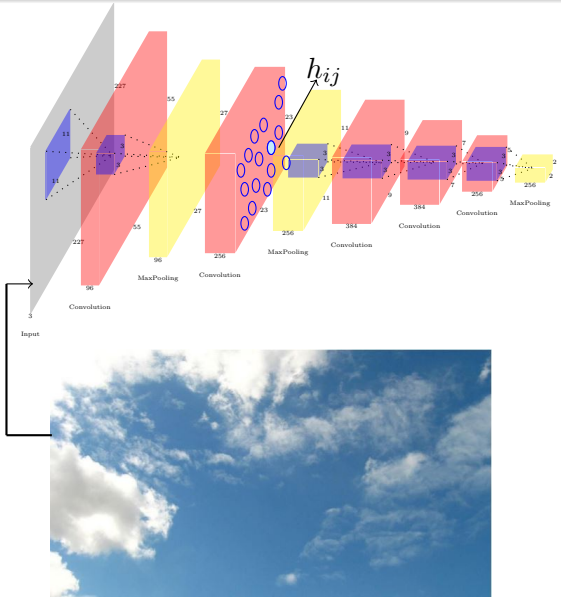
- How would we achieve this?



- How would we achieve this?
- Suppose we want to boost the activation h_{ij} (some neuron in some layer)



- How would we achieve this?
- Suppose we want to boost the activation h_{ij} (some neuron in some layer)
- We can formulate this as the following optimization problem



- How would we achieve this?
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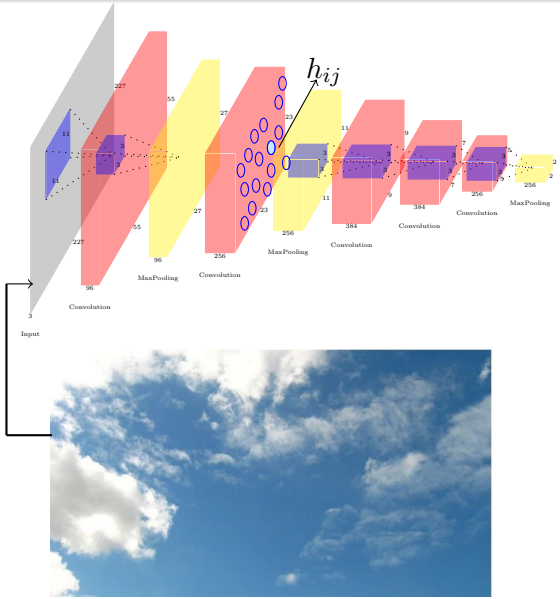
$$\max_I \mathscr{L}(I)$$
$$\mathscr{L}(I) = h_{ij}^2$$

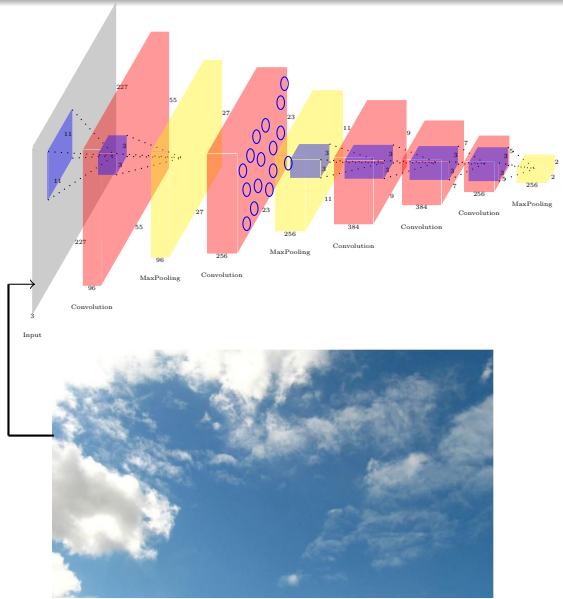


- How would we achieve this?
- Suppose we want to boost the activation h_{ij} (some neuron in some layer)
- We can formulate this as the following optimization problem

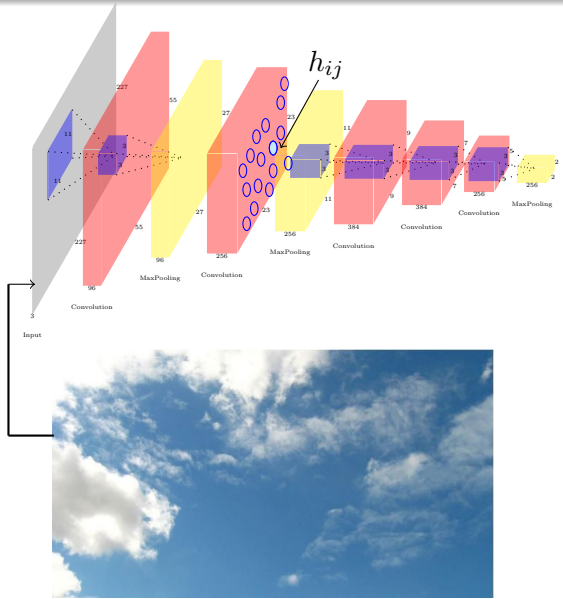
$$\max_I \mathscr{L}(I)$$
$$\mathscr{L}(I) = h_{ij}^2$$

- Consider a pixel i_{mn} in the image

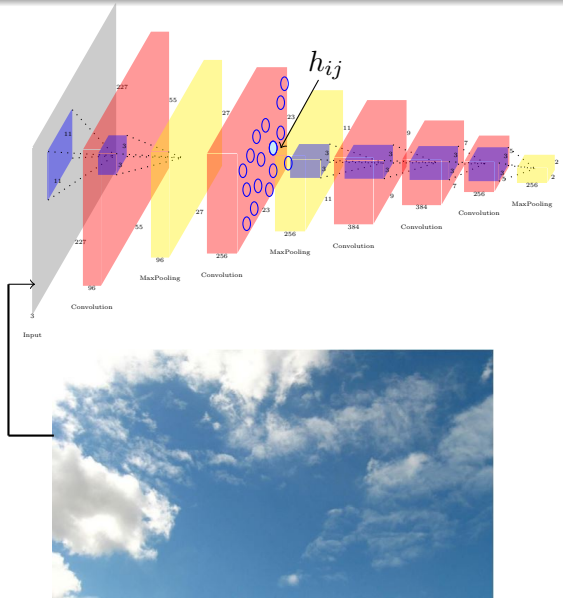
$$\frac{\partial \mathscr{L}(I)}{\partial i_{mn}} = \frac{\partial \mathscr{L}(I)}{\partial h_{ij}} \frac{\partial h_{ij}}{\partial i_{mn}}$$



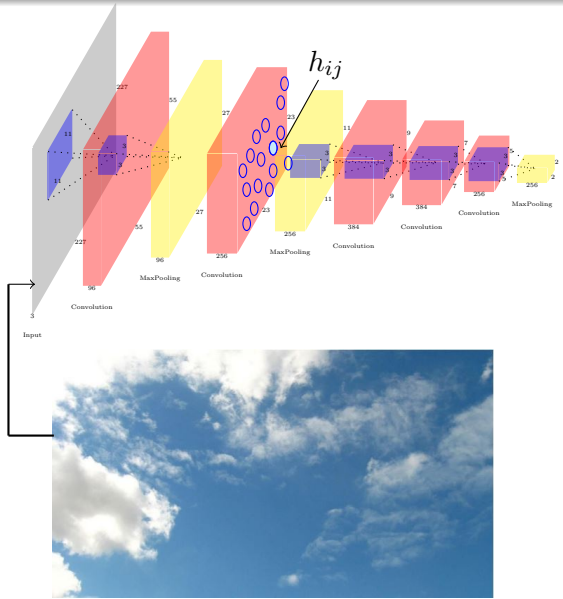
- Once the image is updated $\left(i_{mn} = i_{mn} + \frac{\partial \mathscr{L}(I)}{\partial i_{mn}}\right)$ we feed it back to the network



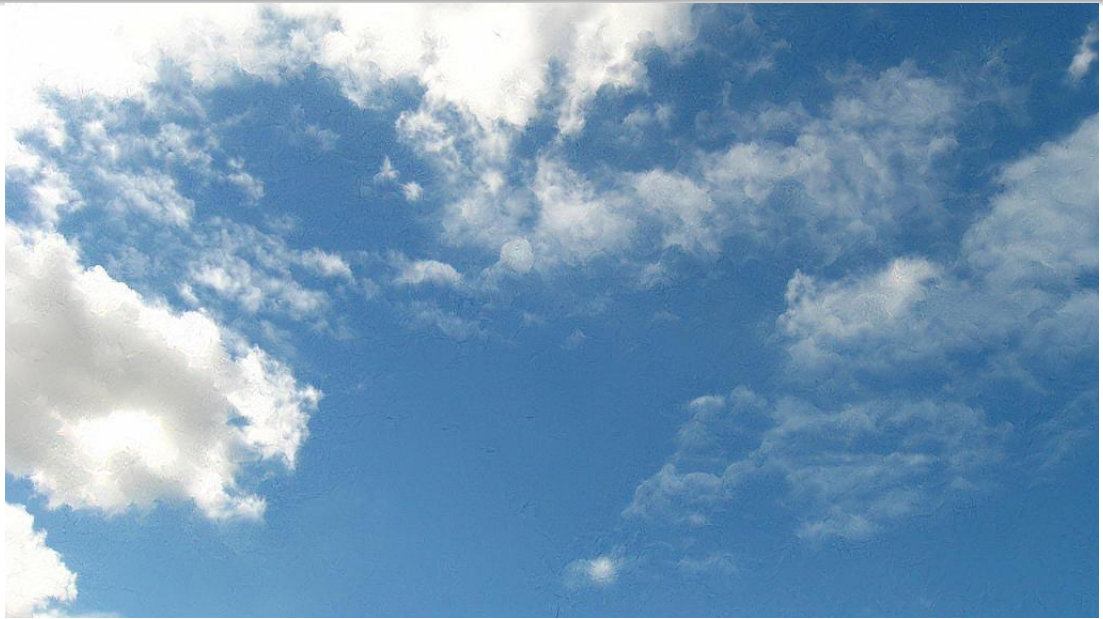
- Once the image is updated $\left(i_{mn} = i_{mn} + \frac{\partial \mathscr{L}(I)}{\partial i_{mn}}\right)$ we feed it back to the network
- This time the target neurons should fire even more (because we have precisely modified the image to achieve this)

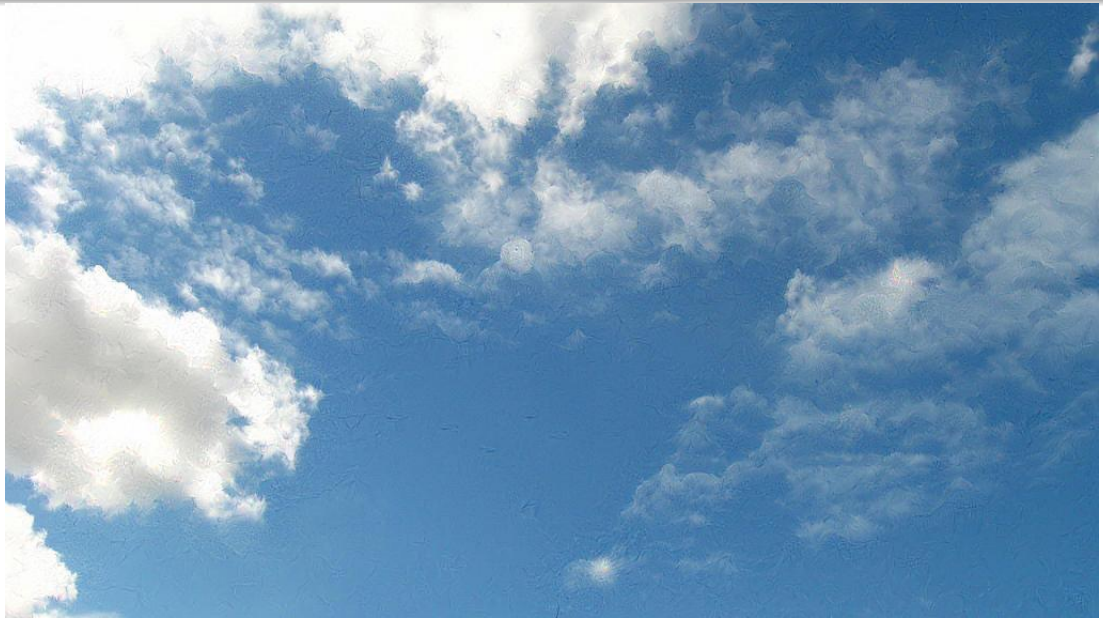


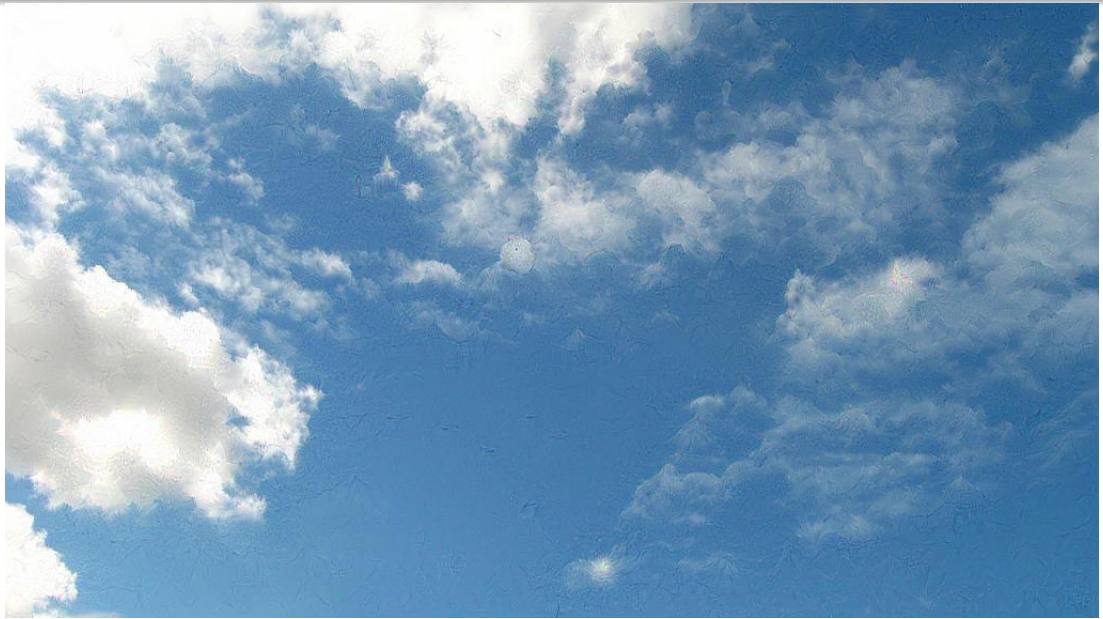
- Once the image is updated $\left(i_{mn} = i_{mn} + \frac{\partial \mathscr{L}(I)}{\partial i_{mn}}\right)$ we feed it back to the network
- This time the target neurons should fire even more (because we have precisely modified the image to achieve this)
- Doing this iteratively would make the image more and more like the patterns that cause the neuron to fire

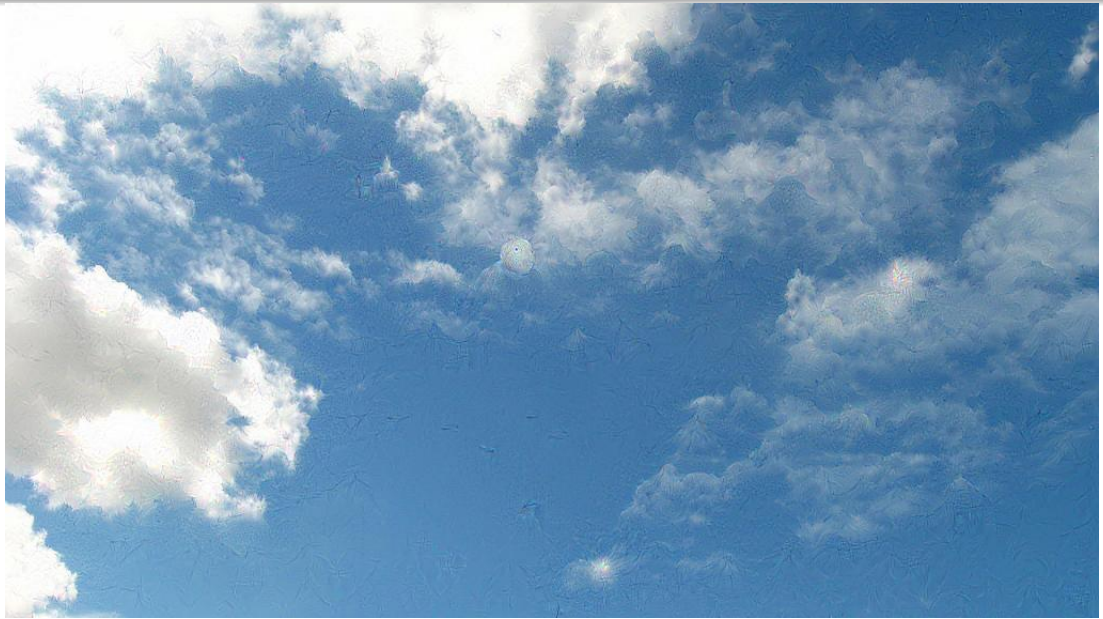


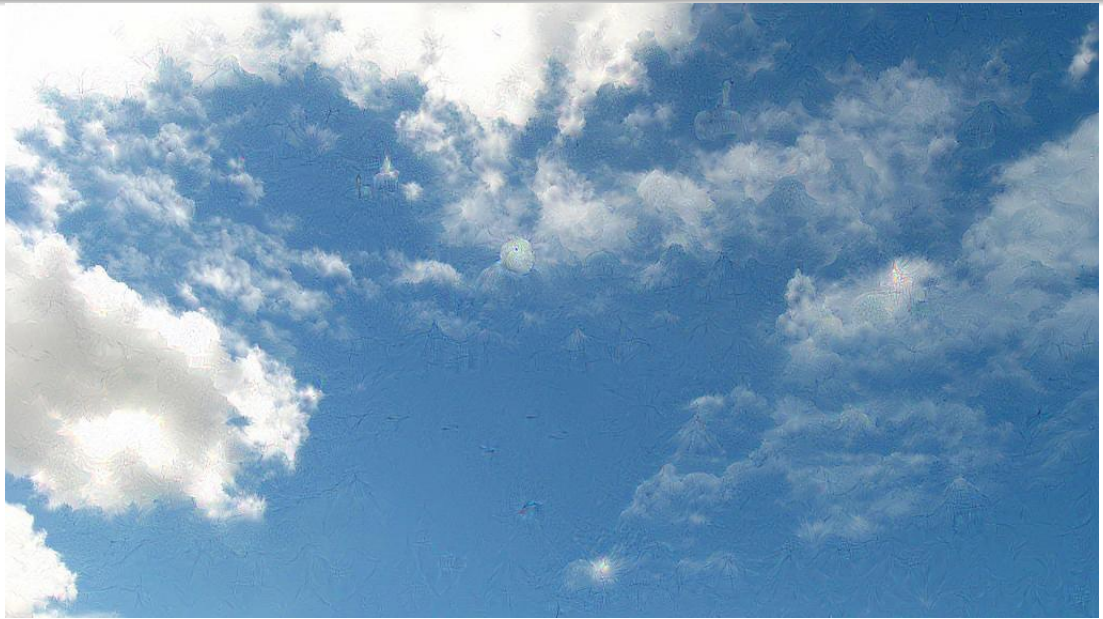
- Once the image is updated $\left(i_{mn} = i_{mn} + \frac{\partial \mathscr{L}(I)}{\partial i_{mn}}\right)$ we feed it back to the network
- This time the target neurons should fire even more (because we have precisely modified the image to achieve this)
- Doing this iteratively would make the image more and more like the patterns that cause the neuron to fire
- Let us run this algorithm

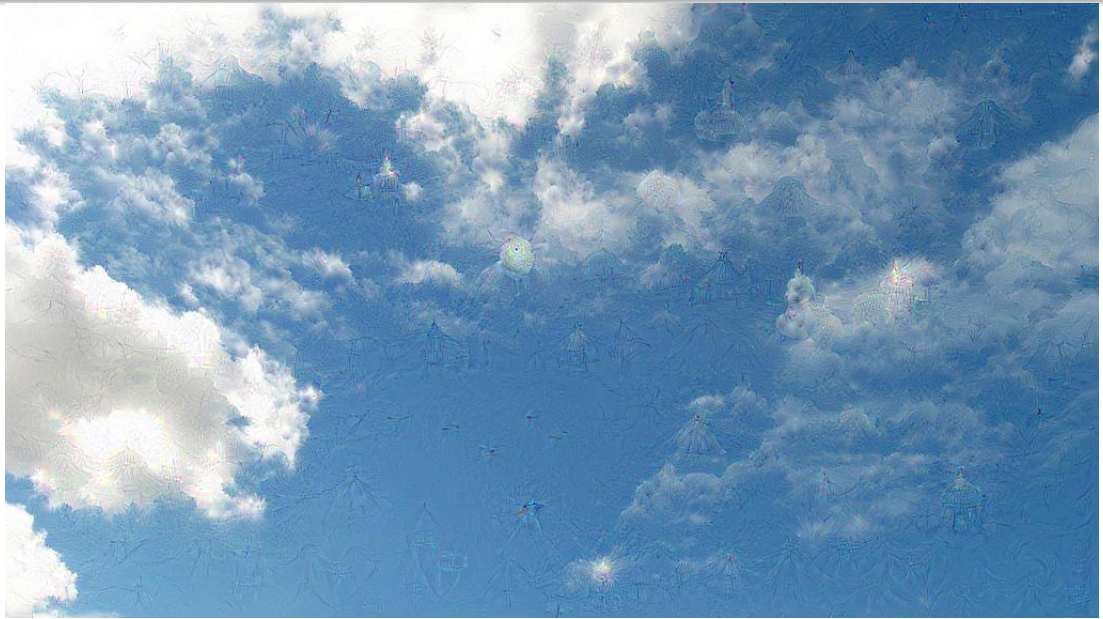


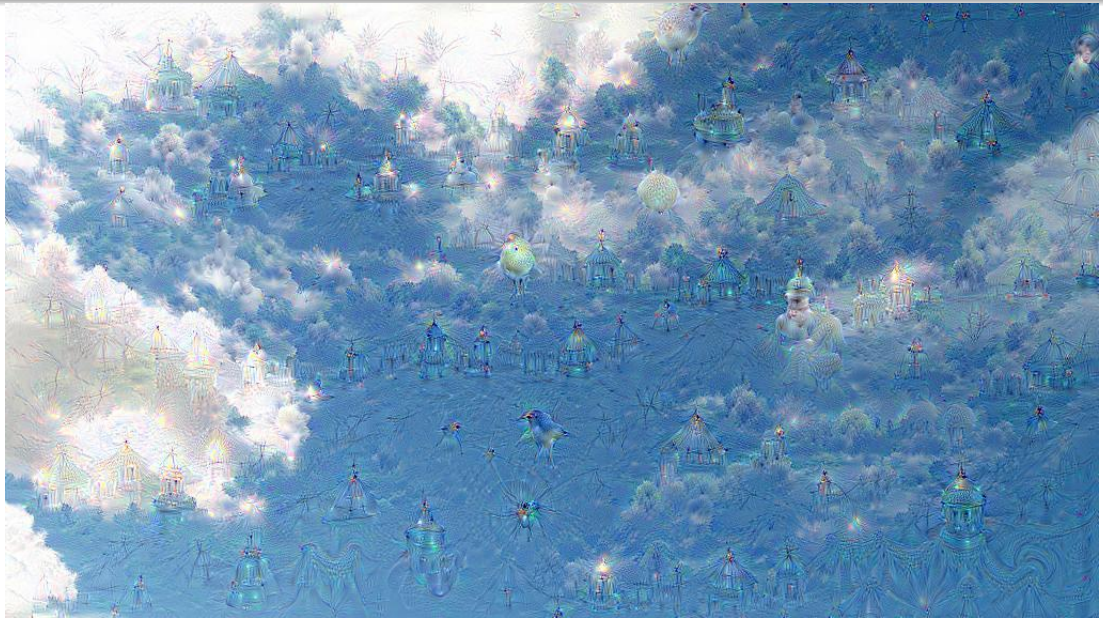


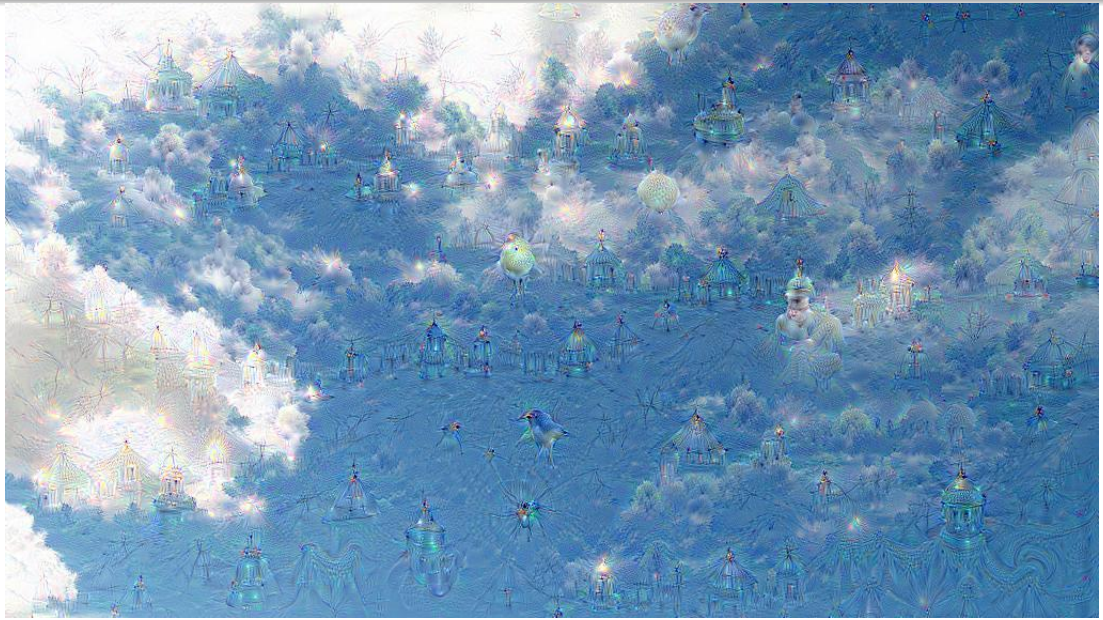


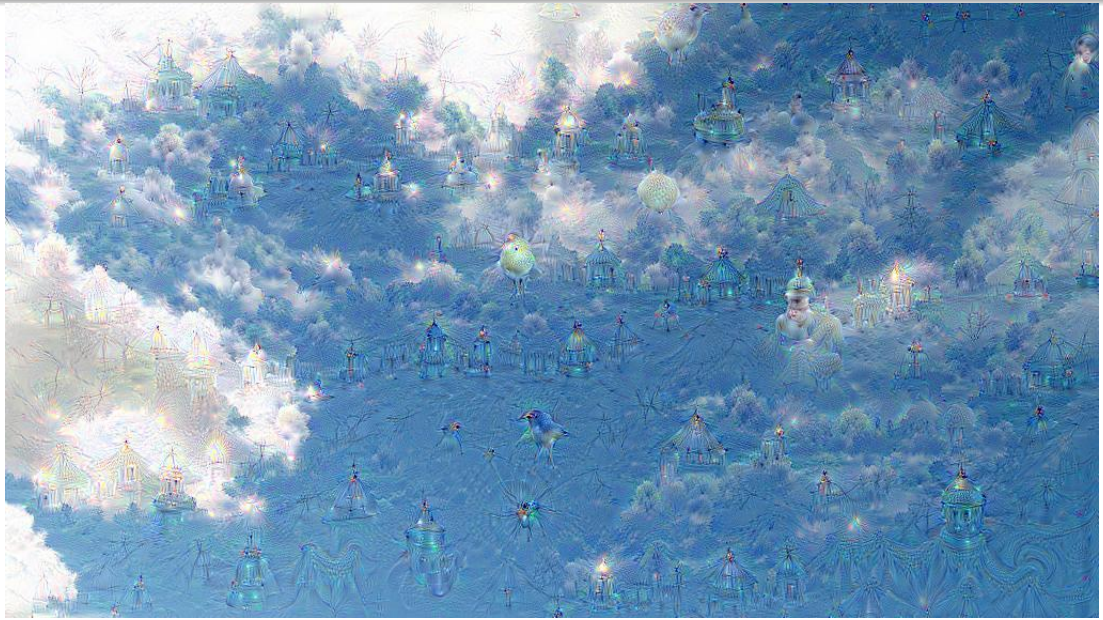


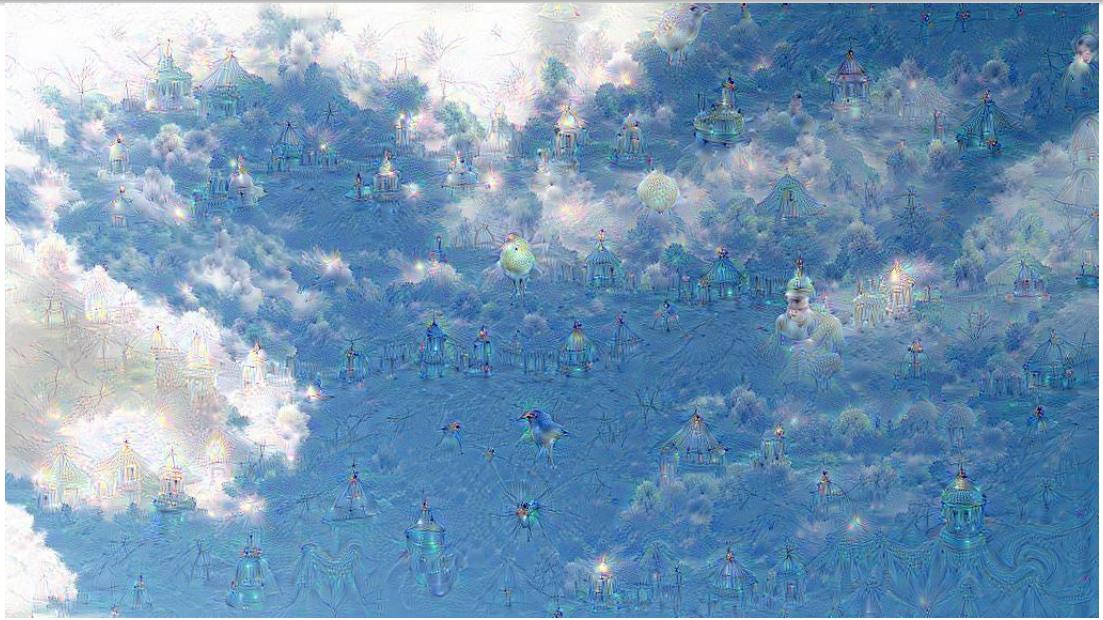


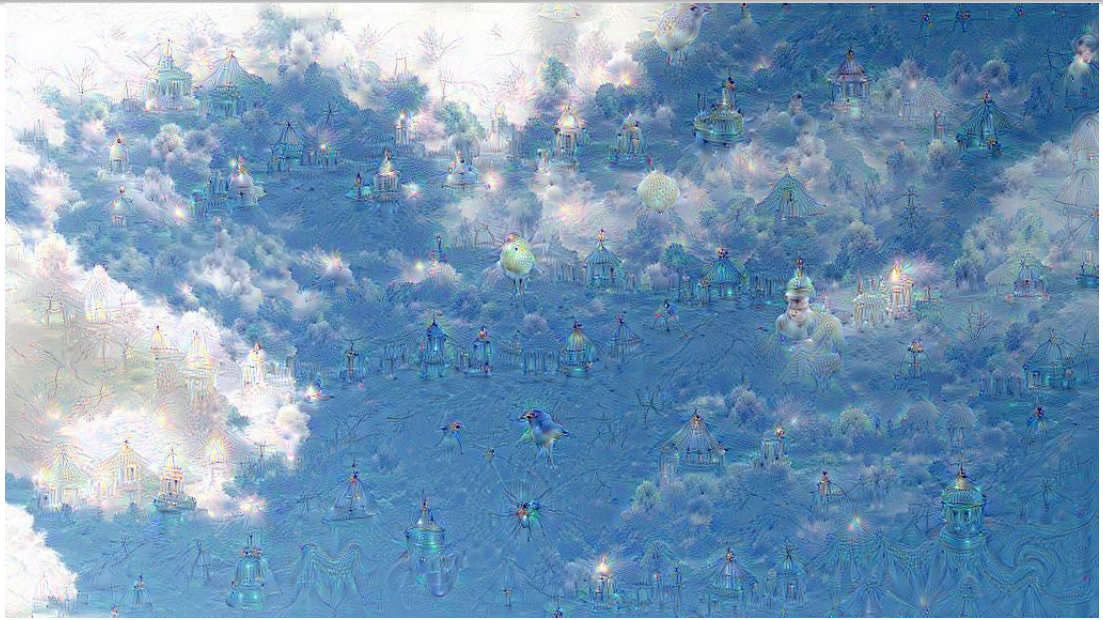


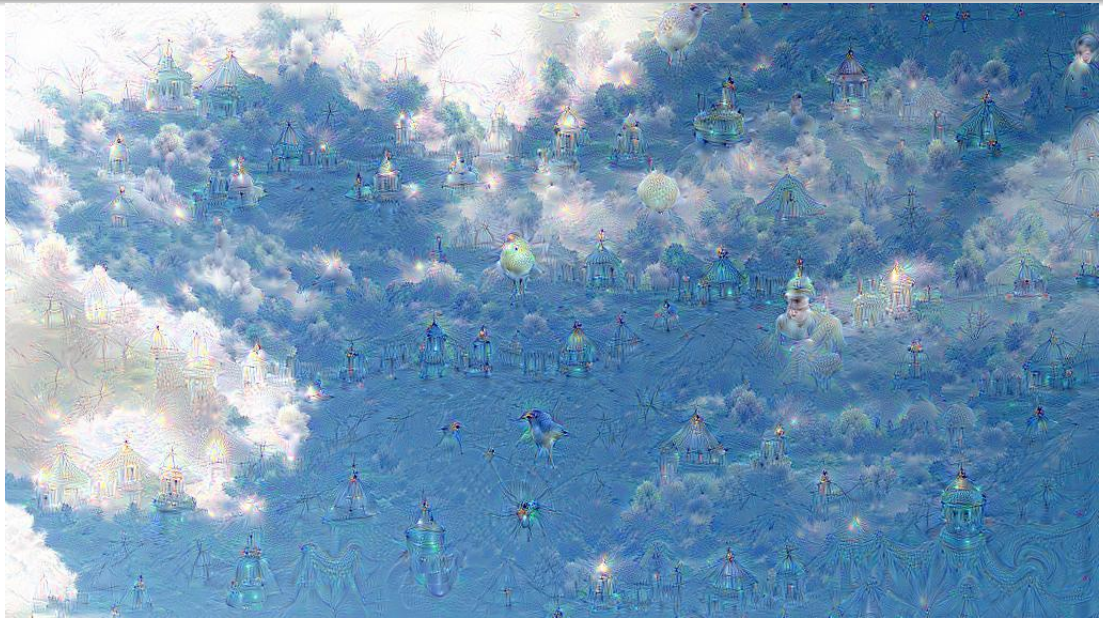


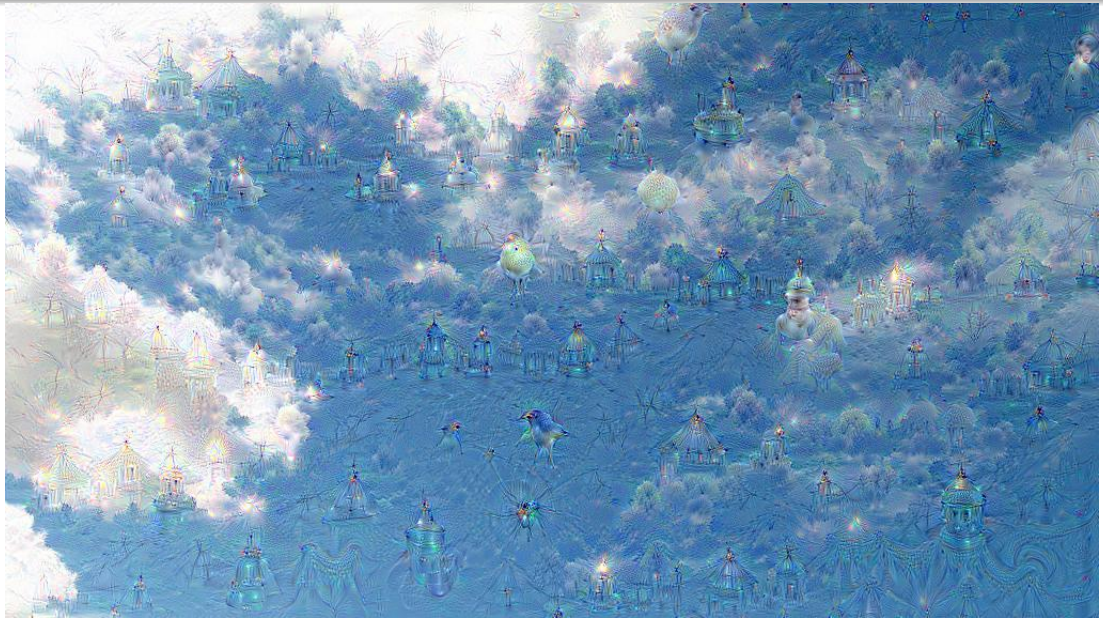


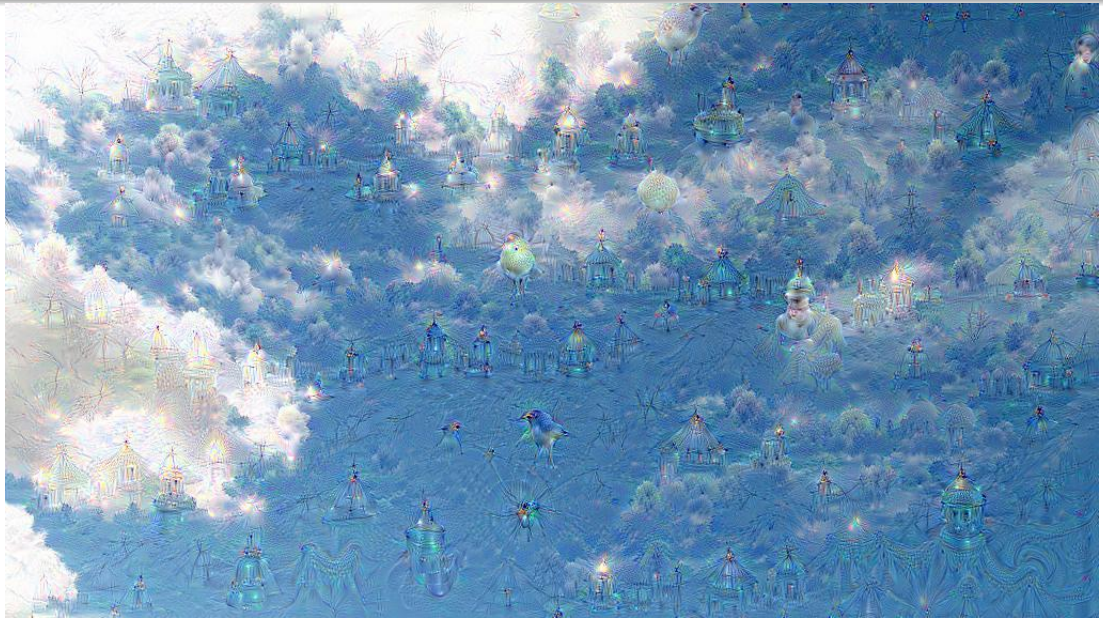


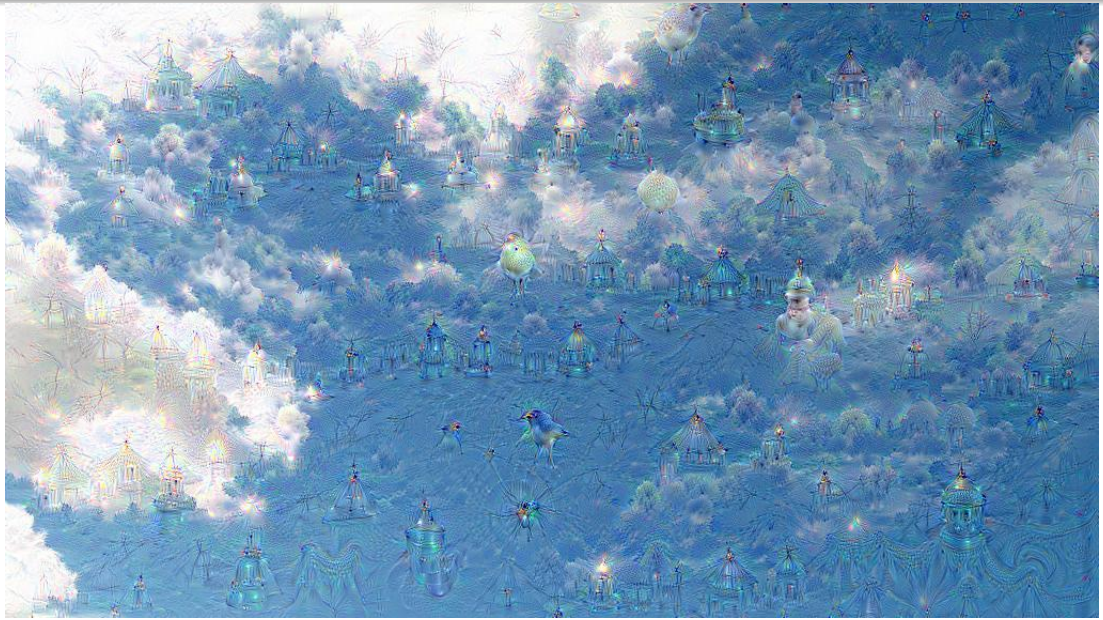


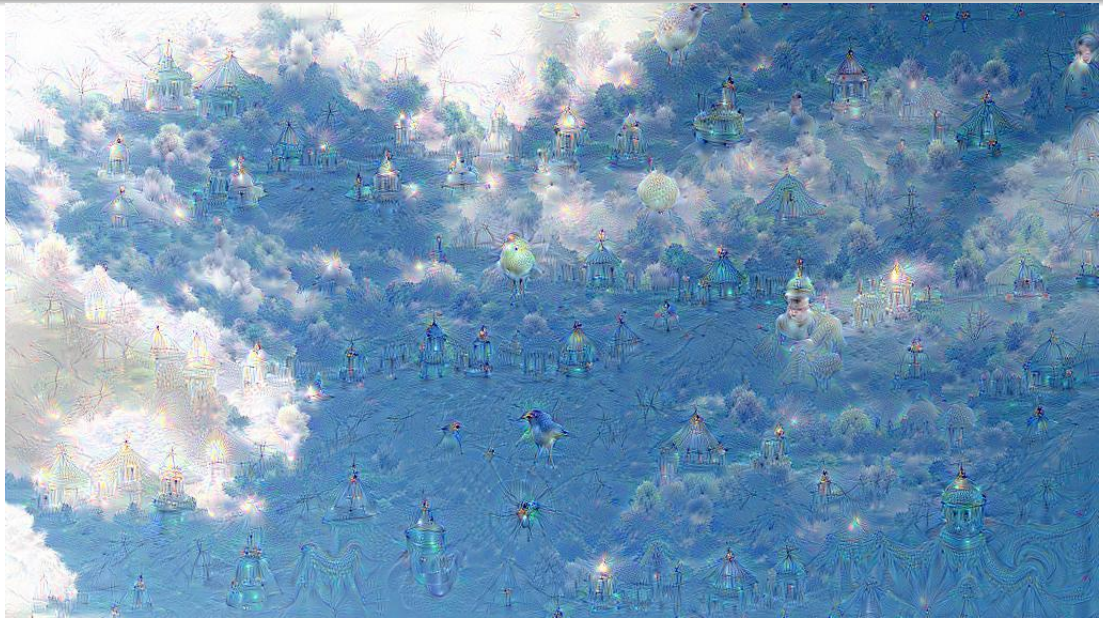


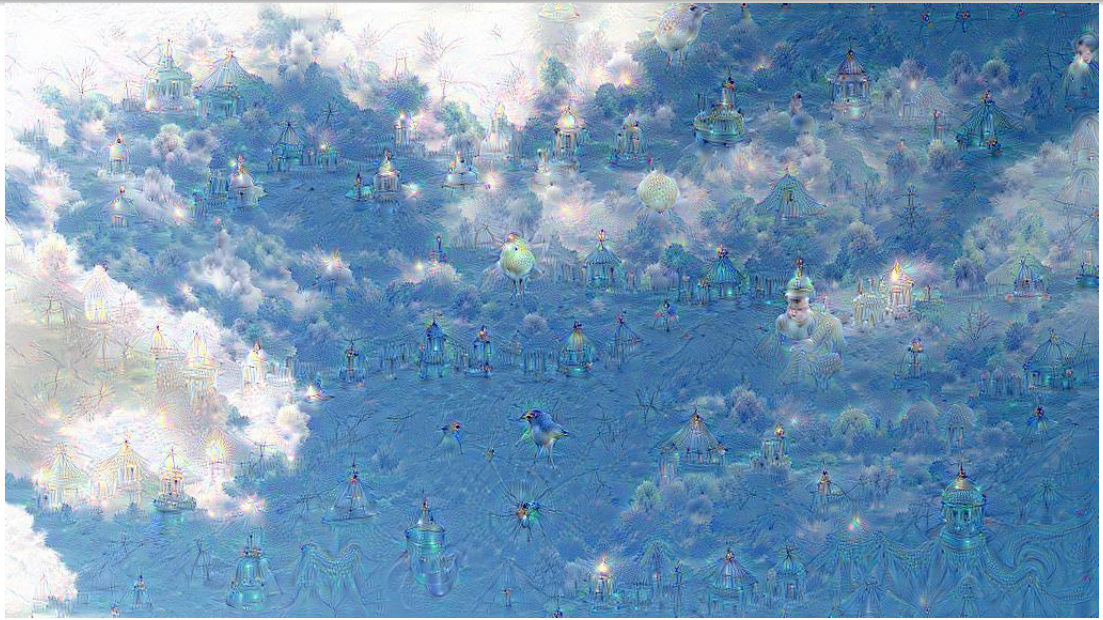


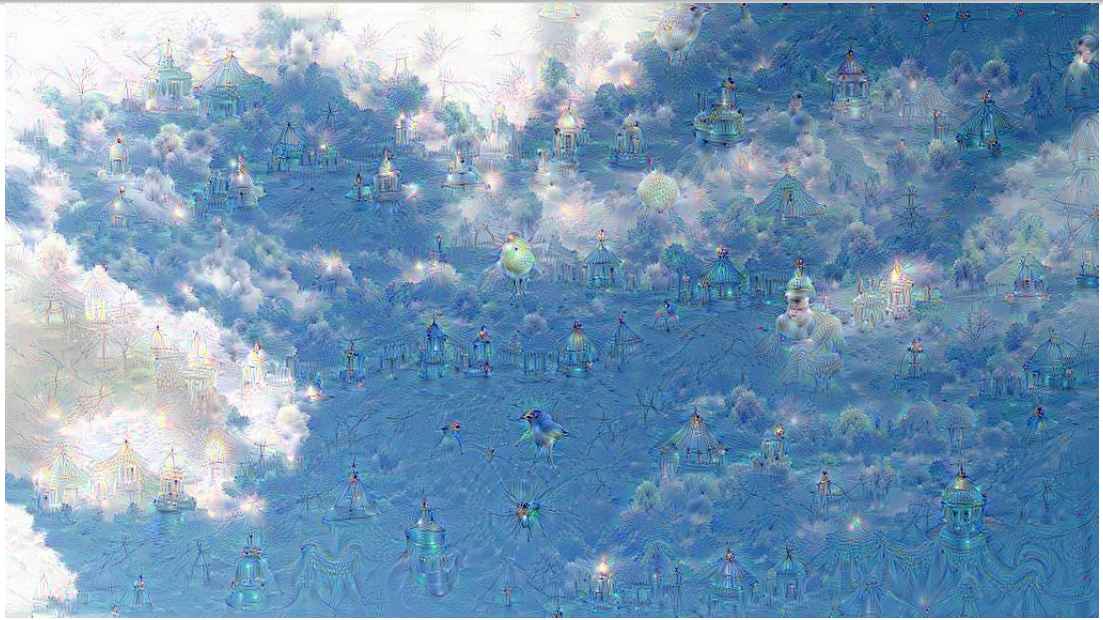


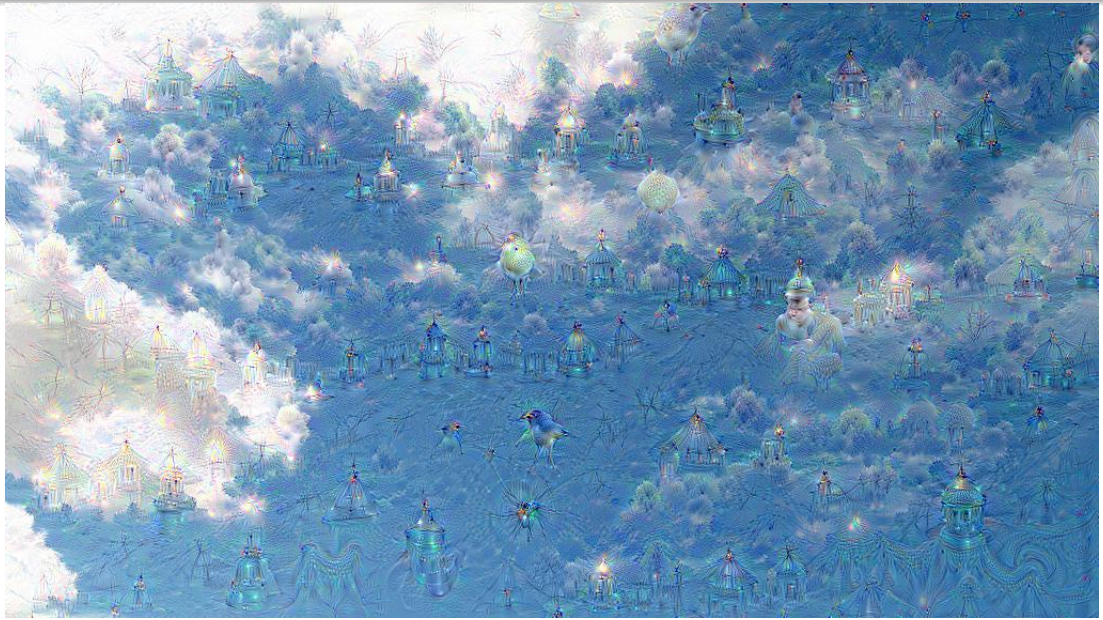


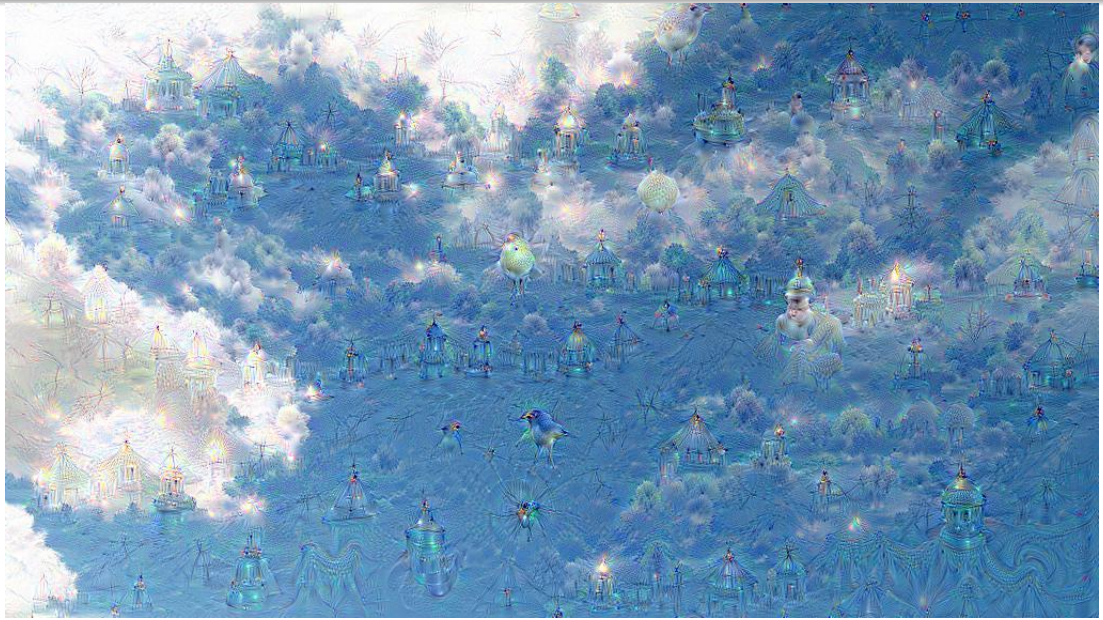


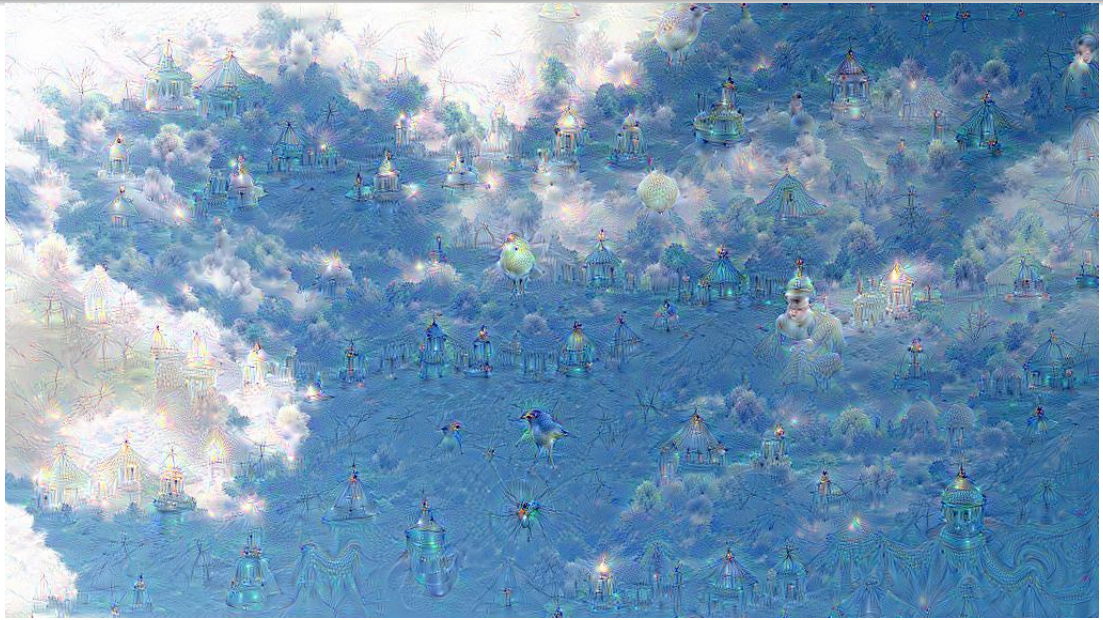


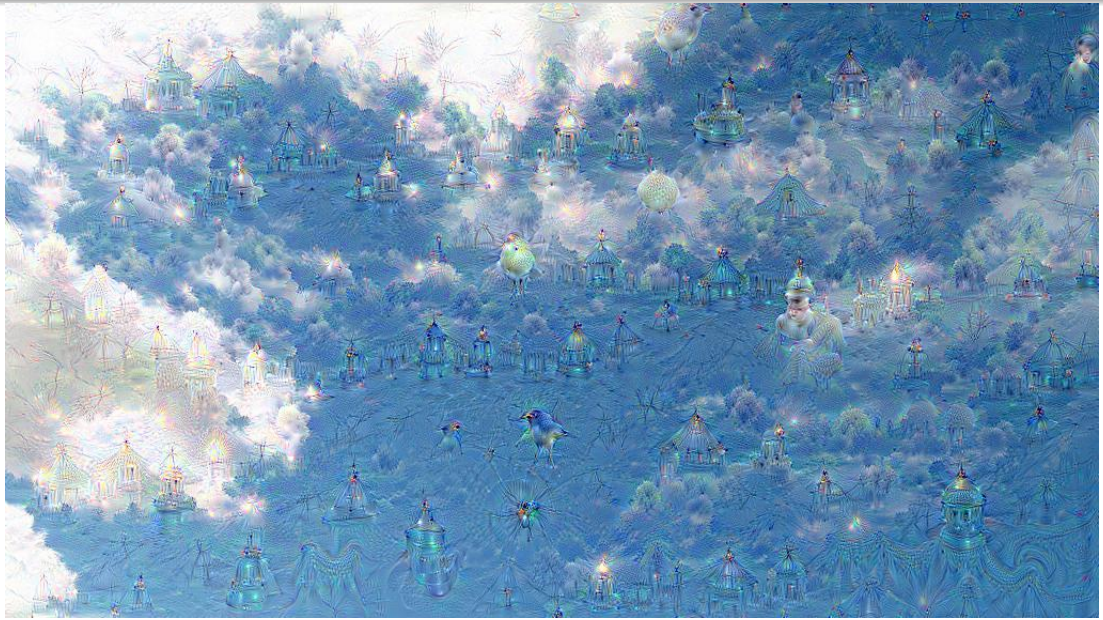


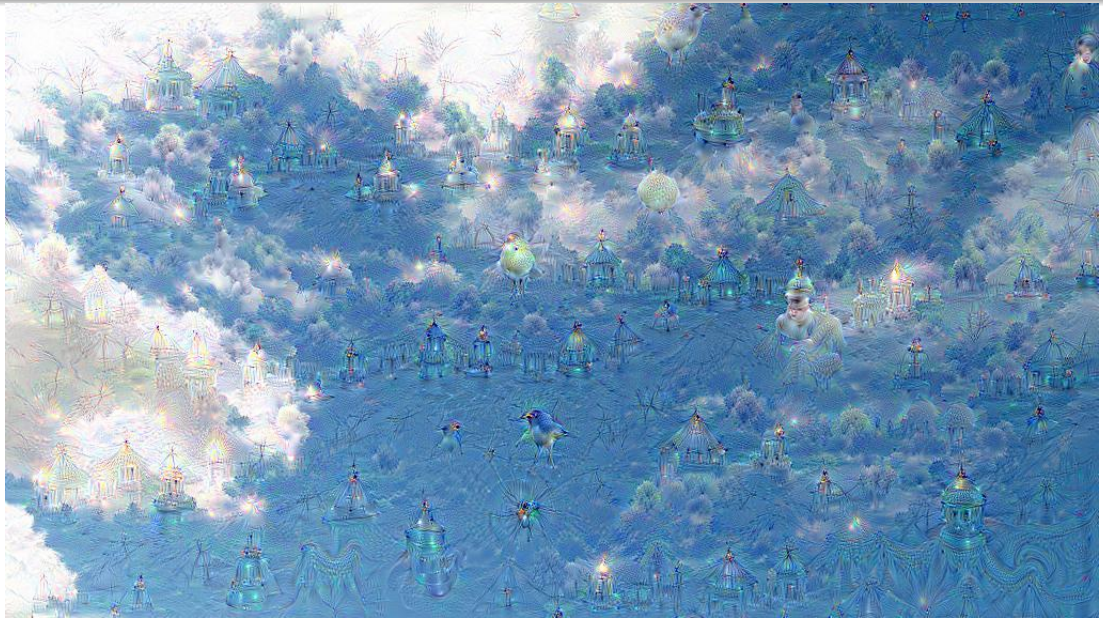


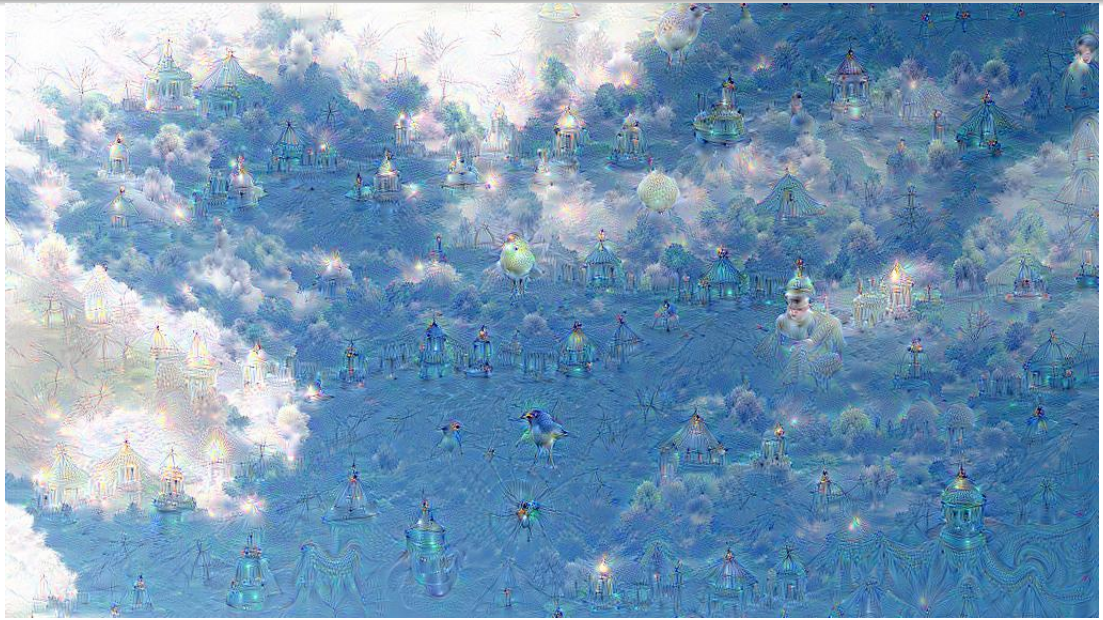


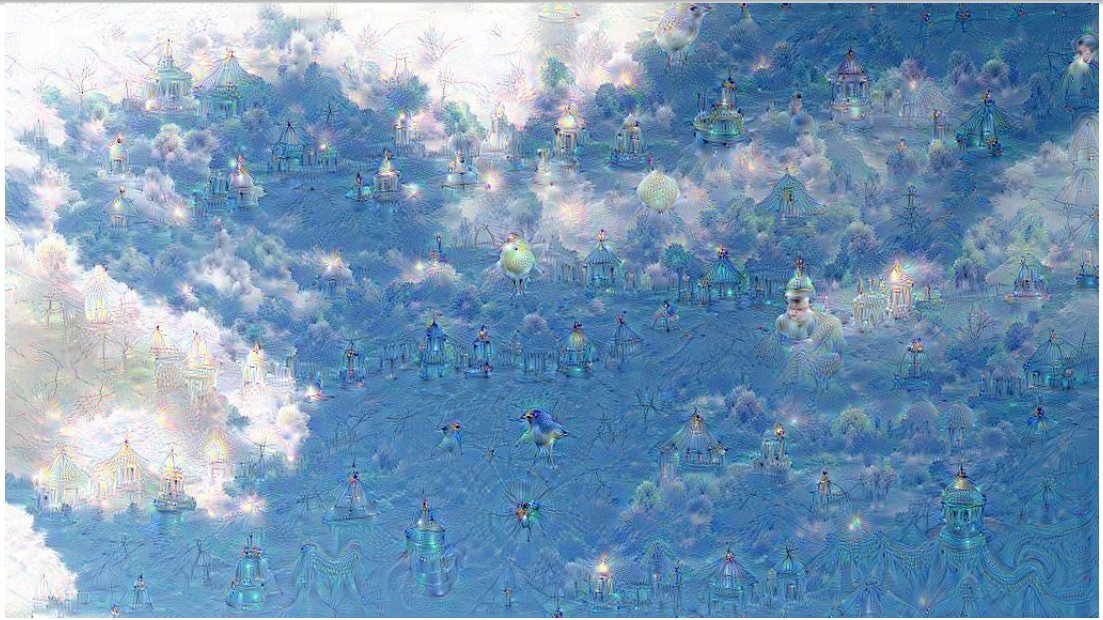


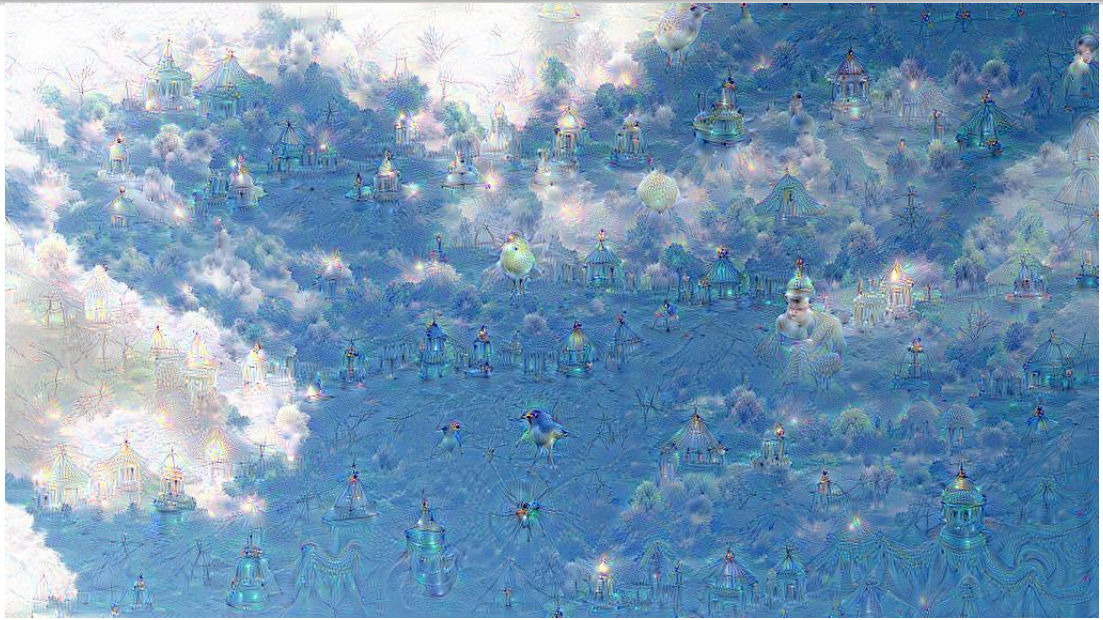


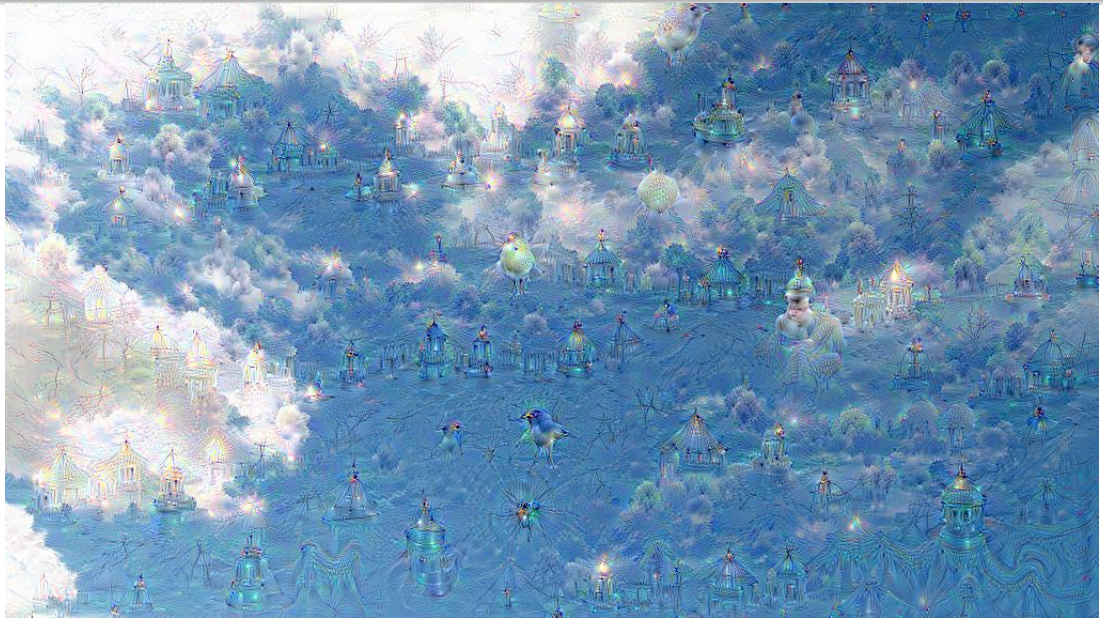


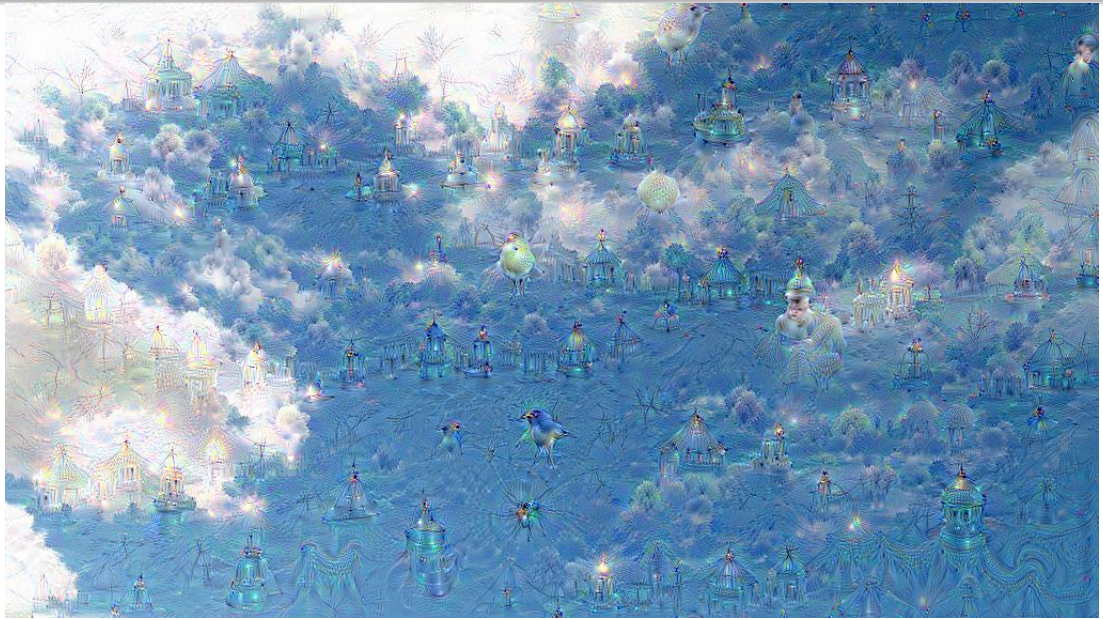


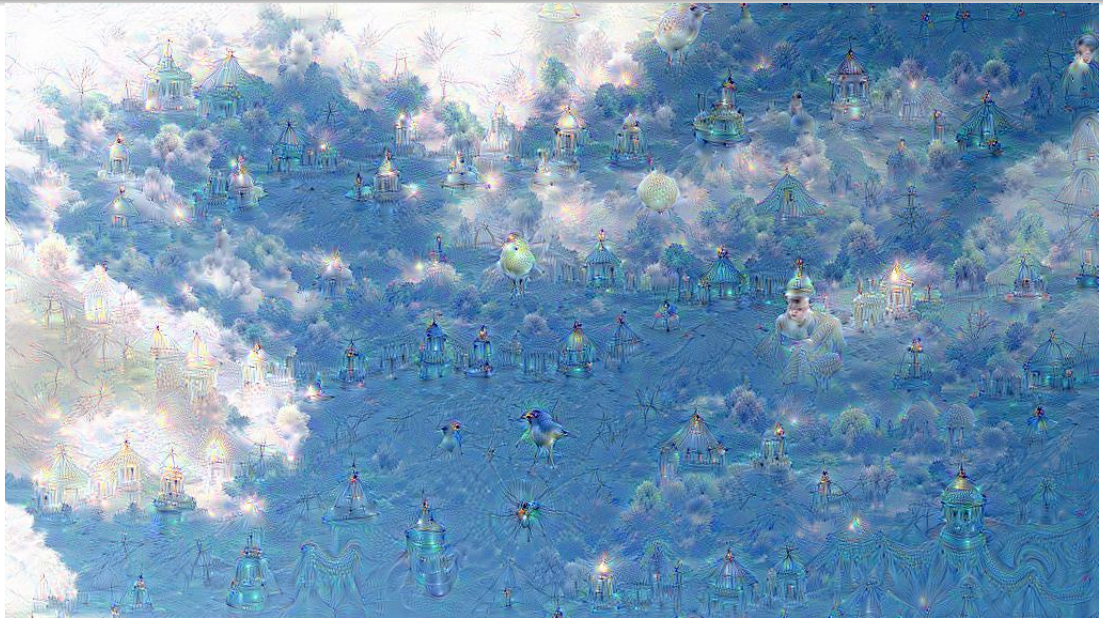


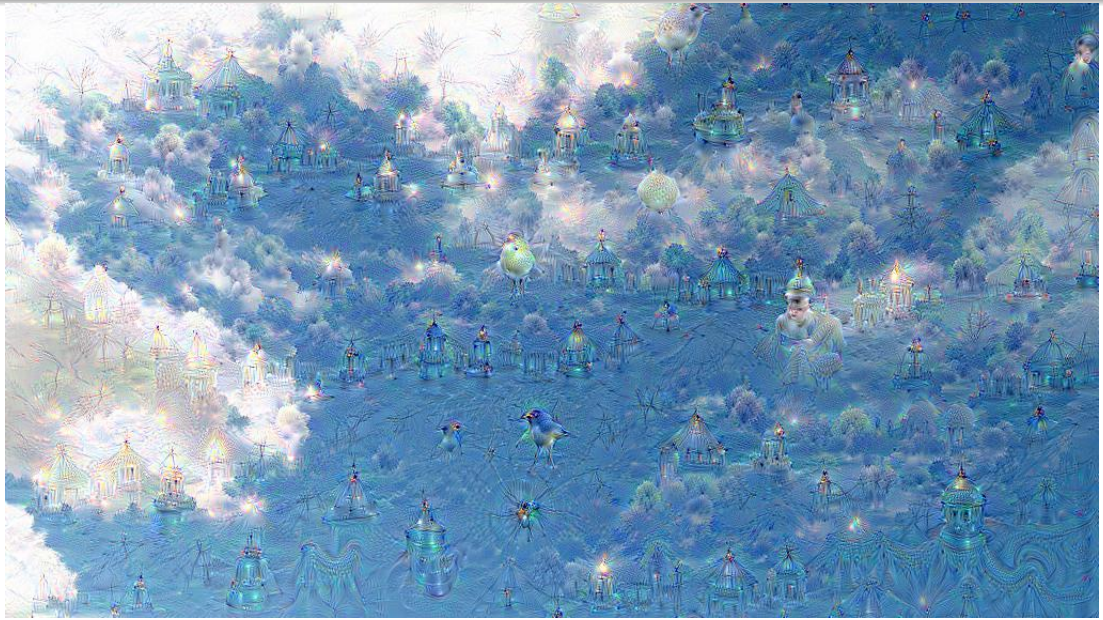


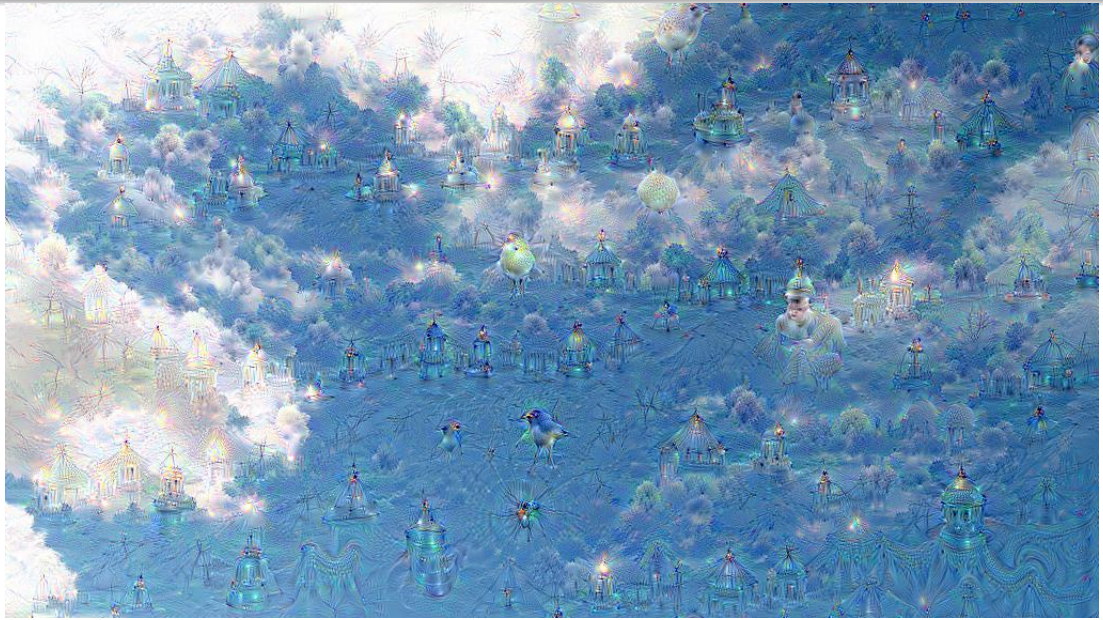


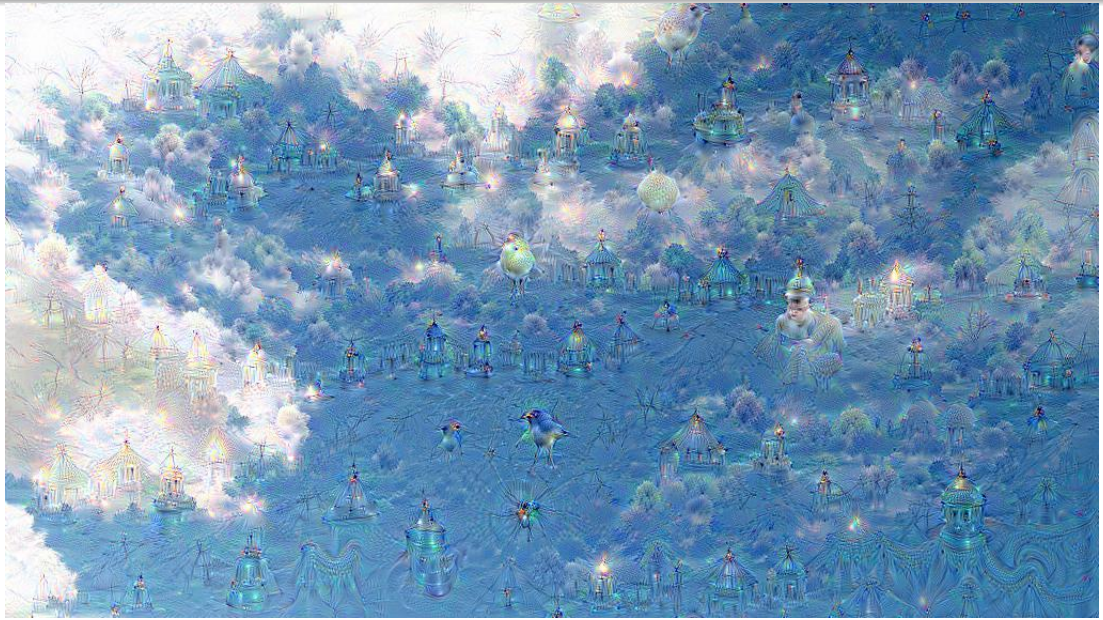


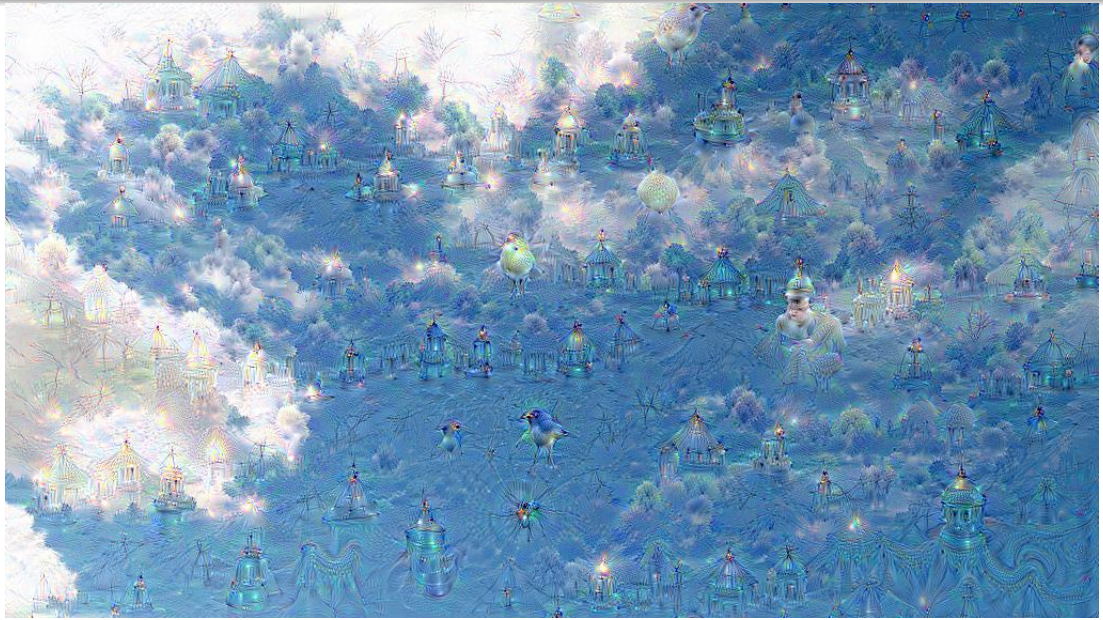


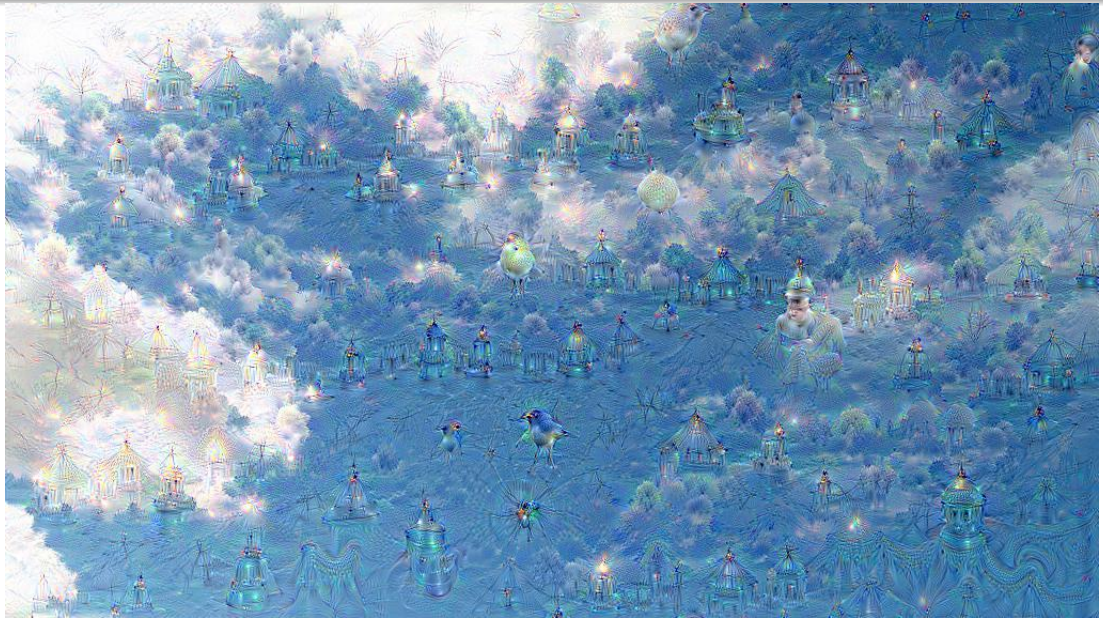


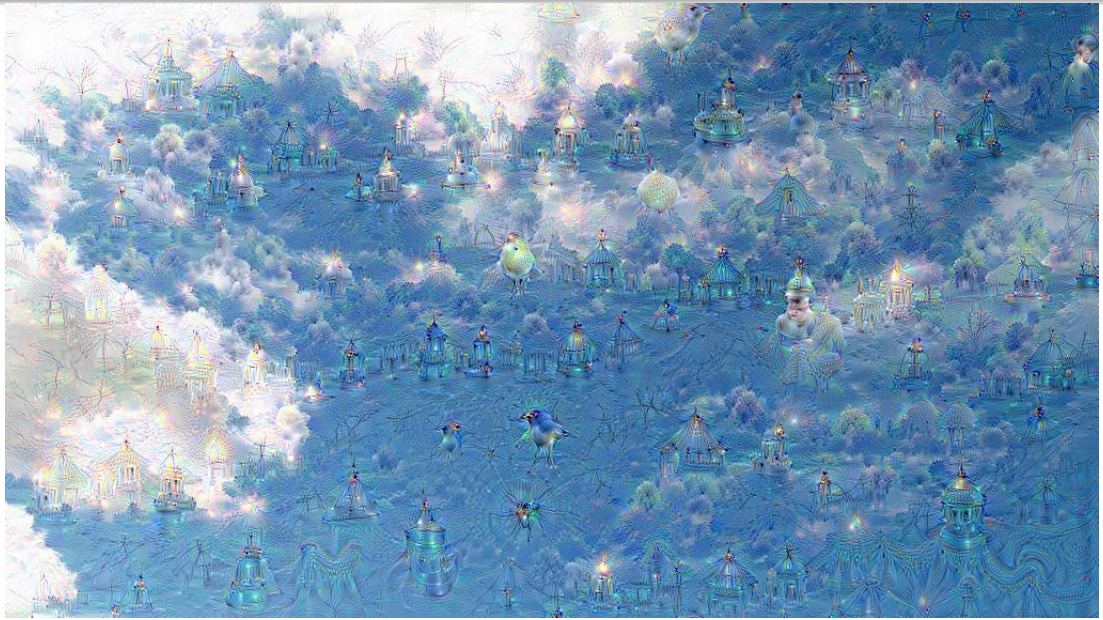


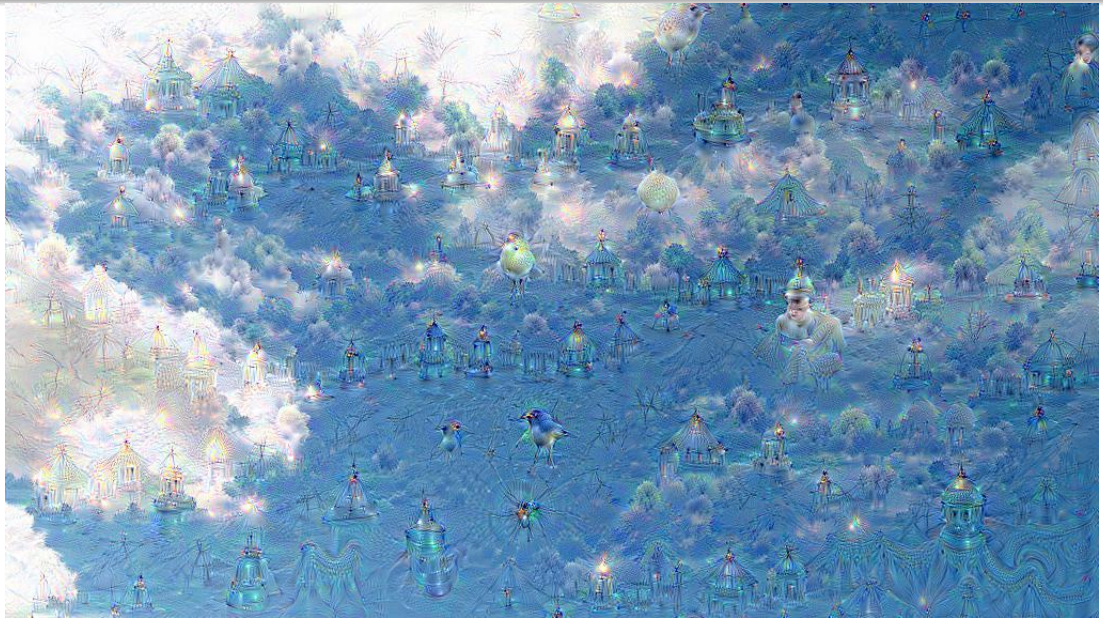


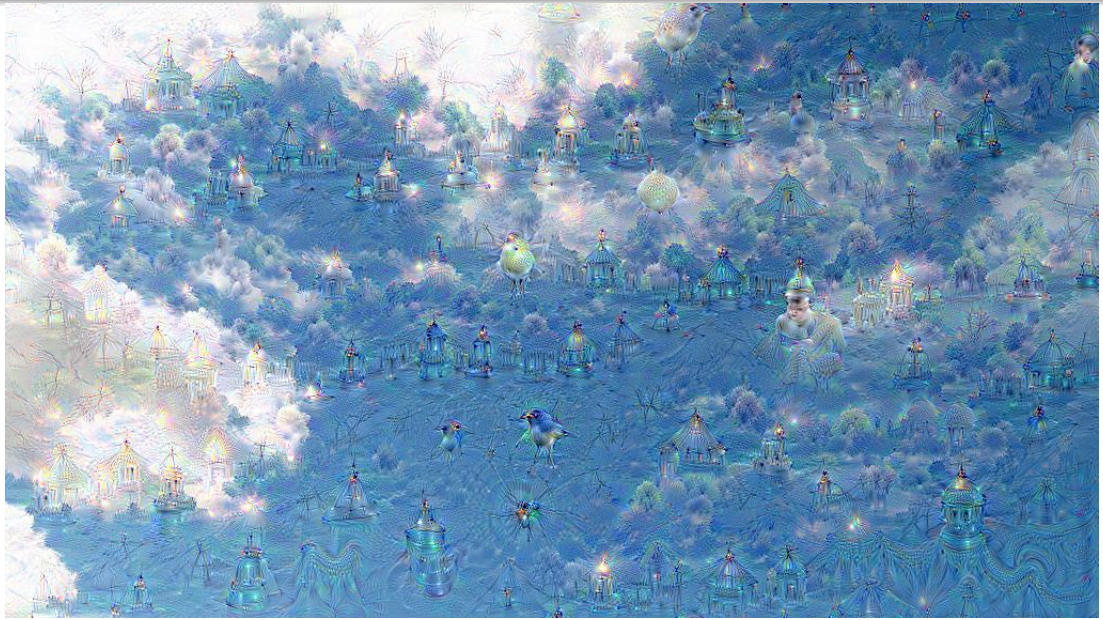


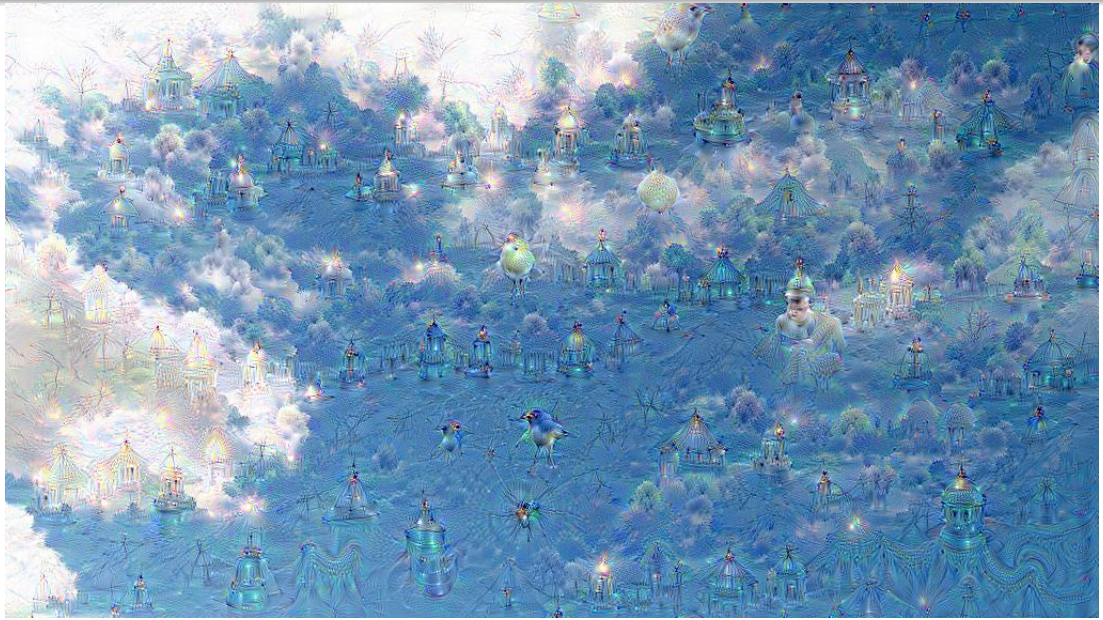


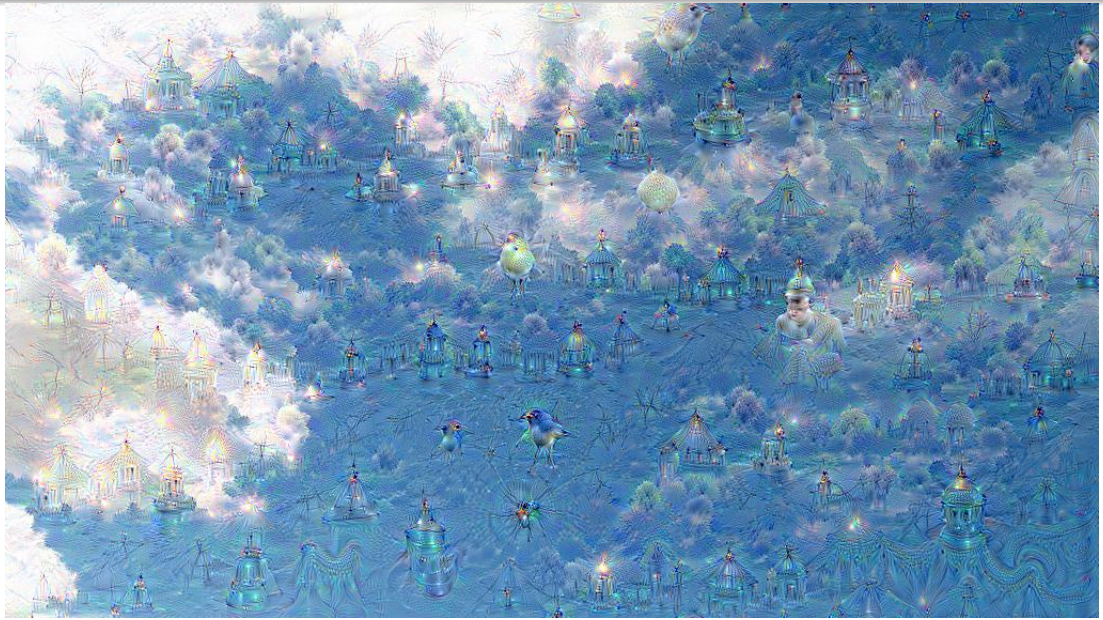


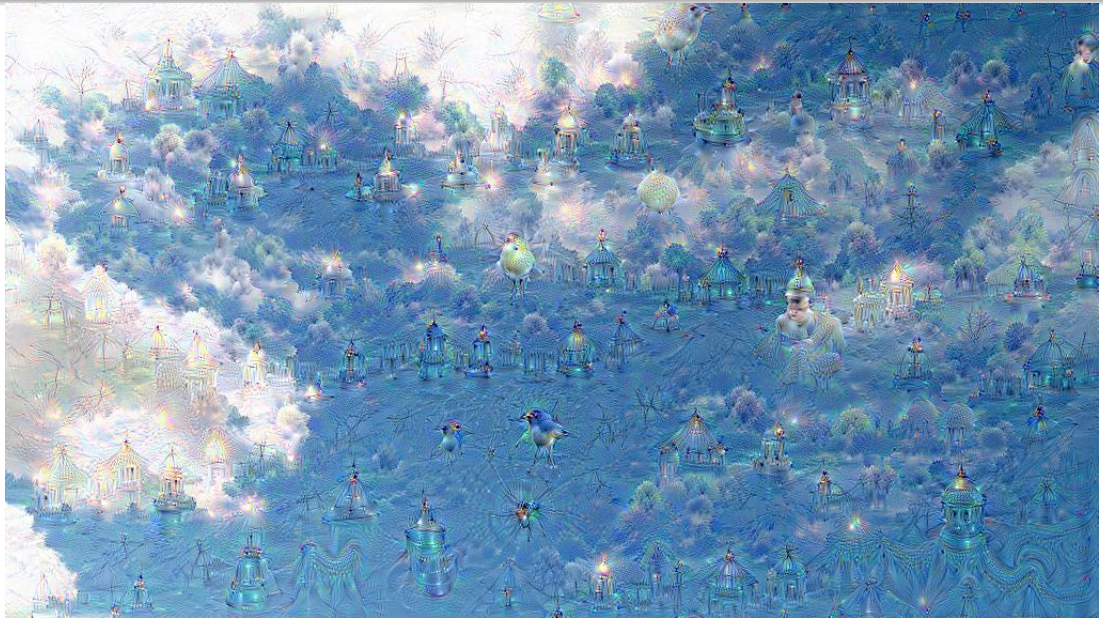


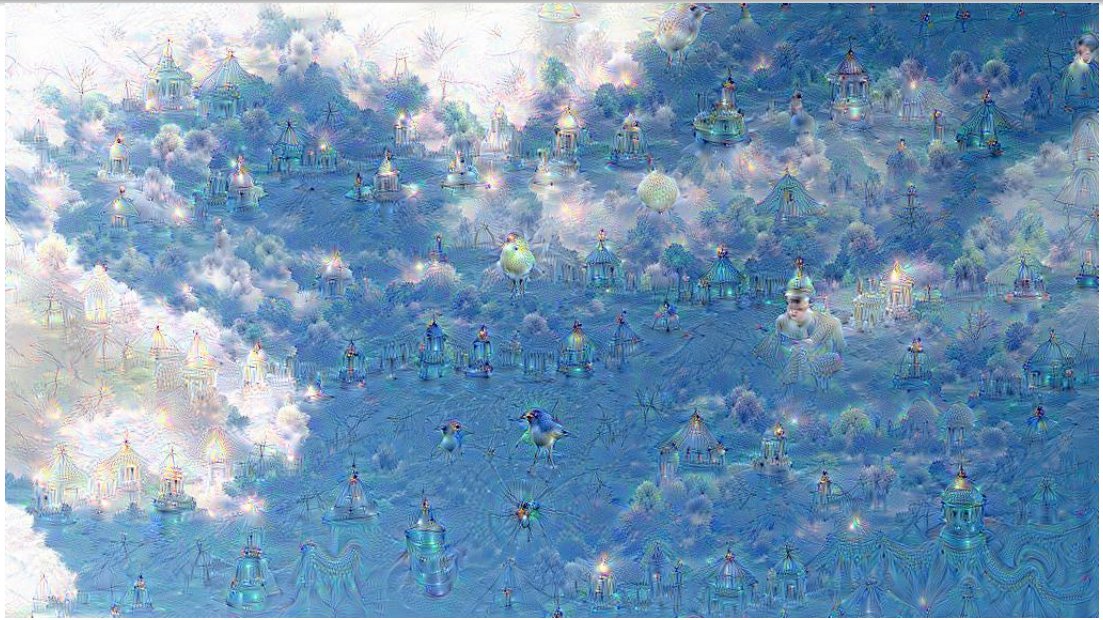


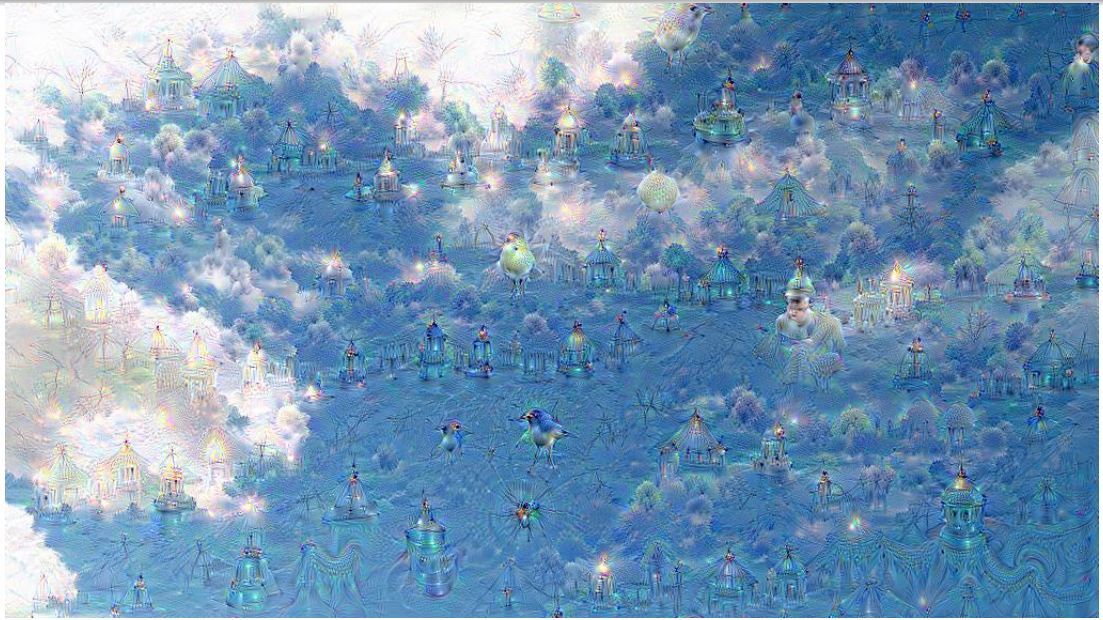


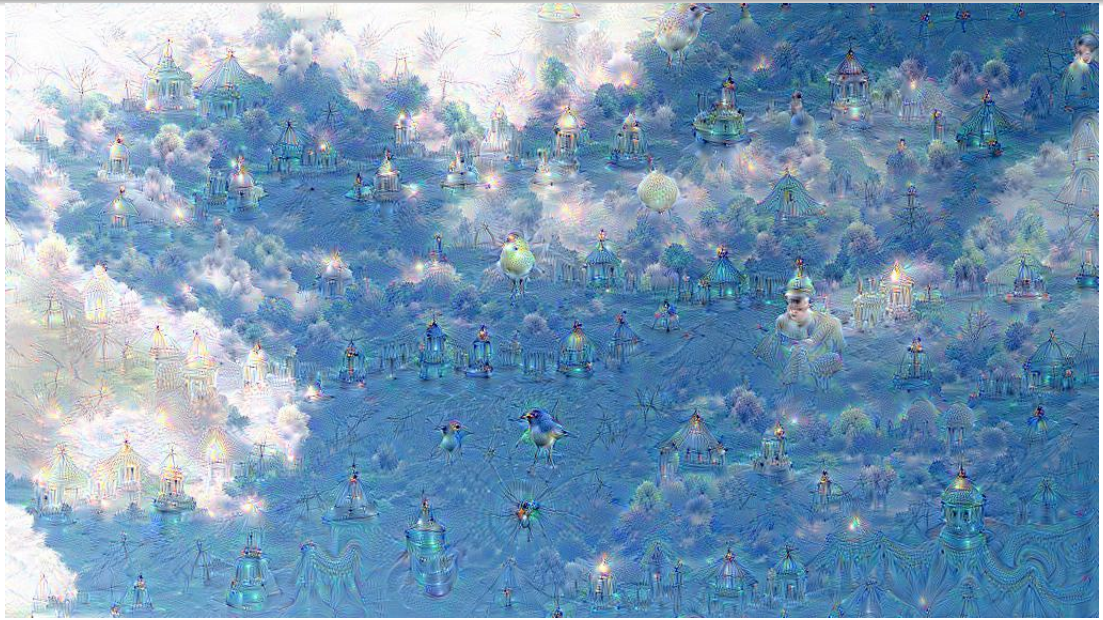


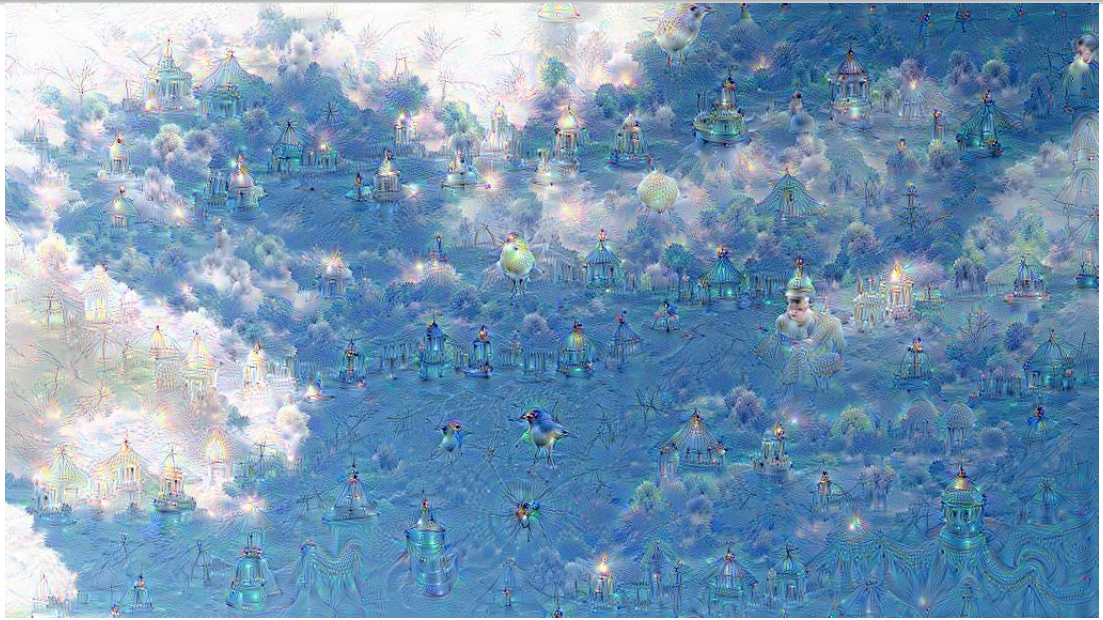


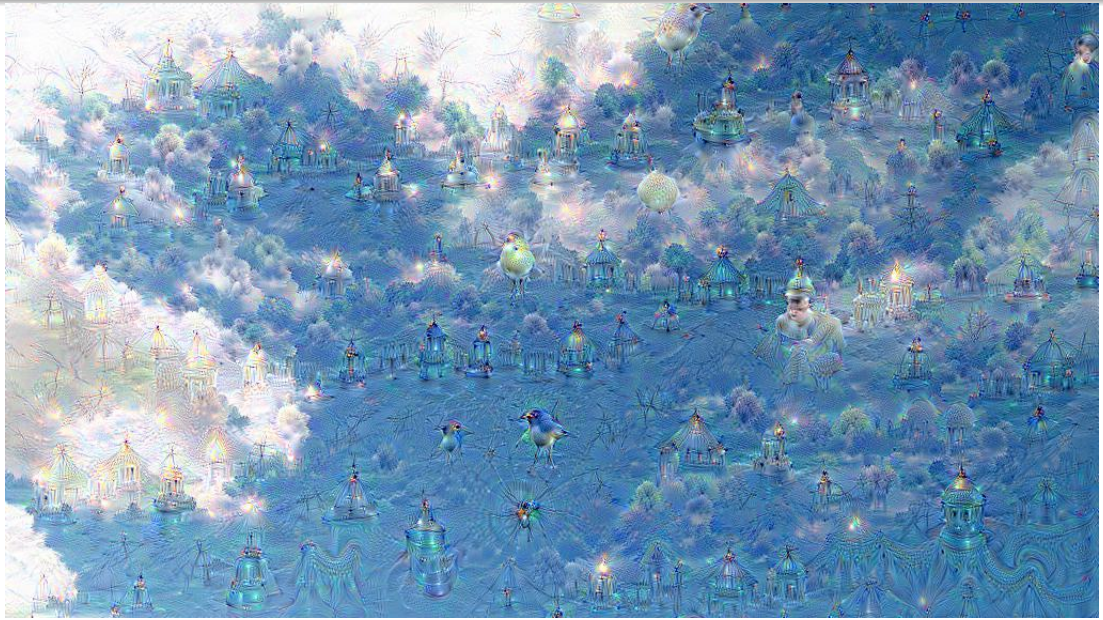


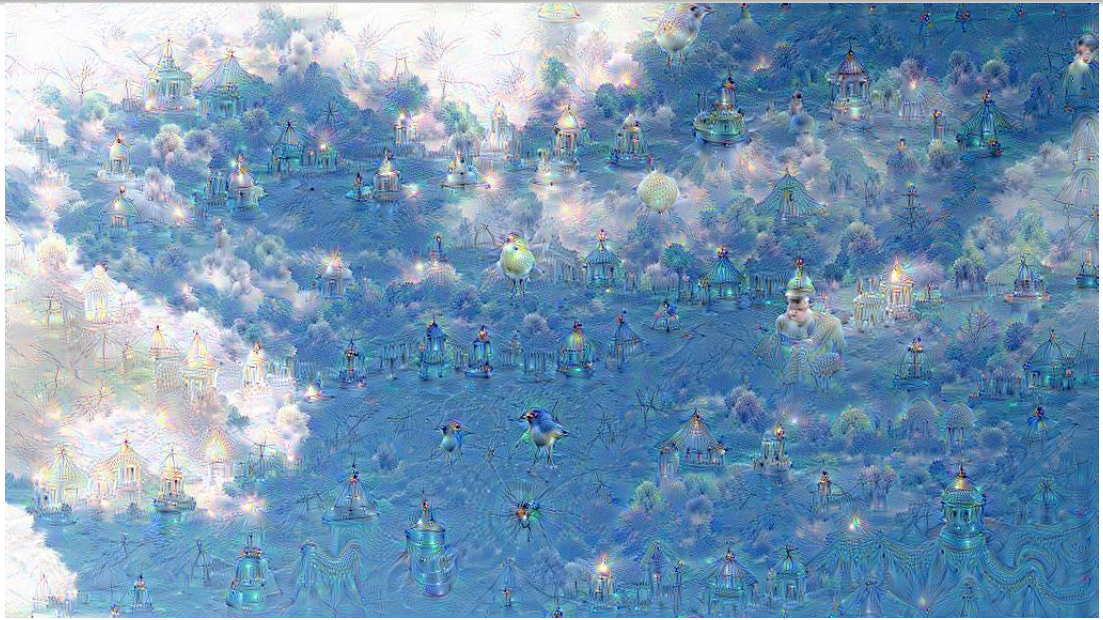


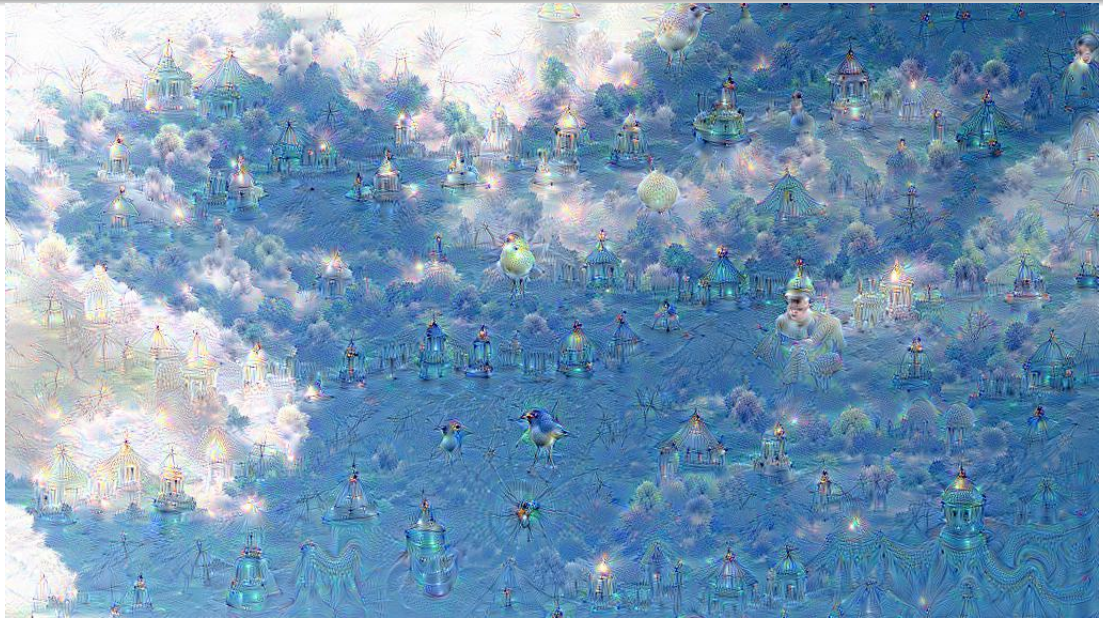


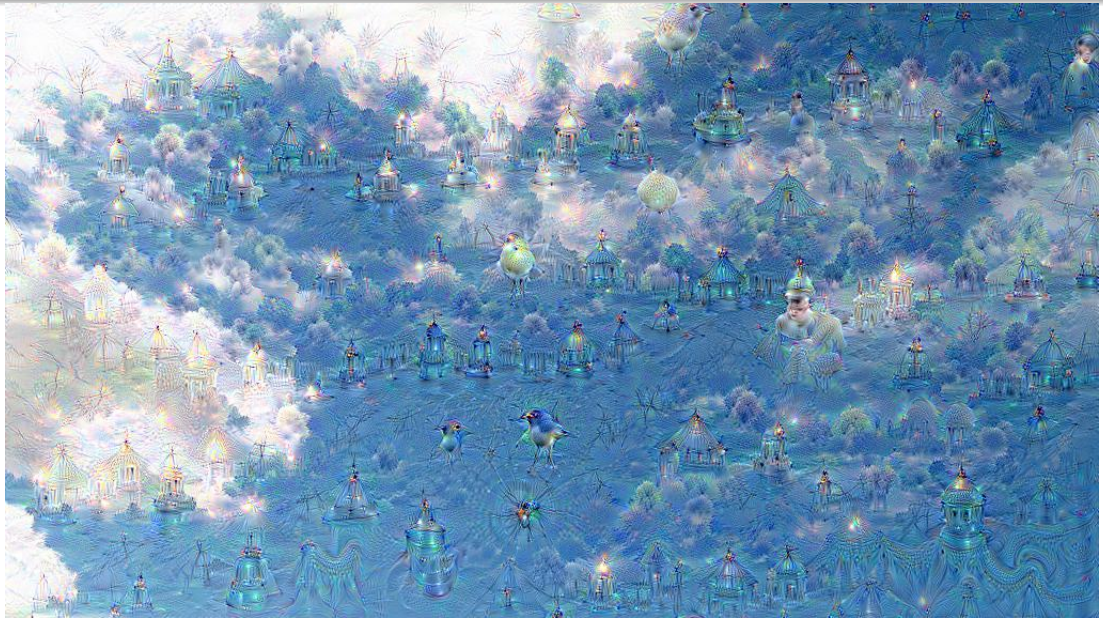


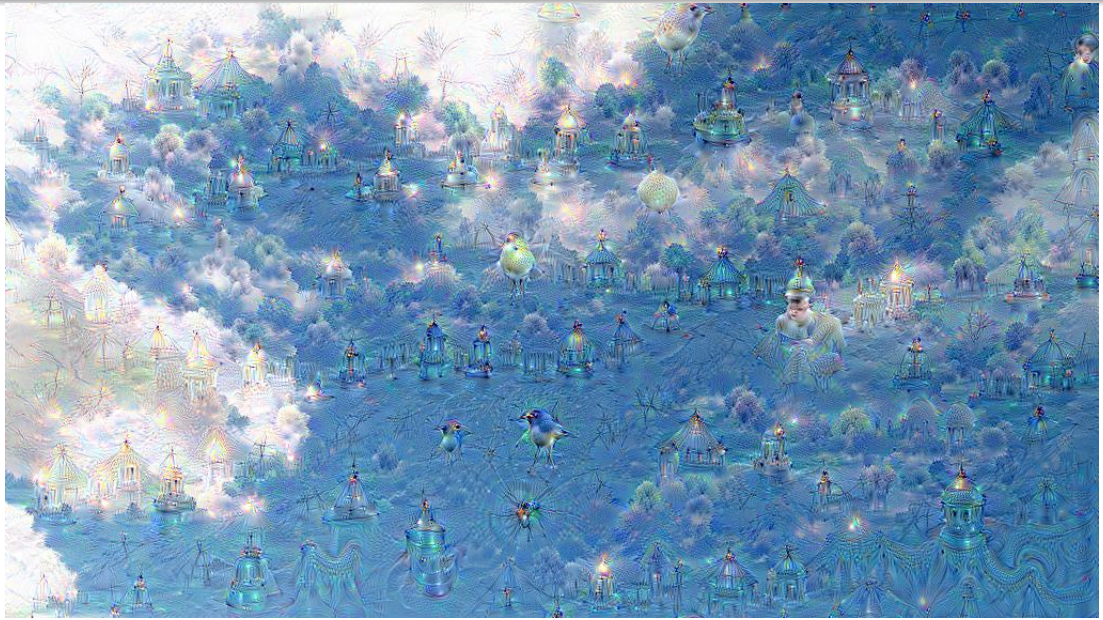


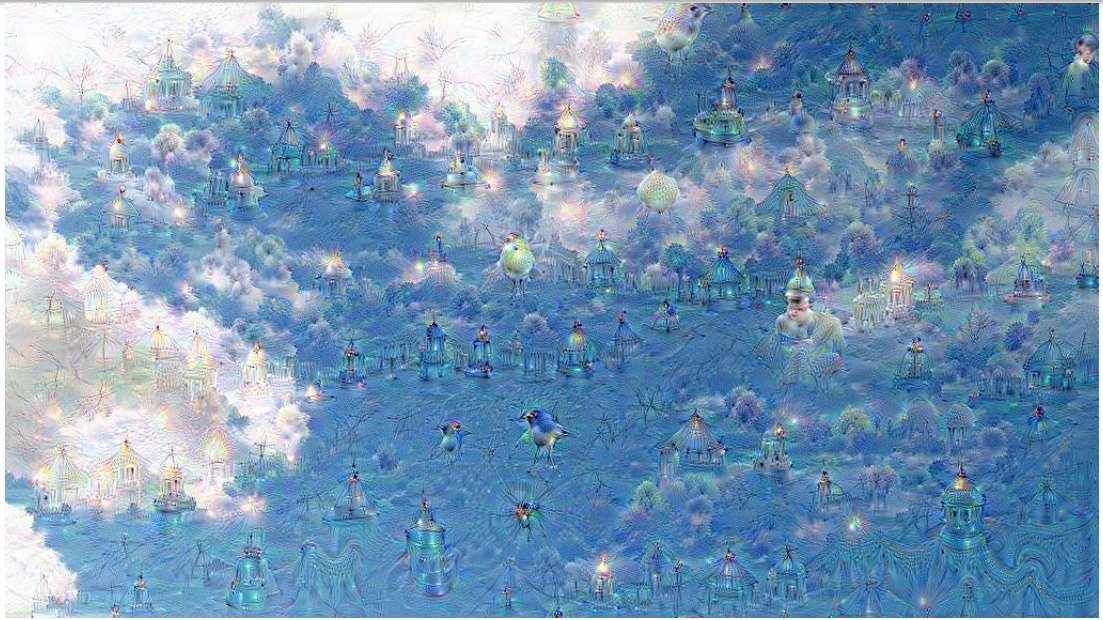


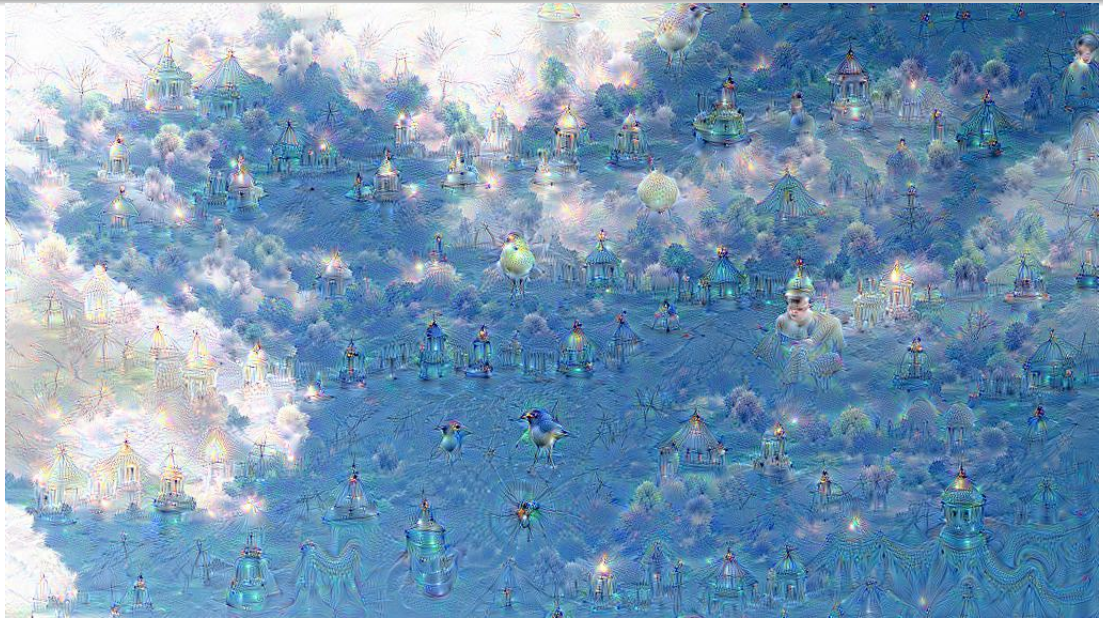


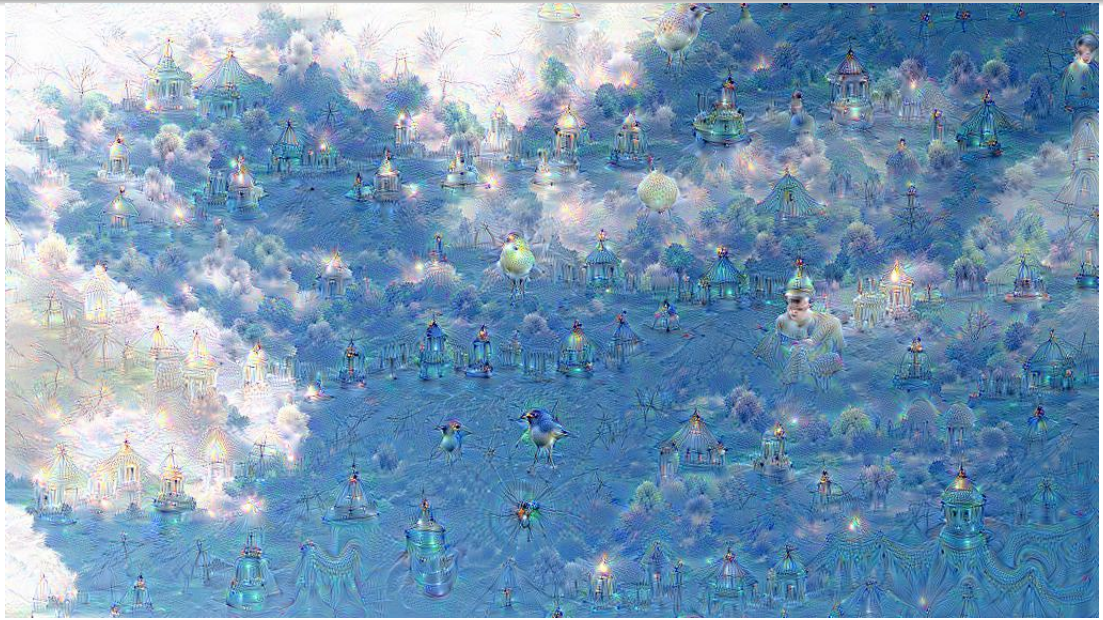


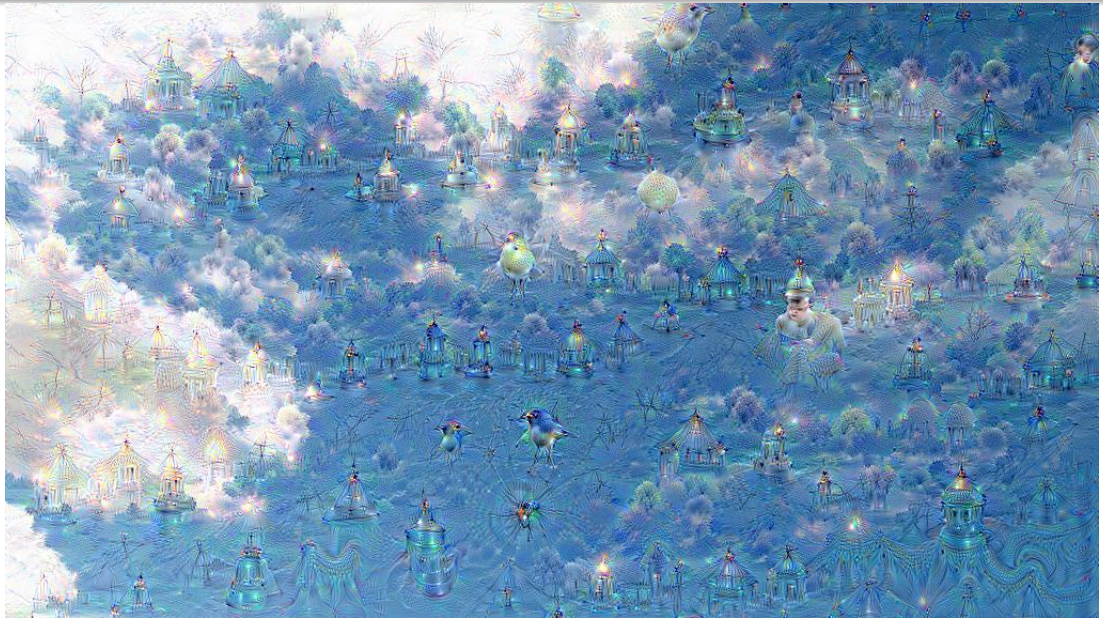


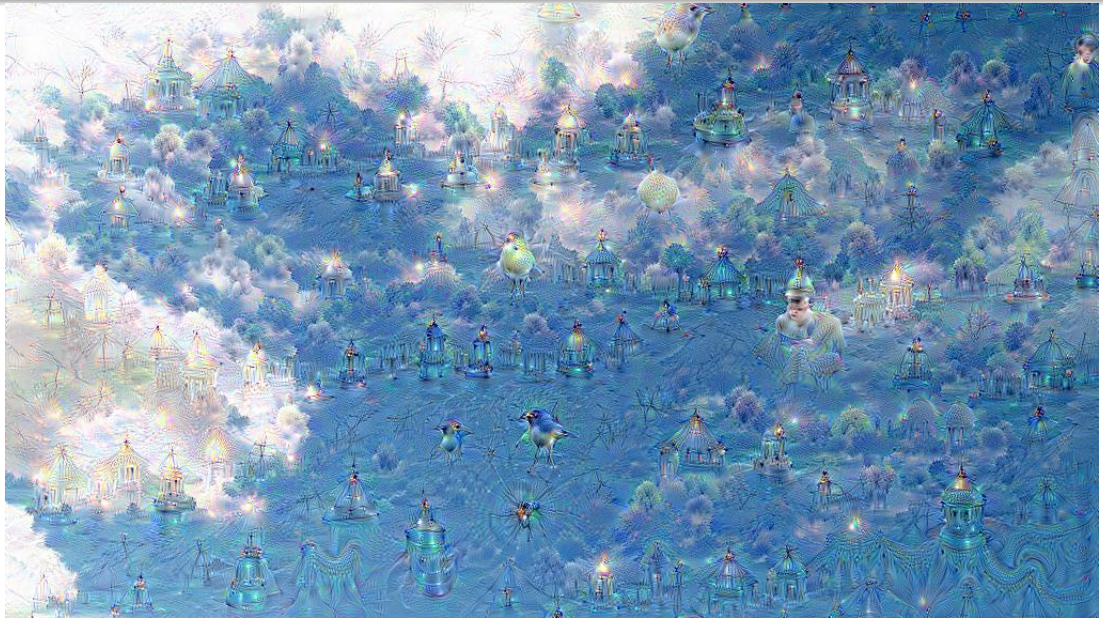


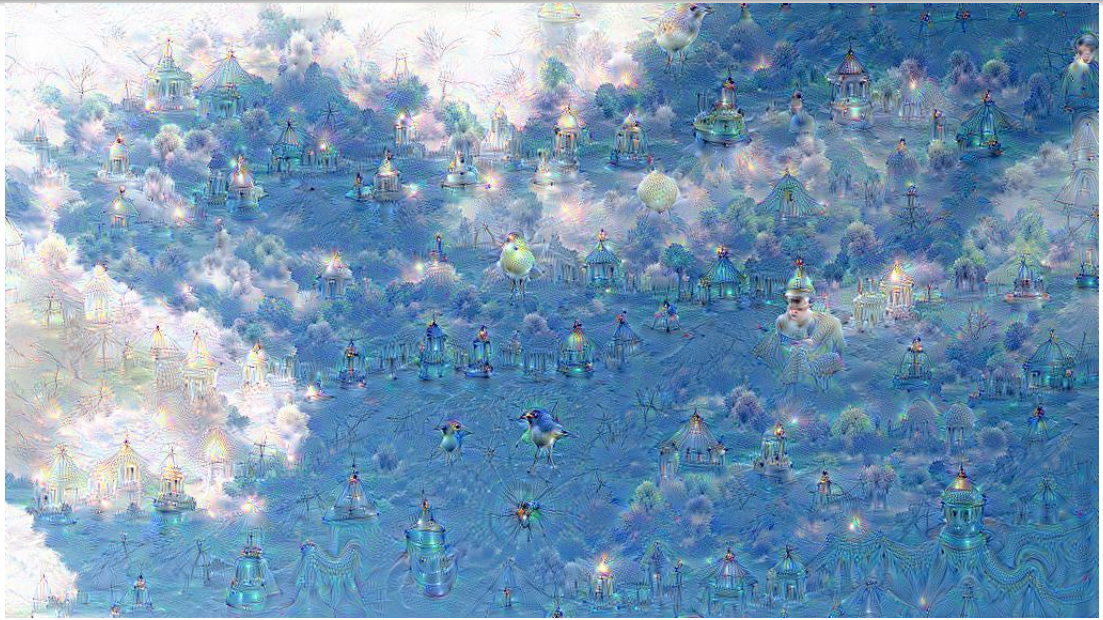


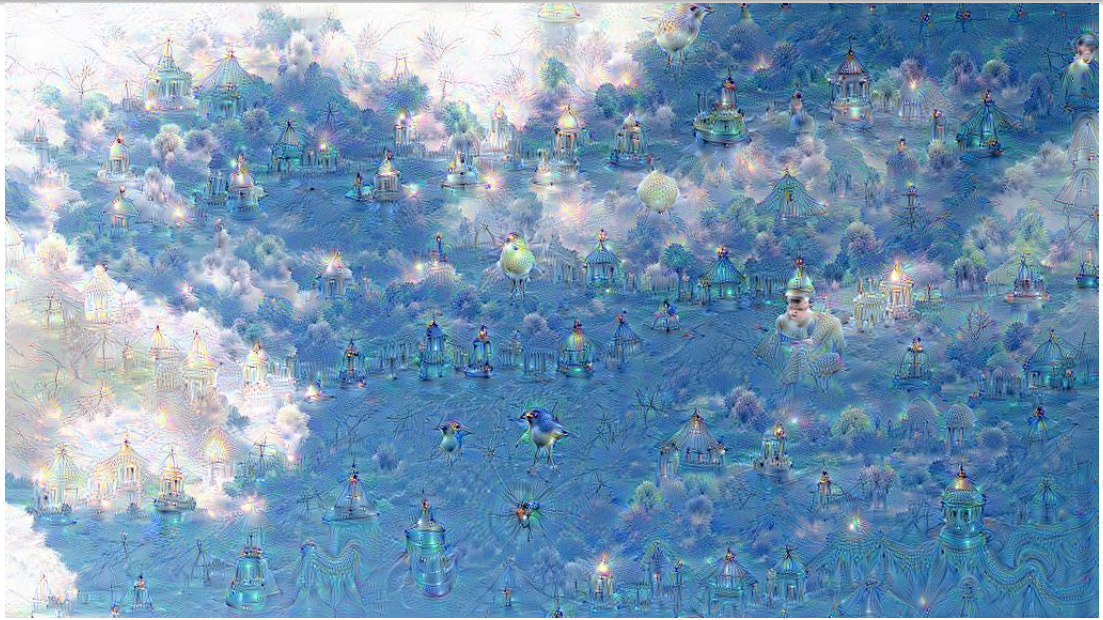


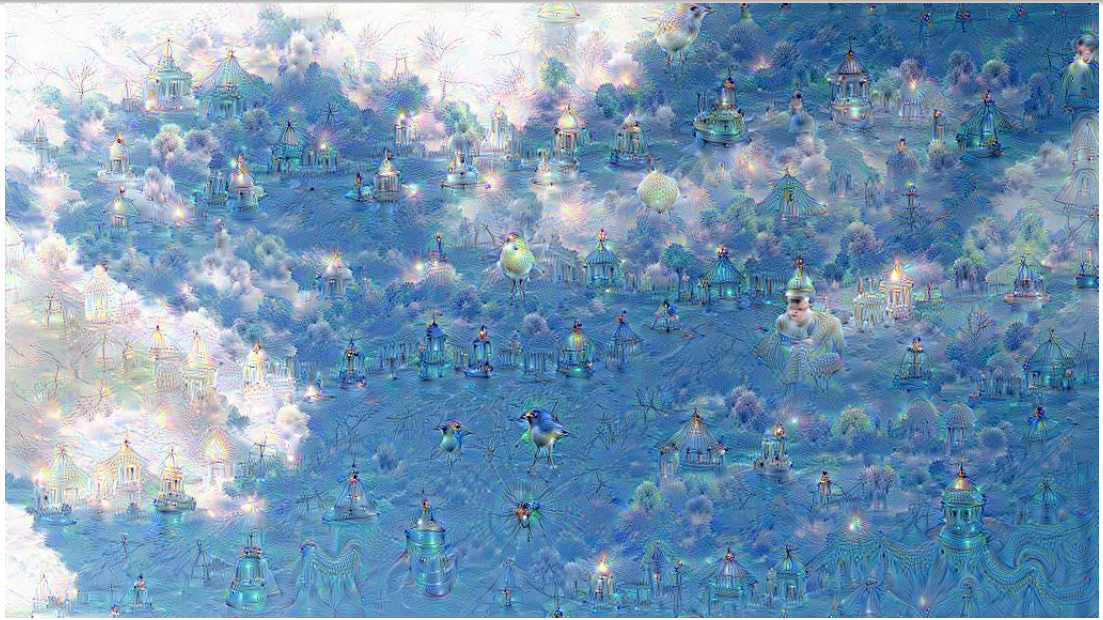


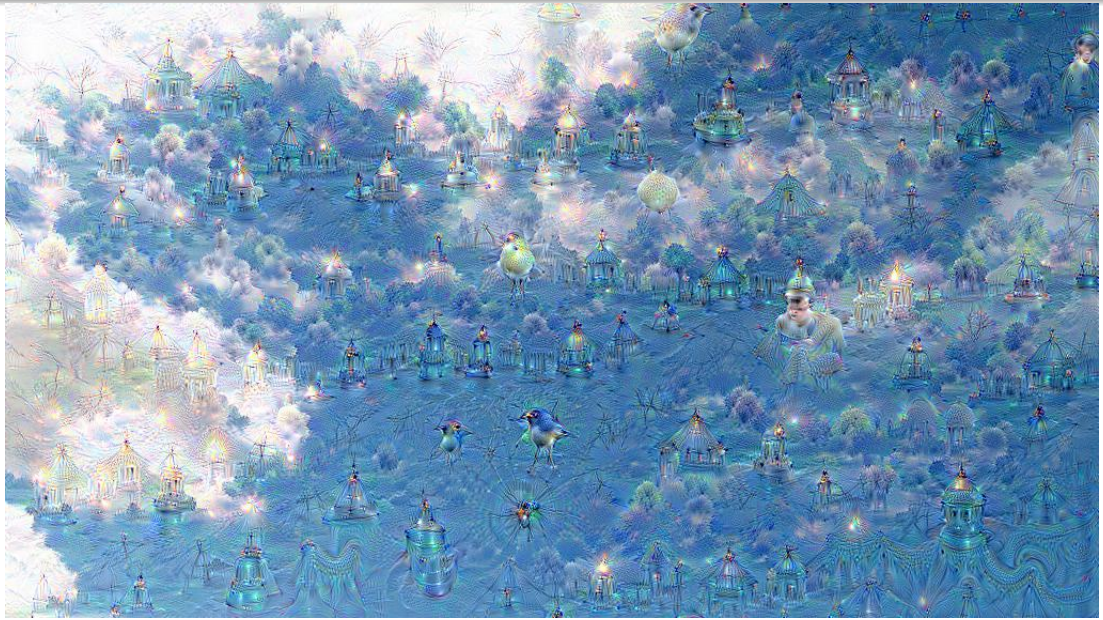


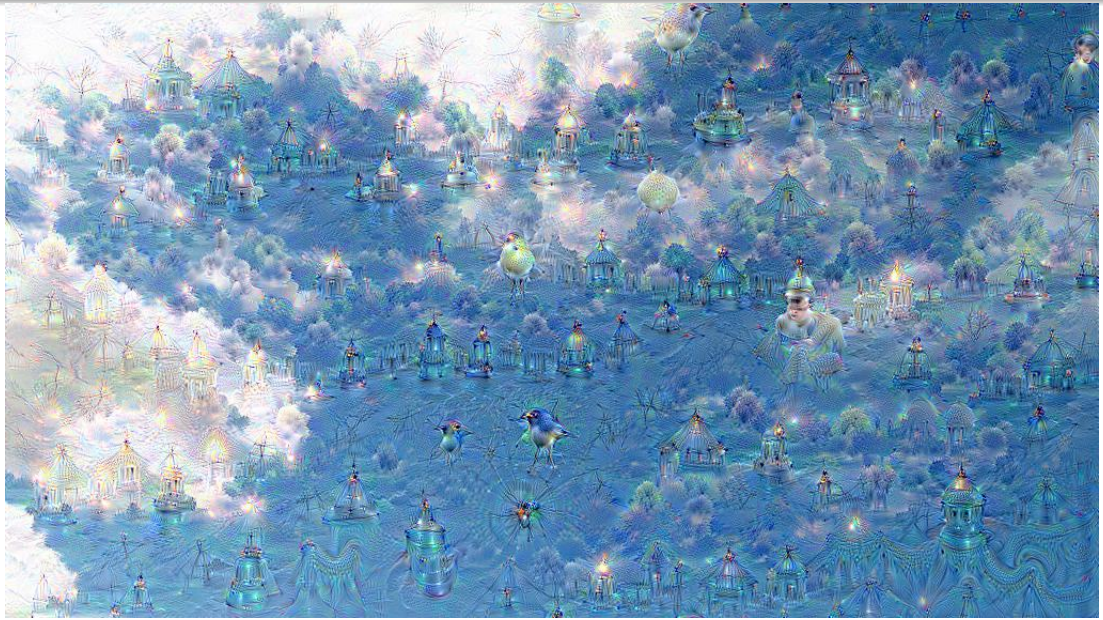


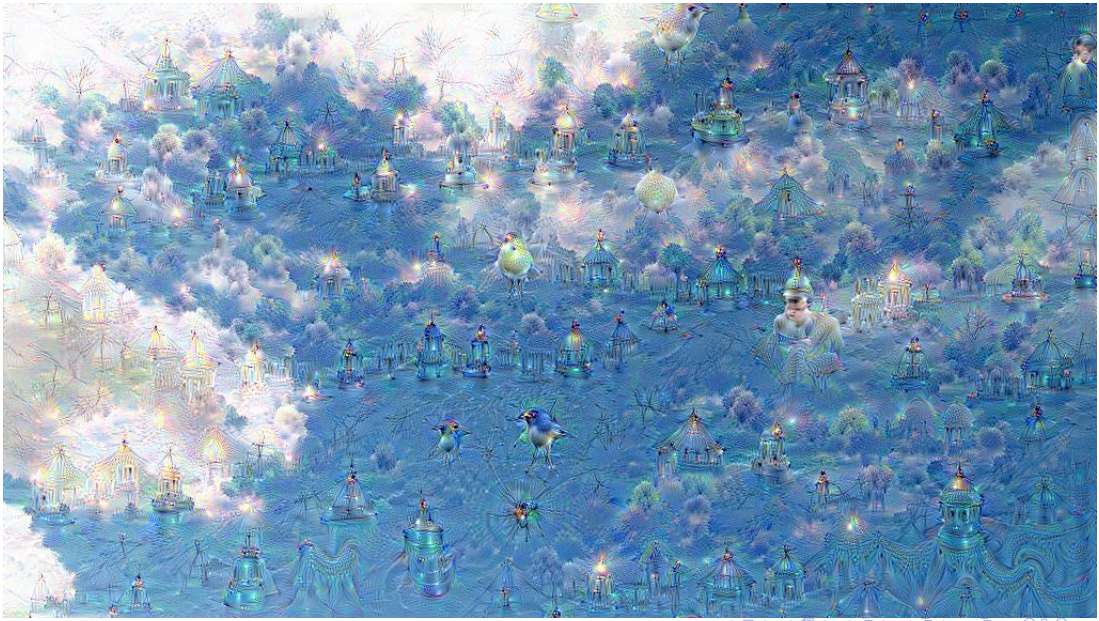












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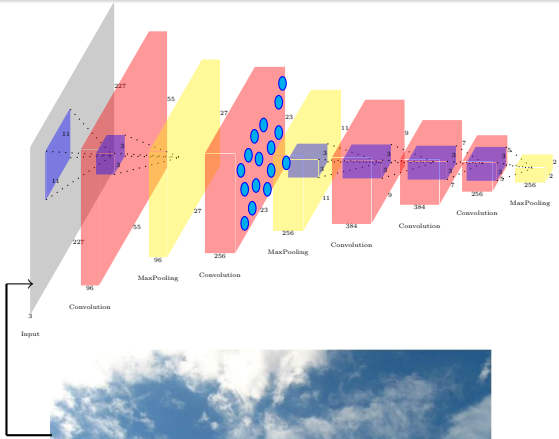
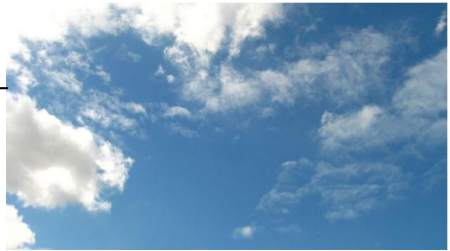
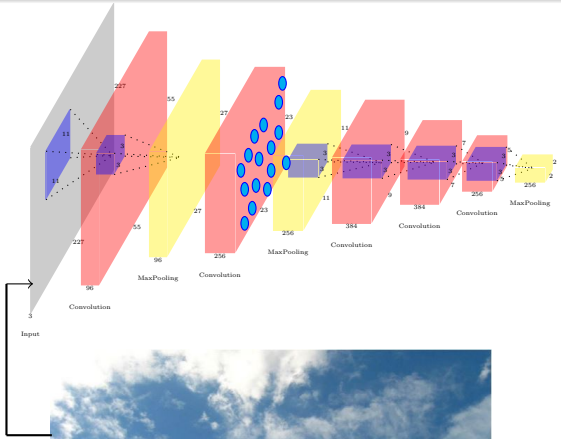
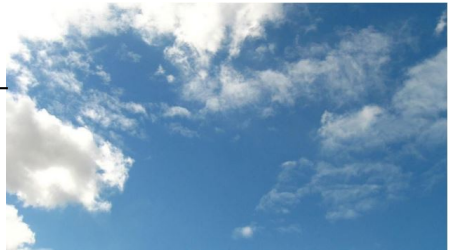




- So what exactly is happening here?

*research.googleblog.com/2015/06/inceptionism-going-deeper-into-neural.html





- So what exactly is happening here?
- The network has been trained to detect certain patterns (dogs, cat, birds etc.) which appear frequently in the ImageNet data

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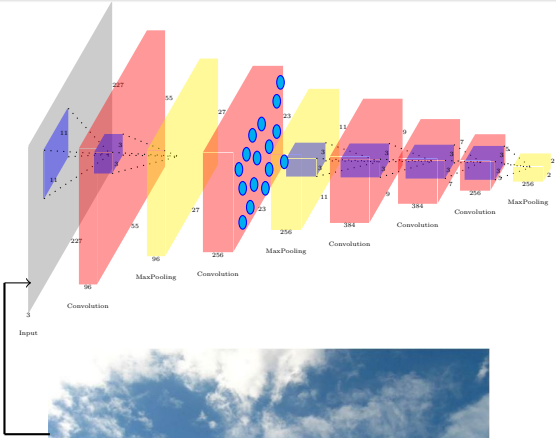
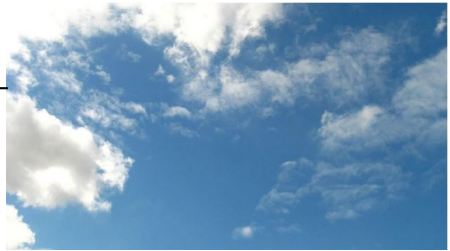
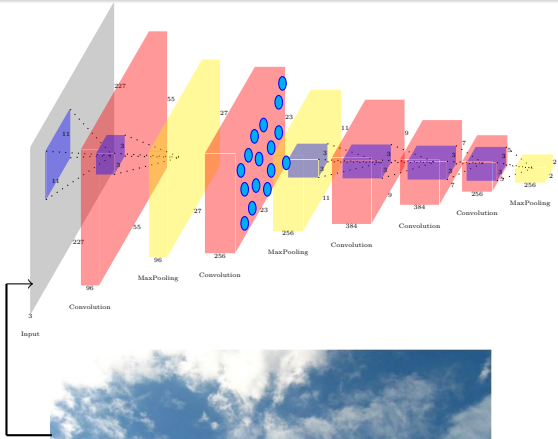
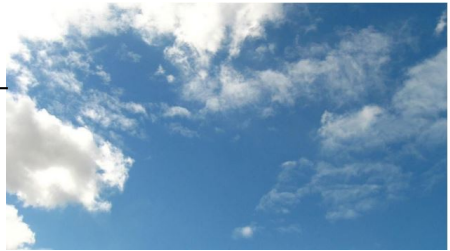




- So what exactly is happening here?
- The network has been trained to detect certain patterns (dogs, cat, birds etc.) which appear frequently in the ImageNet data
- It starts seeing these patterns even when they hardly exist

*research.googleblog.com/2015/06/inceptionism-going-deeper-into-neural.html



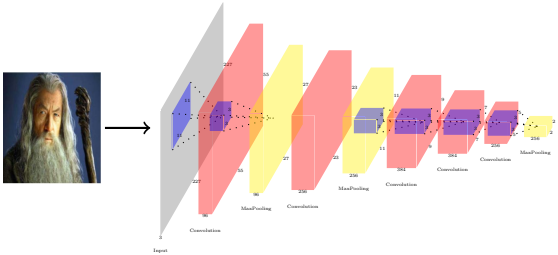


- So what exactly is happening here?
- The network has been trained to detect certain patterns (dogs, cat, birds etc.) which appear frequently in the ImageNet data
- It starts seeing these patterns even when they hardly exist
- *If a cloud looks a little bit like a bird, the network will make it look more like a bird. This in turn will make the network recognize the bird even more strongly on the next pass and so forth, until a highly detailed bird appears seemingly out of nowhere. - Google**

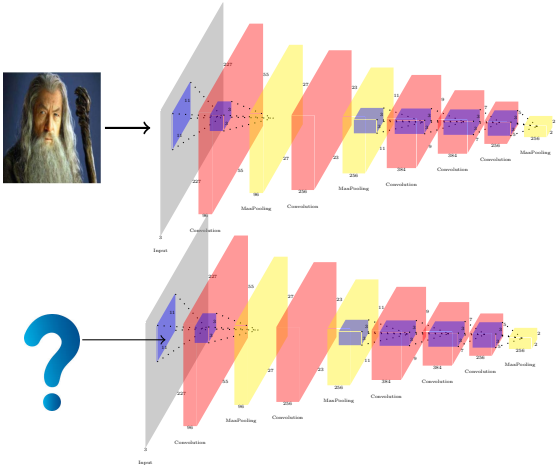
*research.googleblog.com/2015/06/inceptionism-going-deeper-into-neural.html

Module : Deep Art

- To design a network which can do this, we first define two quantities



- To design a network which can do this, we first define two quantities
- **Content Targets** : The activations of all layers for the given content image



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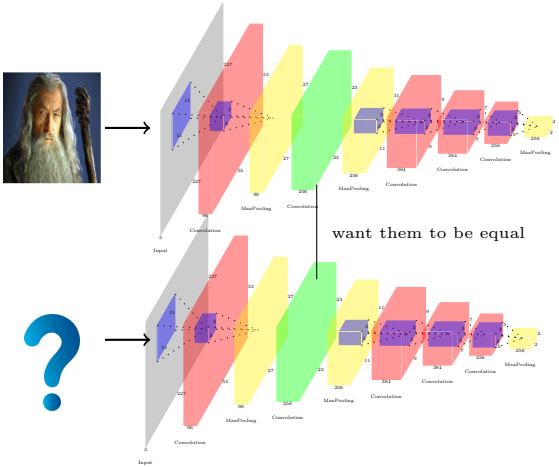

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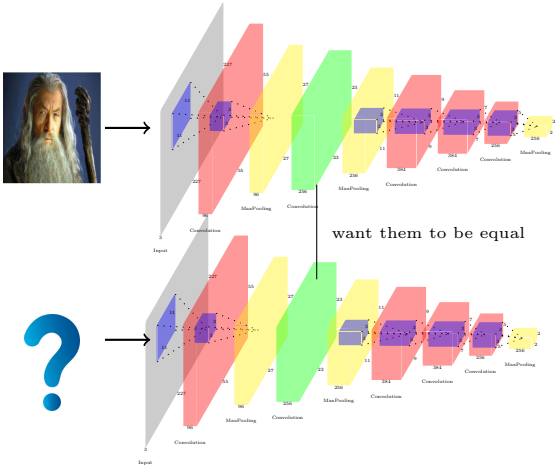

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- **Content Targets** : The activations of all layers for the given content image
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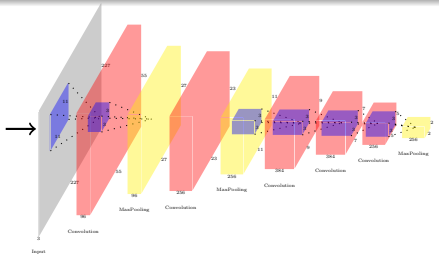
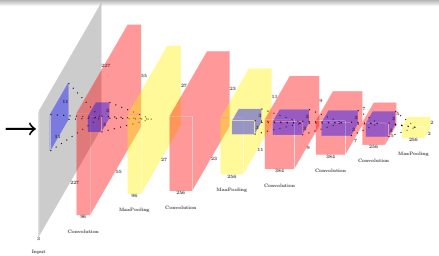
- Let $\vec{p}, \vec{x}$ be the activations of the content image and the new image (to be generated) respectively

$$\mathscr{L}_{content}(\vec{p}, \vec{x}) = \sum_{ijk} (\vec{p}_{ijk} - \vec{x}_{ijk})^2$$



- Next we would want the style of the generated image to be the same as the style image



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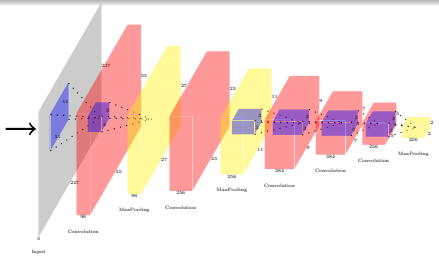
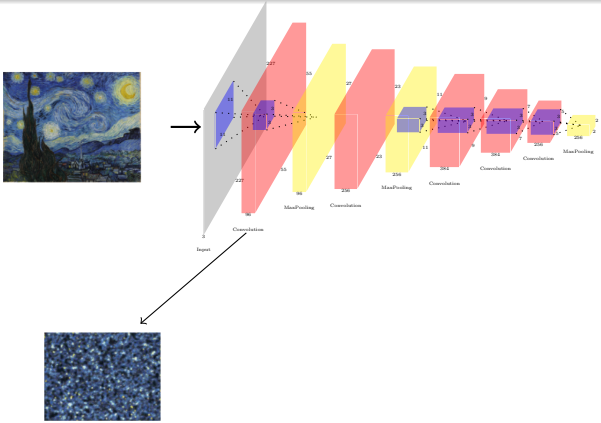
- How do we capture the style of the image?



- Next we would want the style of the generated image to be the same as the style image
- How do we capture the style of the image?
- Turns out that if $V \in \mathbb{R}^{64 \times (256 \times 256)}$ is the activation at a layer then $V^T V \in \mathbb{R}^{64 \times 64}$ captures the style of the image



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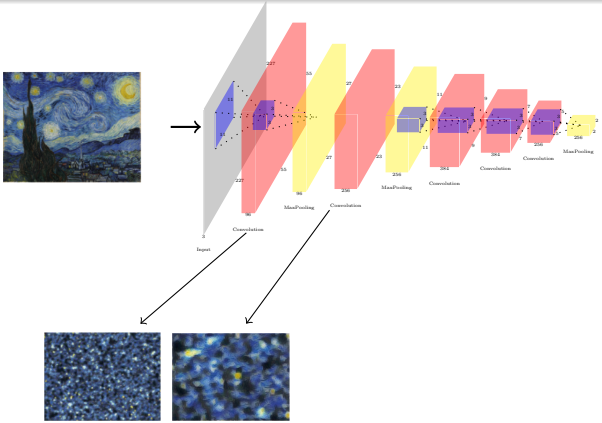
- The deeper layers capture more of this style information



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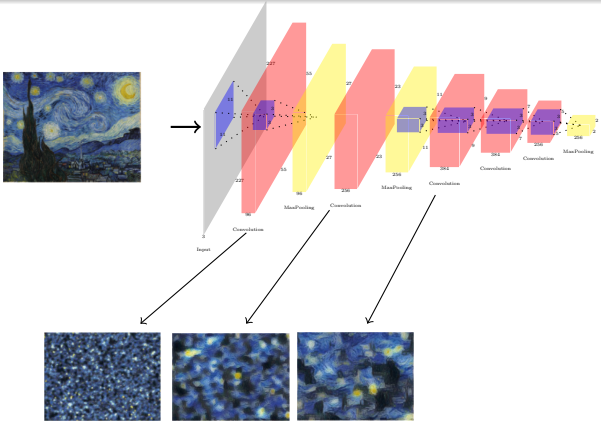

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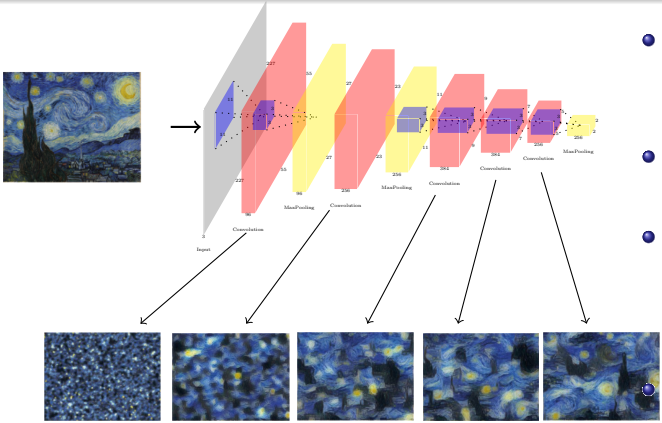

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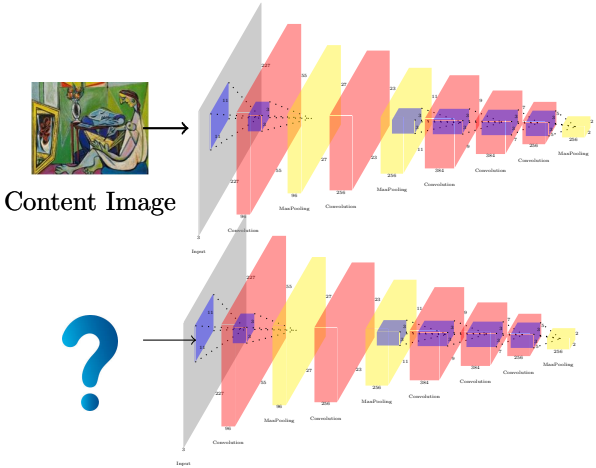
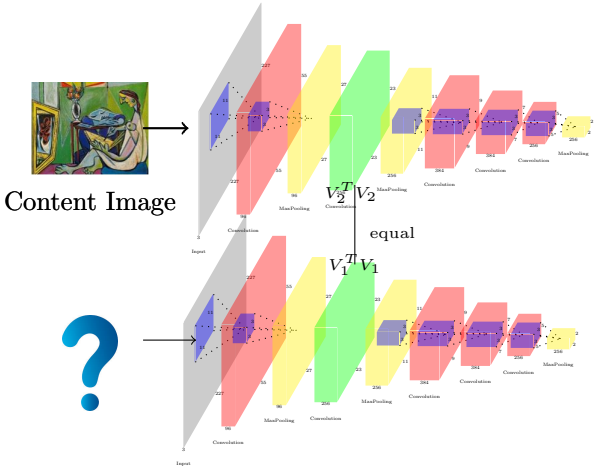


- To ensure that the style of the new image captured by layer ℓ matches the style of the style image, we can use the following objective function :



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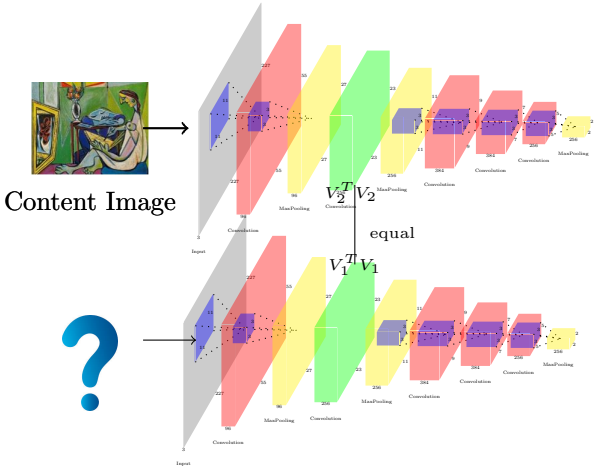
$$E_\ell = \sum_{ij} (G_j^\ell - A_{ij}^\ell)^2$$

where G^ℓ and A^ℓ are the style gram matrices computed at layer ℓ for the style image and new image respectively.



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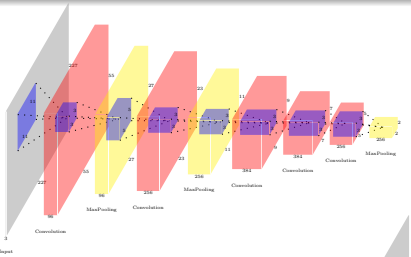
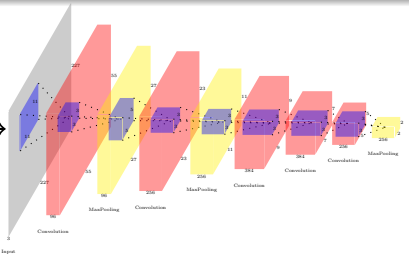
$$\mathscr{L}_{style}(\vec{a}, \bar{x}) = \sum_{\ell=0}^{L} w_\ell E_\ell$$


Input
Convolution
MaxPooling
Convolution
MaxPooling
Convolution
Convolution
Convolution
MaxPooling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Convolution
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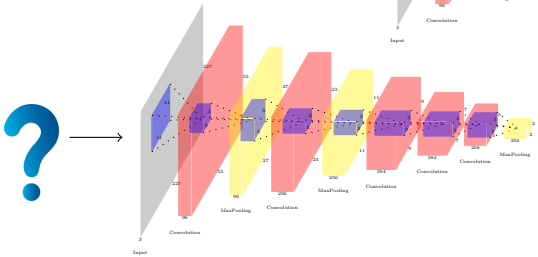
MaxPooling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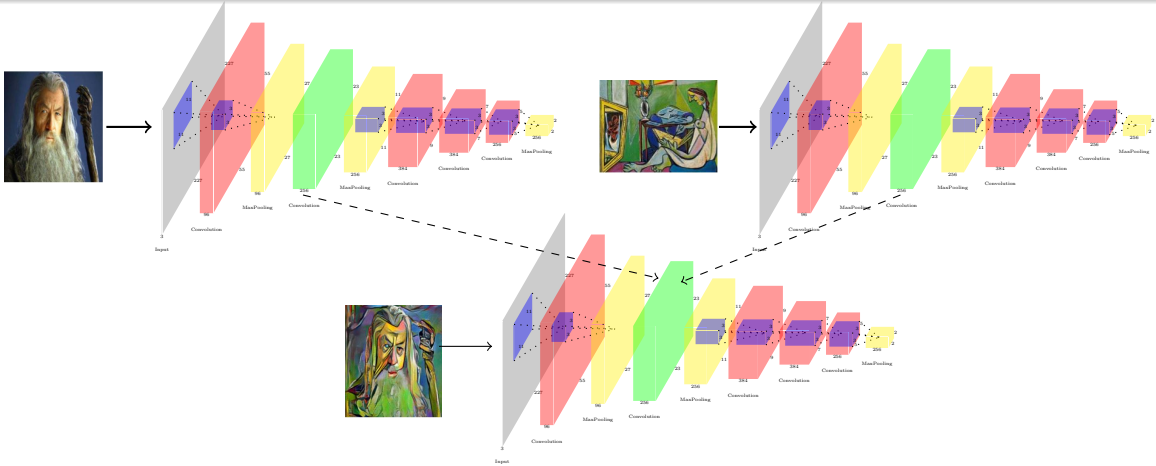
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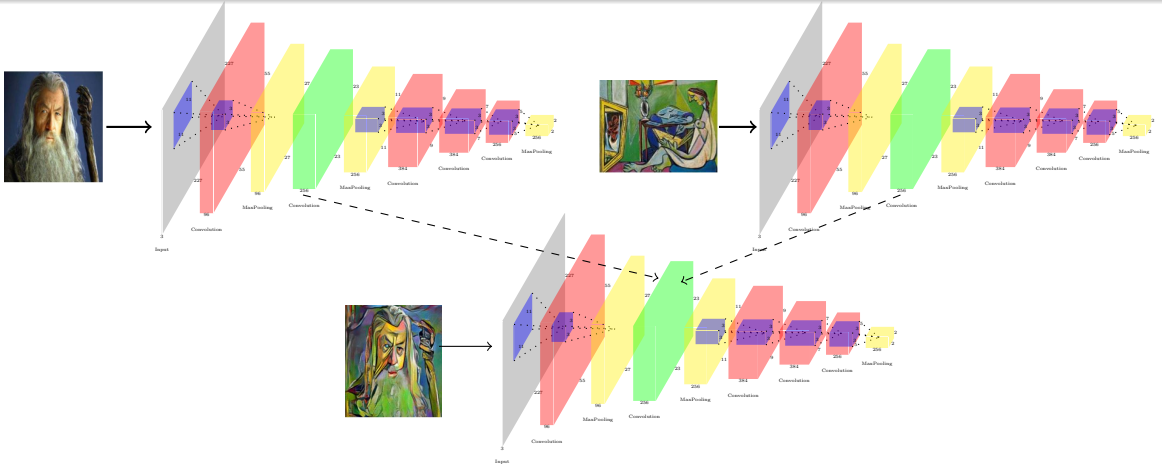

- The total loss is given by :-

$$\mathscr{L}_{total}(\vec{p}, \vec{a}, \vec{x}) = \alpha \mathscr{L}_{content}(\vec{p}, \vec{x}) + \beta \mathscr{L}_{style}(\vec{a}, \vec{x})$$

Module : Fooling Deep Convolution Neural Networks

- Turns out that using this idea of optimizing over the input, we can also "fool" ConvNets

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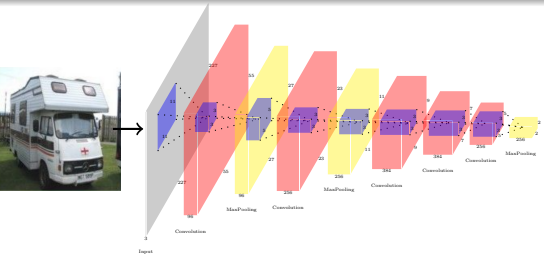
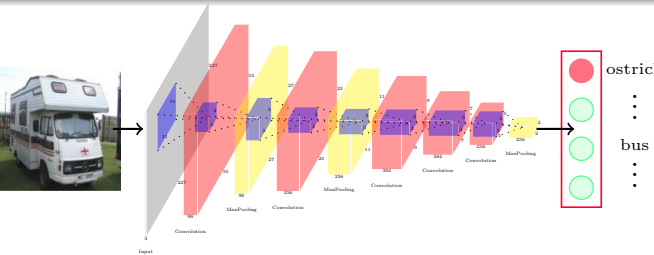
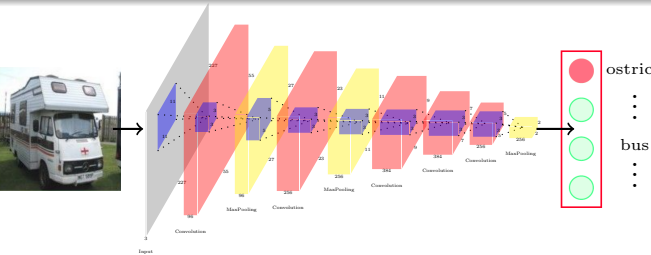
- Let us see how



- Suppose we feed in an image to a Convnet.



- Suppose we feed in an image to a Convnet.
- Now instead of maximizing the log-likelihood of the correct class (bus) we set the objective to maximize some incorrect class (say, ostrich)



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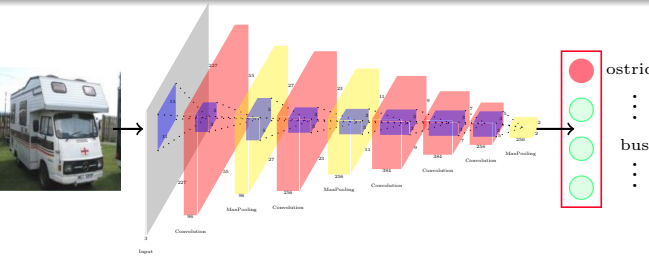
- Turns out that with minimal changes to the image (using backprop) we can soon convince the Convnet that this is an ostrich.



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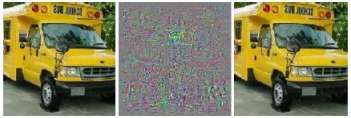
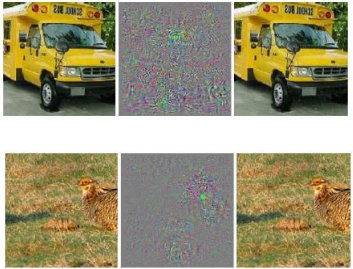
- Let us see some examples



- Notice that the changes are so minimal that the two images are indistinguishable to humans

*Intriguing properties of neural networks, Szegedy et al., 2013



- Notice that the changes are so minimal that the two images are indistinguishable to humans
- But the ConvNet thinks that the third image obtained by adding the first image to the second image is an ostrich

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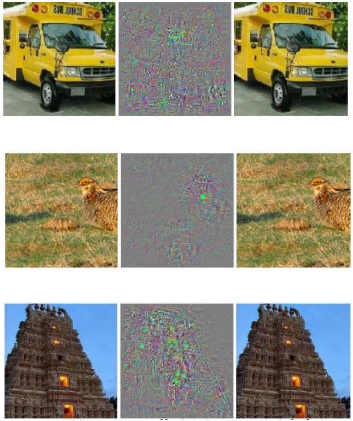

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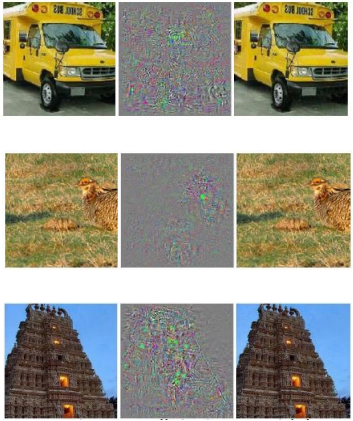

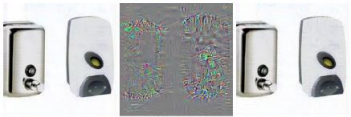

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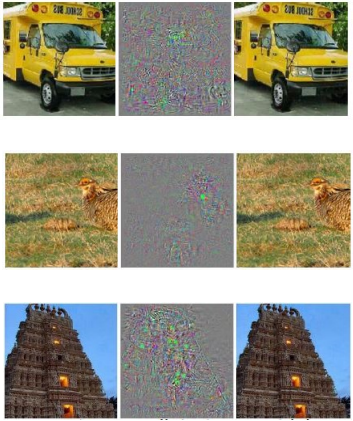

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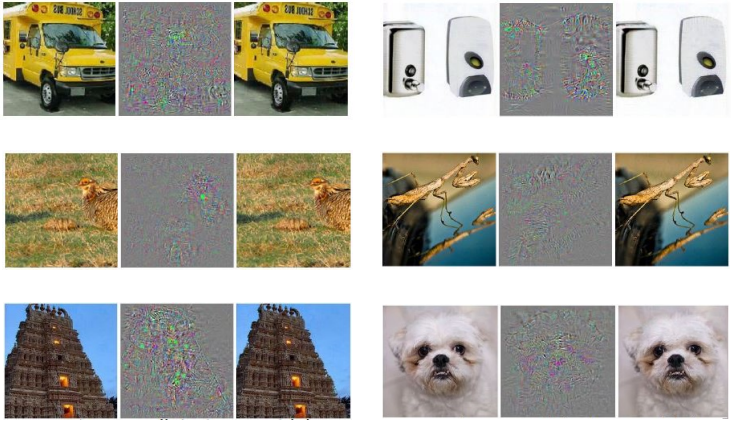

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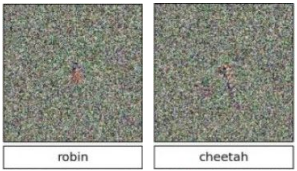
- We can also do this starting with random images and then optimizing them to predict some class.

*Neural Networks are Easily Fooled: High Confidence Predictions for Unrecognizable Images Nguyen, Yosinski, Clune, 2014



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- In all these cases the classifier is 99.6% confident of the class

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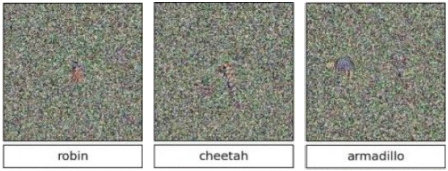

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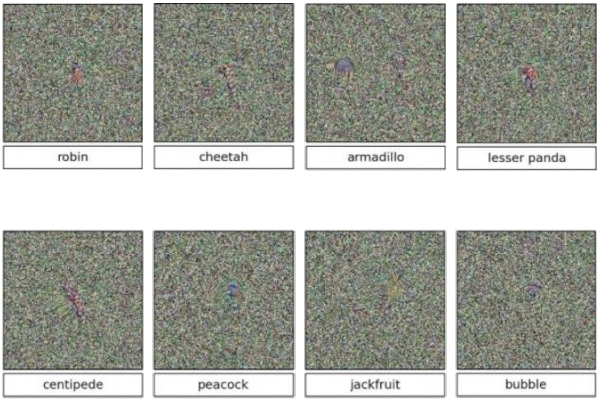

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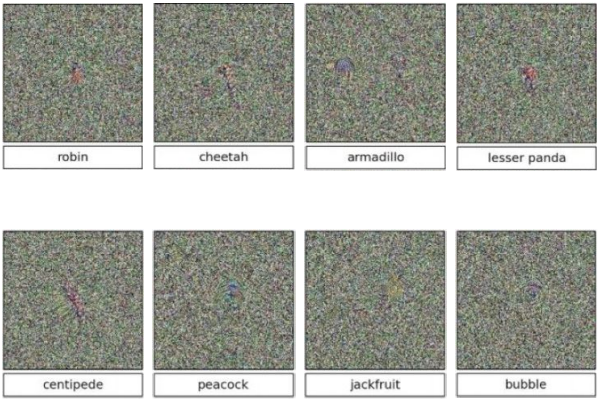

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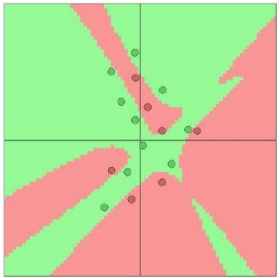
- Let us see an intuitive explanation of why this happens

*Neural Networks are Easily Fooled: High Confidence Predictions for Unrecognizable Images Nguyen, Yosinski, Clune, 2014

- Images are extremely high dimensional objects ($\mathcal{R}^{227\times227}$)



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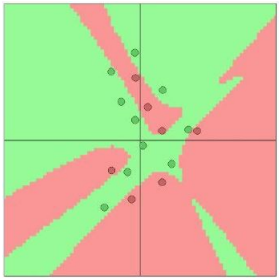

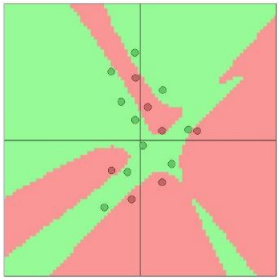

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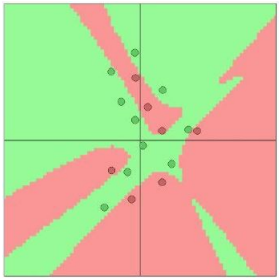
- Of these only a few are images (of which we see some during training)



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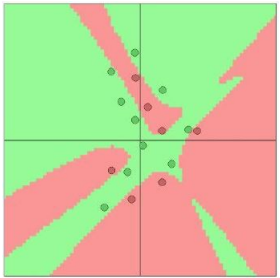

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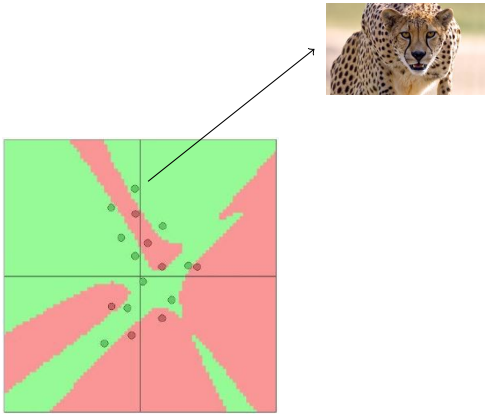

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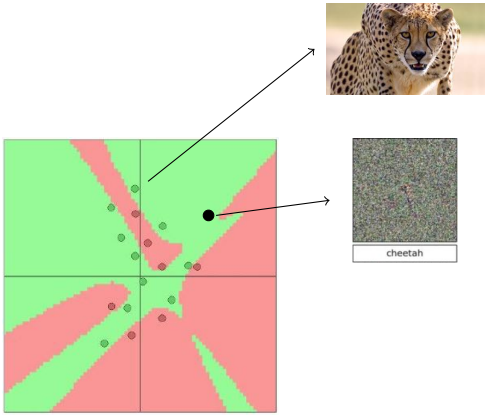

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